

Automated Diabetic Retinopathy Diagnosis — EN manuscript with GOST [N] citations

> **STAGE-G (final pass) — 2026-06-17.** Working author-year citations converted to numbered [N] (GOST 7.32-2001 §6.11, by first appearance). 102 external sources numbered [1]–[102]. Numbers are shared with the Kazakh manuscript (language invariance). Provisional until the experiment-gated chapters (most of Ch 4, Ch 5, Ch 0, Ch 7) join the assembly.

1 PROBLEM DOMAIN ANALYSIS AND CURRENT STATE OF AUTOMATED DIABETIC RETINOPATHY DIAGNOSIS

§1.1.1 Pathophysiology and Clinical Grading Systems

Diabetic retinopathy (DR) is a chronic complication of diabetes mellitus and one of the principal causes of preventable vision loss among working-age adults. Its scale is substantial: in a narrative review of DR pathophysiology, Kusuvara et al. [1] cited a pooled prevalence of any DR of 34.6% among diabetic populations worldwide, and Morya et al. [2], in a further narrative review of pathophysiology and treatment, observed that DR is present in approximately one-third of patients with diabetes. Both estimates are reported here as third-party clinical context rather than as findings of this dissertation, and both derive from non-systematic narrative syntheses whose epidemiological figures are inherited from the primary studies they cite. The clinical importance of DR for an automated screening system lies not only in its frequency but in the fact that its severity is defined by a discrete, ordered set of structural changes visible on the retinal fundus. This section establishes the biological basis of those changes and the standard five-class grading scale (DR 0–4) that discretizes them — the ordinal target taxonomy every classifier in this dissertation must reproduce (per the central thesis, IT-1).

The pathophysiological process underlying these visible changes is, in its classical description, microvascular. Sustained hyperglycemia drives a cascade of metabolic and vascular injury in the retinal capillary bed. Kusuvara et al. [1] identified pericyte dropout from the retinal capillary walls as a consensus mechanism responsible for breakdown of the inner blood–retina barrier (p. 370), and described vascular endothelial growth factor (VEGFA–VEGFR2) signalling as pivotal in the retinal angiogenesis and vascular leakage characteristic of advanced disease (p. 369). The downstream consequences — increased vascular permeability, capillary non-perfusion, and progressive retinal ischemia — culminate, in the most severe form, in pathological neovascularization. Because this is a narrative review without systematic search methodology, and because the authors themselves note that the cellular and molecular mechanisms underlying DR "are not fully determined" given that hyperglycemic animal models reproduce only limited aspects of early disease (p. 364), these mechanistic statements are taken as the review's synthesis of the field rather than as settled fact.

A purely microvascular account of DR is, however, incomplete, and the cited literature converges on a broader framing. Wang and Lo [3] characterized DR as a combination of microvascular damage, inflammation, and retinal neurodegeneration, and reported that retinal neurodegeneration may precede the microvascular changes traditionally regarded as the disease's defining signs. Kusuvara et al. [1] similarly noted clinical and experimental evidence of irreversible neuronal loss preceding vascular lesions (p. 369), and Gettenger et al. [4], reviewing experimental DR models, argued that DR is multifactorial rather than solely hyperglycemia-driven and that retinal neurodegeneration may precede vascular pathology. Taken together, these convergent observations support understanding DR as a disorder of the retinal neurovascular unit rather than of the vasculature alone. This distinction matters for an imaging-based classifier: fundus photography renders the *microvascular* manifestations of the disease, so a model graded against the standard clinical scale necessarily learns the vascular signature of a process whose earliest pathology may be partly neural and not yet fully visible. Gettenger et al. [4] reinforced the heterogeneity of the disease in noting that no single experimental model reproduces the full spectrum of the human DR phenotype (p. 16), which implies a correspondingly heterogeneous range of imaging presentations across patients and stages.

On the fundus image itself, the disease declares itself through a characteristic, roughly ordered progression of lesions. The earliest and smallest are microaneurysms — focal outpouchings of weakened capillaries — followed by intraretinal (dot and blot) hemorrhages as those vessels rupture. Lipid and protein exudation from the leaking barrier produces hard exudates, while focal retinal ischemia produces cotton-wool spots (soft exudates) at the nerve-fibre layer. As the microvasculature deteriorates further, intraretinal microvascular abnormalities (IRMA) and venous beading appear, signalling extensive non-perfusion; in the proliferative phase, neovascularization — fragile new vessels at the disc or elsewhere — supervenes and risks vitreous hemorrhage and tractional detachment. Overlaid on this severity axis, and independent of it, is diabetic macular edema (DME): fluid accumulation at the macula that is a leading cause of central vision loss and, as Wang and Lo [3] noted, the principal indication for first-line intravitreal anti-VEGF therapy (p. 9). For an automated classifier these lesions are the operative image features, and their analytical salience is uneven: the microaneurysms and small hemorrhages that define the earliest, most screening-critical grades are small, low-contrast structures, whereas the neovascular and exudative changes of advanced disease are comparatively conspicuous.

The clinical grading scale partitions this lesion burden into severity classes. At the coarsest level, DR is dichotomized into non-proliferative DR (NPDR) and proliferative DR (PDR), the latter distinguished by the presence of neovascularization [5, p. 699]. Kesharwani et al. [5] is a narrative clinical review with documented citation inaccuracies, published in a journal flagged for questionable editorial practices, and is cited here only for this general disease-background distinction; its specific quantitative claims are not relied upon. Modern screening, however, uses a finer ordinal scale. As summarized by Morya et al. [2], DR severity is graded by

standardized systems — the Early Treatment Diabetic Retinopathy Study (ETDRS) scale and the International Clinical Diabetic Retinopathy (ICDR) scale — that resolve the disease into five ordinal levels: DR 0 (no apparent retinopathy), DR 1 (mild NPDR), DR 2 (moderate NPDR), DR 3 (severe NPDR), and DR 4 (proliferative DR). Progression from DR 1 to DR 3 is defined by the increasing number and extent of microaneurysms, hemorrhages, and microvascular abnormalities within the non-proliferative range, while the transition to DR 4 is marked by neovascularization. This DR 0–4 scale is the canonical five-class taxonomy adopted throughout this dissertation as the classification target (IT-1).

(Figure 1.1)

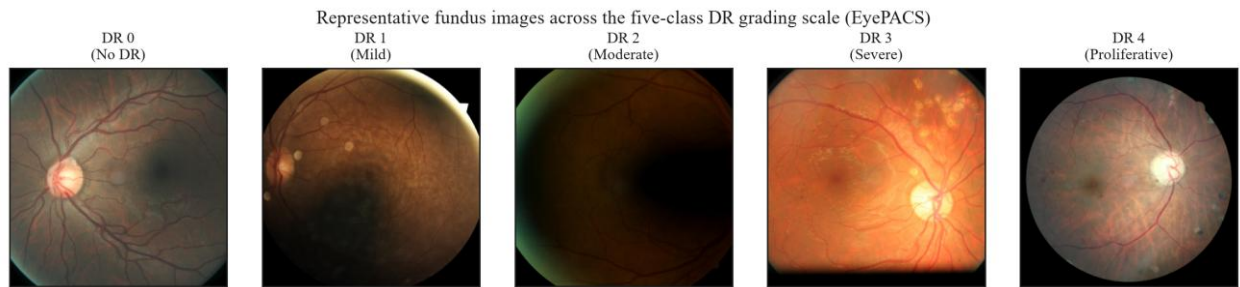


Figure 1.1 – Representative fundus images across DR grades 0–4 (clinical grading context)

Two properties of this taxonomy carry directly into the design of an automated grading system. First, the scale is *ordinal*: the five classes are ranked by severity, and the clinical cost of a misclassification scales with the distance between the predicted and true grade — confusing adjacent grades (for example, mild and moderate NPDR) is far less consequential than confusing no disease with proliferative disease. A grading model is therefore not solving an arbitrary five-way categorical problem but an ordered one with structured error costs, a property that motivates ordinally sensitive evaluation later in this dissertation. Second, the grade boundaries that matter most for early intervention rest on precisely the lesions identified above as small and low in contrast; distinguishing DR 0 from DR 1, or DR 1 from DR 2, can hinge on a handful of microaneurysms occupying a few pixels of the image. The representative grade-stratified fundus images in FIG-1.1 illustrate this difficulty, with the visible lesion burden changing only gradually across the lower grades. The combination of subtle, sparse early features and clinically unequal, ordered errors frames a recurring concern of the chapters that follow: the visibility of diagnostically decisive structures depends heavily on image quality and acquisition conditions, a dependence examined in §1.2.

Early detection is clinically decisive because the disease is, in its earlier grades, treatable. Kesharwani et al. [5] reported — and this figure is treated strictly as cited clinical motivation, subject to the source's noted reliability limitations — that patients with advanced retinopathy retain a high probability of preventing vision loss if treated before the retina is severely damaged (p. 701), and Wang and Lo [3] documented the role of anti-VEGF agents as first-line therapy for DME and PDR (p. 9). The therapeutic window these statements describe exists only if the disease is identified while still in its lower, more subtle grades, which establishes the demand

for accessible, reliable screening. That demand, and the structural barriers to meeting it in resource-limited settings, are the subject of §1.1.2; this section has fixed the disease, its visible signs, and the DR 0–4 grading taxonomy that any such screening system must learn to reproduce.

§1.1.2 Screening Requirements in Resource-Limited Healthcare Settings

The therapeutic window established in §1.1.1 — the fact that diabetic retinopathy is treatable while it remains in its lower, subtle grades — is clinically meaningful only insofar as the disease is actually detected during that window. Because the early grades are frequently asymptomatic, detection depends not on patient-initiated presentation but on systematic, repeated screening of the entire diabetic population. This converts the clinical imperative of §1.1.1 into a logistical one: the population at risk is large and growing, each member requires periodic re-examination, and the examination itself is a specialist task. The gap between the volume of screening required and the capacity available to deliver it is the structural problem this section establishes, and it is the problem that motivates an automated, deployable decision-support classifier as a screening aid.

That the required screening is not, in practice, being delivered is visible in observed compliance. Morya et al. [2], in a narrative review of DR pathophysiology and treatment, reported that only 35–55% of patients with diabetes comply with recommended annual ocular screening. As a non-systematic review with an acknowledged English-only selection bias, the source synthesizes rather than measures this figure, and it is cited here as third-party clinical context rather than as a quantity established by this dissertation; nonetheless, a compliance ceiling of roughly half the at-risk population indicates that a substantial fraction of treatable disease goes undetected under current screening models. The shortfall is not primarily one of patient willingness but of access and capacity, and it is most acute where specialist infrastructure is thinnest.

The capacity constraint is structural rather than incidental. In prior work by the candidate, Yesmukhamedov et al. (2025) — a system-architecture study whose design contribution is developed in Chapter 6 of this dissertation — characterized the Kazakhstan context, in which approximately 1,200 ophthalmologists serve the entire national population, roughly 40% of residents live in rural areas, and an estimated 70% of rural residents have limited access to specialized eye care (pp. 77, 87). These figures are cited as the contextual framing reported in that prior publication. Their analytical significance is that manual grading scales linearly with specialist time: a fixed and geographically concentrated supply of graders cannot, by inspection, meet a screening demand distributed across a large, partly rural population. Under such an arithmetic, increasing screening coverage by adding specialists is not a near-term option, and the binding question becomes whether the grading step itself can be partly automated without removing the clinician from the decision.

The cited literature frames automation precisely as this kind of workload-reduction mechanism. Morya et al. [2] noted that AI-based systems are positioned to

improve screening efficiency and reduce ophthalmologist workload, and relayed reported operating points for deployed tools — for example, the IDxDR system at approximately 80% sensitivity and below 90% specificity, and the Remidio Medios system at 98% sensitivity and 86% specificity. These values are reported by that review without accompanying study-design detail or confidence intervals and are reproduced here only as cited third-party context, not as benchmarks against which this dissertation's eventual system is compared. The same prior candidate publication (Yesmukhamedov et al., 2025) further situated automation within a delivery model: it relayed that the United Kingdom's national DR screening initiative achieved roughly 80% coverage among diabetic patients, and that portable, telemedicine-oriented capture devices — a smartphone-based fundus camera costing on the order of \$5,000 — could extend screening into settings without tabletop ophthalmic equipment (p. 87). Such delivery models are what make a low-resource-deployable classifier operationally relevant, because the images they generate originate outside specialist clinics.

It is important to register what these precedents do and do not establish. The existence of a cleared autonomous system and of a national programme reaching 80% coverage shows that automated and telemedicine-based DR screening is a real, demonstrated capability — a counter-position this section must acknowledge rather than dismiss. But each of these results is a cited third-party outcome, achieved under its own population, device, and regulatory conditions; none is a result of this dissertation, and their portability to resource-limited, device-heterogeneous, low-infrastructure contexts is not entailed by their success elsewhere. Indeed, that portability is the open problem the later chapters address. The same prior publication additionally projected outcomes for a Kazakhstan deployment — access for more than four million rural residents and reductions in late-stage complications and cost — but these are explicitly third-party projections cited for contextual framing, not demonstrated results of this research, and they are treated as such throughout this dissertation.

Two boundary conditions follow from this deployment framing and bind the chapters that follow. First, the operative deployment context is a *resource-limited environment*, defined for this dissertation as a setting characterized by at least two of: absence of GPU acceleration for inference; available memory below 16 GB; near-real-time clinical throughput constraints; and network connectivity insufficient for continuous cloud reliance. The screening contexts described above — mobile units, portable cameras, rural clinics — instantiate this definition, and it is the computational envelope against which the dissertation's models are positioned. Second, automation here means decision support, not replacement: the system is conceived as an aid operating within a physician-in-the-loop paradigm, in which the clinician interprets, audits, and retains responsibility for the diagnosis. Neither the cited operating points nor the projected coverage figures are taken to establish standalone diagnostic reliability.

Finally, the data on which such screening depends are themselves geographically and demographically specific. Publicly available screening-research datasets are assembled from particular populations and devices — the IDRiD

database, for instance, was compiled from an Indian population for DR screening research [6] — which both supplies the material for developing automated graders and signals that performance is bound to the data on which it is established. This dataset-bound character of screening data is a recurring constraint in this dissertation and is examined directly once the technical preconditions for automated screening are in view. What this section has fixed is the system-level demand: population-scale screening against a structural shortage of specialists, to be met by a low-resource-deployable, decision-support classifier. Whether such a classifier can function on the images that real-world, often non-specialist, screening actually produces depends on the quality of those images — the subject of §1.2.

§1.2.1 Sources of Image Degradation in Clinical Practice

The screening contexts established in §1.1.2 — mobile units, portable cameras, and non-specialist operators working outside dedicated ophthalmic clinics — are precisely the conditions under which fundus images are least likely to be acquired ideally. Before the relationship between image quality and automated performance can be examined empirically, the acquisition-side sources of degradation must themselves be identified and organized. This section provides that taxonomy as a structured analysis grounded in the dissertation's operational definition of image quality; it asserts no measured degradation rate, deferring all quantified quality–performance relationships to §1.2.2. Under that operational definition (OD-1), image quality is not a free-standing aesthetic property but the measurable capacity of a fundus image to support automated detection of the microvascular features relevant to DR staging, assessed through downstream classification performance under matched conditions. Degradation, correspondingly, is any acquisition-side phenomenon that erodes that capacity — and, as developed below, the phenomena that do so are neither random nor diagnostically neutral.

The first and most direct axis of degradation is optical. Defocus, arising from imperfect focusing on the retinal plane, and motion blur, arising from involuntary eye movement or camera instability during the brief exposure, both act as low-pass filters on the image: they attenuate precisely the high-spatial-frequency content that distinguishes fine retinal structure from background. Because the earliest DR lesions — microaneurysms and small intraretinal hemorrhages — are themselves small, high-frequency features established in §1.1.1 as occupying only a few pixels, even mild optical blur can render them indistinguishable from the surrounding capillary bed. Optical degradation is therefore not a uniform loss of fidelity but a selective erasure of the most screening-critical signal.

A second axis is photometric. Fundus photography illuminates a curved, semi-reflective interior surface through a small pupil, and the resulting illumination is rarely uniform: peripheral vignetting, central reflex, and shading gradients impose large, low-frequency intensity variations across the field that are unrelated to retinal pathology. Compounding these are exposure errors — under-exposure that compresses dark detail and over-exposure that saturates bright regions — and globally low contrast. The diagnostic consequence is twofold. Uneven illumination

can mask genuine lesions in shadowed regions, and it can also mimic pathology, since a bright shading artifact is locally indistinguishable from a hard exudate and a dark gradient from a hemorrhage. Photometric degradation thus threatens both sensitivity and specificity, and it does so through structured, spatially correlated distortions rather than additive noise.

A third axis is geometric. Off-axis capture, misalignment between the optical axis and the pupil, truncation of the circular field of view, and variation in magnification all change where retinal structures appear and how large they are, without changing the underlying anatomy. For an automated grader, this undermines the spatial consistency on which feature learning depends: the optic disc and fovea — the natural landmarks anchoring lesion localization — may sit at inconsistent positions and orientations, and a truncated field may simply exclude peripheral lesions from the image altogether. Geometric degradation is the axis least visible to a human reader, who compensates effortlessly, yet most disruptive to a model that has not been explicitly normalized against it.

A fourth axis is patient- and media-related. Adequate imaging depends on a sufficiently dilated pupil and a clear optical media; both are frequently compromised in the diabetic population that screening targets. Insufficient mydriasis narrows the effective aperture and darkens the periphery, while media opacities — cataract in particular, but also vitreous opacity and corneal haze — scatter and attenuate light along the whole imaging path. Corneal reflections, lens flare, and dust on the optics add localized artifacts that, like shading, can be mistaken for or can obscure pathology. The distinguishing feature of this axis is that its causes lie partly in the patient rather than the instrument, so it cannot be eliminated by better camera engineering alone and will persist in any real screening population.

These axes are not independent of the deployment context. Each is aggravated in the resource-limited, portable, and non-specialist-operated settings established as the dissertation's target screening context in §1.1.2: handheld and smartphone-based cameras have smaller, less stable optics; non-specialist operators are less able to optimize focus, alignment, and illumination; and undilated, opportunistic screening encounters maximize media- and pupil-related degradation. The very settings in which automated screening is most needed are therefore the settings in which image degradation is most prevalent and most severe — a coupling that sharpens, rather than softens, the technical problem.

A reasonable objection is that modern cameras and automatic quality-control gating already discard unusable images at capture, marginalizing the problem. Quality-control gating does exist and is valuable, but it reframes the problem rather than removing it. Gating discards images rather than recovering their signal, which lowers effective screening coverage in exactly the resource-limited contexts where re-acquisition is costly or impossible; and sub-threshold degradation that passes the gate still erodes the small-lesion signal that early grading depends on. The analytical conclusion of this section is therefore that fundus image degradation is target-correlated: across all four axes, it preferentially attacks the small, low-contrast features that define the earliest and most treatable DR grades. This characterization is corroborated by the retinal-imaging literature, in which the same degradation factors

— uneven illumination, blur, and imaging artifacts — have been modeled explicitly and used to drive dedicated fundus-enhancement networks [7]; that study is an enhancement method whose degradation model is evaluated through vessel-segmentation and optic-disc detection rather than through DR-grade prevalence, so it grounds the taxonomy of this section without supplying a measured degradation rate. This correlation is what elevates degradation from a cosmetic nuisance to a diagnostic threat, and it is what motivates a preprocessing stage that explicitly normalizes geometry, illumination, and contrast before classification — the design developed in Chapters 2 and 3. The *measured* extent to which degradation translates into lost classification performance is taken up next, in §1.2.2.

§1.2.2 Impact of Image Quality on Diagnostic Model Performance

The degradation taxonomy of §1.2.1 describes how fundus images lose diagnostic signal; this section asks the quantitative question that follows from it and is answered in the empirical literature: how far does that loss translate into measurably lower automated-classification performance? Under the operational definition adopted in this dissertation (OD-1), the question is not rhetorical but constitutive — image quality is *defined* as an image's measurable capacity to support automated detection of DR-relevant features, so any demonstrated link between graded quality and downstream metrics is, in OD-1's terms, a direct observation of the construct itself. The cited evidence, examined below, indicates that image quality and data provenance are first-order determinants of measured performance — in places by margins that rival the effect of architecture choice — while also showing, importantly, that the relationship is not the simple monotone "discard the bad images and gain accuracy" that a naive reading might assume.

The most direct evidence holds architecture fixed and varies only quality. Fu et al. [8] re-annotated 28,792 EyePACS images into a three-level quality scale (Good, Usable, Reject) to form the EyeQ dataset, and reported that downstream DR-detection accuracy fell monotonically as quality declined — for a ResNet-50 classifier, from 0.9154 on Good images to 0.9150 on Usable and 0.9068 on Reject images — leading the authors to state that the performances of automated diagnostic systems are highly dependent on image quality. This is the cleanest available statement of the relationship, but its scope is bounded: EyeQ is derived from a single source (EyePACS), the evaluation uses an internal train/test split with no external validation, and no confidence intervals are reported, so the magnitudes are indicative rather than generalizable. Zago et al. [9] corroborate the link across databases, reporting strong inter-database generalization for a retinal image-quality classifier (DRIMDB, ELSA-Brasil), though it assesses overall gradability rather than DR-grade accuracy directly. That production DR systems treat quality as performance-critical is shown by Dai et al. [10], whose DeepDR system embedded an explicit image-quality-assessment stage ahead of lesion detection and grading across 466,247 images, albeit on largely private Chinese data.

A second line of evidence treats image quality as the parsimonious explanation for performance differences *between* datasets. Rakhlin [11], evaluating a modified

VGG-style network for binary referable-DR detection, reported an AUC of 0.967 on Messidor-2 against 0.923 on a withheld Kaggle/EyePACS partition, and attributed the Messidor-2 advantage not to superior generalization but to image gradability — approximately 100% of Messidor-2 images being gradable versus roughly 75% for Kaggle — while cautioning that images from different hardware or unrepresented populations "may sometimes result in decreased accuracy." This source occupies the P1 position in the dissertation's paradigm map: it applies a normalization pipeline but treats image quality as an exogenous property of the data rather than as a formalized model component, and no theoretical claim about preprocessing's importance is attributed to it. Its limitations are non-trivial — a non-peer-reviewed preprint with an underspecified pipeline, evaluating a binary referable-DR task (ICDR 1–4) rather than the five-class DR 0–4 scale of this dissertation — so the gradability attribution is read as the author's interpretation, not as an isolated experimental result.

The strongest evidence that data conditions can rival architecture comes from holding the architecture *identical* across data sources. Voets et al. [12], reproducing the Gulshan et al. [13] pipeline on public data with an InceptionV3 network, obtained an AUC of 0.951 on a Kaggle/EyePACS test set but only 0.853 on Messidor-2 — the same network, the same training procedure, a 0.098 AUC gap traceable to data provenance and labeling rather than model design. That a single architecture produces materially different performance across data sources is, in itself, evidence that data and quality conditions exert leverage comparable to architectural variation. The reproduction is cited here for its own finding, not as a comparison against Gulshan, which functions in this dissertation only as a paradigmatic reference and not as a numerical benchmark; and its authors are explicit that four entangled factors (distribution, single- versus multi-grading, per-patient versus per-image labels, and hyperparameters) cannot be separated, so the gap is attributed to data conditions collectively rather than to image quality alone.

That same study supplies the first of two nuances that prevent an over-strong reading. Voets et al. [12] found that excluding the roughly 19.9% of Kaggle images they judged ungradable did not significantly change algorithm performance. If image quality were a simple monotone lever, removing the worst images should have helped; that it did not indicates that aggregate quality-gating — discarding images below a threshold — is not a dependable route to better performance, and that the quality–performance relationship operates at the level of fine signal preservation rather than coarse inclusion or exclusion. A field study reinforces the cost of gating from the deployment side: Beede et al. [14], across eleven Thai clinics, found that automatic quality rejection of field-captured images reduced coverage and disrupted workflow — a socio-technical observation, not an accuracy measurement. The second nuance concerns task dependence. In the DeepDRiD benchmark, Liu et al. [15] reported that automated image-quality assessment itself reached only a quadratic-weighted-kappa and accuracy of roughly 0.65–0.70, which the authors characterized as insufficient for clinically feasible automatic screening; meanwhile, the team that won the regular-fundus DR-grading sub-challenge did so with minimal preprocessing, relying instead on training-strategy innovations such as multi-gate mixture-of-experts and online hard-example mining. These findings — drawn from

single-center Shanghai data, without confidence intervals, and using an accuracy-only image-quality metric — establish that preprocessing is necessary but not sufficient, and that its marginal payoff depends on the task and on what the rest of the pipeline already does.

The candidate's own prior work contributes a further data point, reported here as previously published results. In Sapakova, Yesmukhamedov and Sapakov (2025), image-enhancement preprocessing (resizing and augmentation) was found to raise validation accuracy from 71% to 86% on a small custom convolutional network applied to APTOS 2019 and private clinical images, with the enhanced configuration reaching a ROC-AUC of 0.9638. This result is consistent with the external evidence above, but its weight must be bounded precisely: it derives from a single small architecture and cannot be generalized to the broader class of efficient CNNs without explicit comparison; it is one of the candidate's prior studies that move toward, but predate, the integrated preprocessing-CNN formulation this dissertation develops; and it is reported once, since a companion card describes the same article and the two may not be treated as independent confirmation. The present dissertation formalizes and extends this line of prior work into the eight-stage pipeline specified in Chapter 3.

Read together, and with each source's limitations and differing tasks, datasets, and metrics held in view, the evidence converges on a precise position rather than a slogan. Image quality and data provenance are measured, first-order determinants of automated DR-classification performance, occasionally rivaling architecture in their effect; yet performance is not reliably recovered by discarding low-quality images, and preprocessing's benefit is task- and pipeline-dependent. This is exactly the reasoning that motivates the dissertation's design choice — to *condition images in* by normalizing geometry, illumination, and contrast as an integral part of the model, rather than to *gate images out* or to assume that any enhancement guarantees a gain. None of this licenses a universal claim that preprocessing improves performance on every dataset, nor an attribution of any future observed effect to preprocessing in isolation; those constraints are carried forward into the experimental chapters. One source of the quality and distribution variation evidenced here is not random at all but systematic and structured — the imaging device — and it is examined next, in §1.2.3.

§1.2.3 Device-Specific Variability in Fundus Imaging

The image-quality variation examined in §1.2.2 was treated there as a single distribution-shift problem, but it has two components of very different character. Much of the within-image degradation catalogued in §1.2.1 — defocus, motion, transient reflections, patient-dependent media opacity — is partly stochastic, varying from capture to capture even on one instrument. A second component is not stochastic at all: it is determined by the imaging hardware itself and is therefore systematic and reproducible. This section isolates that component, because a systematic shift behaves differently from a random one. A model can, in principle, average over random degradation seen across enough training images, but a systematic difference between the device that produced the training data and the

device that produces the deployment data is a coherent, repeatable transformation of the input distribution — precisely the situation that the machine-learning literature studies as domain shift.

Fundus cameras differ from one another along several reproducible axes of image formation. They differ in color rendition, because illumination spectra, sensor color filters, and on-board color processing are not standardized across manufacturers; in illumination geometry, which sets the vignetting and central-reflex pattern characteristic of a given optical design; in field-of-view angle, which determines how much retina surrounds the posterior pole; and in sensor resolution and optics, which set the finest detail recoverable. None of these is a transient capture error; each is a stable property of the camera model, so two cameras imaging the same retina yield images that differ in consistent, predictable ways. For an automated grader, this means that the appearance statistics a model learns to associate with a healthy or diseased retina are partly entangled with the device that produced the training images.

The dissertation's dataset architecture is deliberately constructed to span this device heterogeneity rather than to hide it. The primary training corpus, EyePACS, was acquired predominantly with Canon fundus cameras (Canon CR-1, per its dataset documentation); the external evaluation datasets introduce other manufacturers — RFMiD with Topcon (TRC-NW300) and Kowa cameras, DDR with Canon and Topcon, and ODIR-5K with Canon and Zeiss. These camera attributions are cited as documented acquisition properties of the public datasets rather than as findings, and two of the external sets, RFMiD and ODIR-5K, carry multi-disease taxonomies from which only the diabetic-retinopathy subset is used, so their labels must be mapped to the five-class DR scale before any cross-device comparison and their demographic and equipment differences acknowledged. The point of assembling such a device-diverse corpus is that cross-device generalization can then be observed directly rather than assumed.

That observation is anchored in an established theoretical frame. The performance loss a model suffers when its deployment distribution differs from its training distribution is the subject of domain generalization. Zhou et al. [16], in a survey of the field, noted that most statistical learning algorithms rest on an oversimplified assumption that source and target data are independent and identically distributed, and that a learning agent trained only with source data will typically suffer significant performance drops on an out-of-distribution target domain. As a general (non-retinal) survey without original experiments, this source supplies conceptual grounding rather than retinal evidence, but the grounding is directly applicable: a fundus camera defines a domain, and a model trained on one camera's distribution is, with respect to another camera, exactly the out-of-distribution learner the survey describes. The same literature catalogues the strategies proposed to mitigate such shift; Wang and Deng [17], surveying deep domain adaptation, organized these into discrepancy-based, adversarial-based, and reconstruction-based method families — again a general taxonomy, cited here to situate device shift as an instance of a well-characterized problem rather than as a novel or intractable one.

(Figure 1.2)

Cross-Dataset Comparison — 8 Active Datasets
All datasets used in dissertation experiments (H-1 through H-7)

Dataset	Size	Camera	Resolution	FOV	Population	Taxonomy	Tier
EyePACS	~35,126	Canon CR-1	3888×2592-5184×3456	45°	US (California)	5-class ICDR	Training
APTOS 2019	3,662	Various (mixed)	Various	Var.	Indian (rural)	5-class ICDR	Transfer
IDRID	516	Kowa VX-10a	4288×2848	50°	Indian (Nanded)	5-class + masks	Clinical
Messidor-2	1,748	Topcon TRC NW6	1440×960-2240×1488	45°	French	Referable / Non-ref.	Degradation
DDR	13,673	Canon, Topcon	Various	Var.	Chinese	6-class (excl. gr. 5)	Domain
ODIR-5K	5,000 pts	Canon, Zeiss	Various	Var.	Chinese (Beijing)	Keyword → DR grade	Domain
RFMiD	3,200	Topcon, Kowa	Various	Var.	Indian	Binary DR (0/1)	Domain
Clinical (KZ)	60	Mixed (regional)	Various	45°	Kazakh (Almaty)	5-class ICDR	Local val.

Camera coverage: Canon Topcon Kowa Zeiss — 4 manufacturers across 8 datasets
Total labeled images: ~63,589 · Populations: US, Indian, French, Chinese, Kazakh.

Figure 1.2 – Cross-dataset / multi-camera landscape comparison

Framing device variability as a structured domain-shift problem points to two distinct responses, both developed later in this dissertation, and it is important to state precisely what each can and cannot claim. The first response is preprocessing as a standardization mechanism: normalizing geometry, illumination, and contrast narrows the appearance differences that separate one camera's images from another's, which is part of the rationale for the pipeline specified in Chapters 2 and 3. A reasonable objection is that such normalization, applied aggressively, might already remove device signatures and render the problem moot at training time. The objection is only partly correct: standardization can narrow color and illumination differences, but it cannot provably erase shift that lives in resolution, optics, and field of view, and whether the residual shift still degrades classification is an empirical question rather than one that can be settled by assertion. That is why the second response is an explicit, controlled device-domain-shift evaluation (Experiment 6, §4.7), in which a model trained on EyePACS is tested across the camera groups represented in DDR, ODIR-5K, and RFMiD. This section motivates that experiment; it does not pre-judge its outcome, and no cross-device performance magnitude is claimed here.

A boundary must be drawn around what such a device-shift evaluation can ultimately establish. Demonstrating that a model maintains, or loses, performance across camera groups is an empirical observation of cross-device behavior; it is not a certification of device-agnostic deployment readiness, nor does it constitute regulatory compliance for use with any particular instrument. The dissertation's device-shift results, when they arrive in §4.7, are bounded to the specific camera models represented in the tested datasets and carry no certification claim — a boundary set here so that the motivation of this section is not later mistaken for a stronger guarantee.

With device-specific variability identified as the systematic, theory-grounded component of the distribution-shift problem, the clinical and technical characterization of the screening task is complete: the disease and its grading taxonomy (§1.1.1), the screening burden and deployment context (§1.1.2), the sources and measured impact of image-quality loss (§1.2.1–§1.2.2), and now the

structured device heterogeneity that any cross-site system must withstand. The chapter accordingly turns from the *problem* to the *methods* proposed to meet it. Section 1.3 begins that turn with the convolutional neural network architectures that constitute the learning machinery of automated DR classification.

§1.3.1 CNN Architectures for Medical Imaging

With the screening problem characterized — the disease and its grading taxonomy, the burden of population-scale detection, and the image-quality and device-shift conditions under which detection must operate — the chapter turns to the learning machinery proposed to meet it. Automated diabetic retinopathy classification is, in the present generation of systems, overwhelmingly a deep-learning task, and the workhorse model class is the convolutional neural network — the dominant approach across medical-image analysis more broadly, as surveyed by Litjens et al. [18]. A CNN learns a hierarchy of spatial features by composing convolution and pooling operations, building from local edge- and texture-detectors in early layers toward increasingly abstract, task-specific representations in deeper ones; FIG-2.2 schematizes this feature hierarchy. The relevant question for this dissertation is not whether CNNs can grade DR — they demonstrably can — but which architecture to adopt, and on what grounds, given a literature whose reported numbers are high but, as shown below, not directly comparable.

The architectures used in medical imaging today descend from a clearly traceable ImageNet-era lineage, and the dissertation's own backbones sit within it. Krizhevsky et al. [19] established the modern template with AlexNet, an eight-layer network whose combination of ReLU activations, dropout regularization, and PCA-based color augmentation reduced ImageNet top-5 error to 15.3%. Depth and multi-scale design were advanced by VGG's stacked 3×3 filters [20] and by the Inception family [21, 22], whose Inception- became the backbone of several landmark DR and dermatology systems. He et al. [23] addressed the degradation that afflicted deeper networks by introducing residual connections, allowing networks such as ResNet-50, -101, and -152 to be trained at previously impractical depths and reaching a 3.57% top-5 error in ensemble; ResNet-50 is one of the two backbones adopted in this dissertation. Huang et al. [24] extended the connectivity idea with DenseNet's dense feature reuse, and Tan and Le [25] introduced EfficientNet, whose compound scaling of depth, width, and resolution produced a family (B0–B7) reaching 84.4% ImageNet top-1 accuracy at B7; EfficientNet-B3 is the dissertation's second backbone. The subsequent EfficientNetV2 [26] improved training efficiency further and was reported to surpass a ViT-L/16 model pretrained on ImageNet-21k. These are ImageNet-classification references, cited for the architectural innovations that the dissertation inherits rather than for any DR-specific result.

Applied to diabetic retinopathy, this architectural family achieves strong but heterogeneous performance, and the heterogeneity is the analytically important fact. Pratt et al. [27], among the first to apply a CNN to five-class DR grading on the Kaggle/EyePACS data, reported 75% accuracy and 95% specificity but only 30% sensitivity — an early and instructive illustration that high headline accuracy under

class imbalance can coexist with poor detection of disease. Rakhlin [11], already discussed for its image-quality findings, reported an AUC of 0.967 on Messidor-2 for a VGG-style network, but on a binary referable-DR task rather than the five-class scale, and as a non-peer-reviewed preprint. Xu et al. [28] reported accuracy and AUC of 0.97 for an EfficientNet–Swin-Transformer hybrid on APTOS 2019, again five-class but on a different dataset. Liu et al. [15] found EfficientNet to be the dominant backbone among top DeepDRiD teams, with a leading quadratic-weighted kappa of 0.9303. Gargeya and Leng [29] reported an externally-validated AUC of 0.94 on Messidor-2, and Quellec et al. [30] detected referable DR while producing weakly-supervised lesion heatmaps — each on its own datasets and protocols. These figures span different datasets, class taxonomies, and validation protocols, and per the metric-consistency discipline adopted here they cannot be lined up as a ranking; what they jointly establish is that the architecture class is capable, not that any one configuration is best.

That heterogeneity extends to validation rigor, and several widely cited results require explicit caveats. Arora et al. [31] applied EfficientNetB0 to a Kaggle DR set and reported an average accuracy of 0.8653, but with a maximum training accuracy of 97.11% — an approximately eleven-point train–test gap indicating substantial overfitting — and reported no AUC or external validation; its result cannot be generalized to the broader class of efficient CNNs. Sharma et al. [32] proposed a ViT-CapsNet hybrid reporting 94% accuracy on EyePACS, but its per-class AUC values of 0.44–0.56 are internally incompatible with that accuracy, so the accuracy figure is cited here only with that inconsistency noted and the AUC figures are not relied upon. These are not peripheral blemishes; they are characteristic of a literature in which impressive single-number results frequently rest on single-dataset, internally-validated, or under-reported evaluations.

The same architectures generalize in form to adjacent retinal tasks and modalities, which both demonstrates their flexibility and marks the boundary of what such results say about DR-fundus grading. Wan et al. [33] applied a CNN to pixel-level lesion segmentation on IDRiD, a different task from image-level grading, as does the FGADR benchmark [34], which couples pixel-level lesion masks with grades. Khosravi et al. [35] reported strong external-validation AUCs (FastViT 0.9811, ResNet18 0.9626) but for retinal-hemorrhage classification rather than DR grading, and on a severely imbalanced set without confidence intervals. Ryu et al. [36] reported ResNet101 AUCs of 0.960–0.976 internally and 0.938–0.962 externally, but on optical coherence tomography angiography rather than fundus photography — outside this dissertation's modality scope. Beyond the retina, the same transfer-learning template reached dermatologist-level skin-cancer classification [37] and age-related macular degeneration grading [38] — landmark medical-imaging CNNs, but on tasks outside DR-fundus grading. Each shows the architecture transferring to a related problem; none is a fundus-DR result, and they are cited as such.

A second architectural current, the vision transformer, now runs alongside the CNN and must be addressed rather than ignored. Dosovitskiy et al. [39] showed that a pure transformer applied to image patches could reach 88.55% ImageNet top-1

accuracy with sufficient pretraining, framing the trade-off as one in which large-scale training can outweigh the convolutional inductive bias. Several DR and ophthalmic studies report transformers matching or exceeding CNNs: Goh et al. [40] found a Swin transformer [41] outperforming CNN baselines on Kaggle, SEED, and Messidor-1 in their experiments, and González-Díaz et al. [42] evaluated vision transformers on a four-class age-related macular degeneration task — a different disease from DR, cited only as a transformer-architecture reference. Geetha and Hema [43] reported 98.4% accuracy for a ViT-based hybrid, but disclosed neither the datasets used nor any external validation or statistical testing, so it is admitted only as a transformer-era architectural reference and not as evidence of generalization.

These transformer results are genuine findings on their respective datasets, and they constitute the strongest counter-position to a CNN-centred design. The dissertation engages it directly rather than dismissing it. The reported transformer advantages vary widely in validation rigor; the inductive-bias trade-off that favors transformers is most pronounced under the very large pretraining corpora that this dissertation does not assume; and EfficientNetV2 indicates that well-scaled CNNs remain competitive with transformer models. The dissertation's commitment to CNN backbones is therefore a deliberate design choice — one that enables CNN-native self-supervised pretraining, a four-channel input incorporating the FOV mask, and replication across two architecture families — and not a claim that CNNs are superior to transformers, a claim this dissertation does not make and its experiments are not designed to support.

This reasoning fixes the architectural commitment. Rather than adjudicate an unresolved architecture debate, the dissertation holds the architecture deliberately constant — adopting two established, well-characterized CNN backbones, ResNet-50 and EfficientNet-B3, chosen for their documented behavior and for the replication value of spanning two design families, with no claim that either is optimal or that the architecture space has been searched. This converts architecture from a contested variable into a controlled one, so that the experimental factor of interest becomes the preprocessing and pretraining configuration applied to these fixed backbones. How a backbone is initialized and adapted — by transfer learning from natural images or by in-domain self-supervised pretraining — is the next determinant of performance, and the one on which the dissertation's full-pipeline arm turns. It is the subject of §1.3.2.

§1.3.2 Transfer Learning and Self-Supervised Pretraining in Ophthalmic Diagnostics

The architectural commitment of §1.3.1 leaves open the question that most directly shapes a fixed backbone's performance: how its weights are initialized and adapted before and during fine-tuning on diabetic-retinopathy data. Initialization is not a neutral starting point. A randomly initialized deep network rarely converges to competitive performance on the moderate-sized labeled datasets available for DR, so essentially every modern system begins from pretrained weights and adapts them to the task. The choice of *which* pretrained weights — and of what learning process produced them — is therefore a design decision with first-order consequences, and it

is the axis along which the dissertation's full-pipeline configuration most fundamentally departs from its baseline.

The field's default answer is transfer learning from ImageNet. The landmark systems of the DR literature initialize a convolutional backbone with weights learned on the ImageNet natural-image classification task and then fine-tune on fundus data: Gulshan et al. [13] initialized an Inception-ensemble from ImageNet weights before training on their development set, and Saxena et al. [44] likewise transferred an ImageNet-pretrained InceptionResNetV2, reporting cross-dataset AUCs of 0.958 on Messidor-1 and 0.92 on Messidor-2 for binary DR detection. More recent hybrid designs follow the same template; Xu et al. [28] built an EfficientNet-Swin-Transformer model, reporting accuracy and AUC of 0.97 on five-class APTOS 2019, on a pretrained-then-fine-tuned basis. These figures arise on different datasets, class taxonomies, and validation protocols and are reported here only as evidence that ImageNet transfer is the prevailing initialization practice, not as a comparison among the systems.

That this practice works at all is notable, because it bridges a substantial domain gap. ImageNet consists of natural photographs of objects, scenes, and animals; a fundus image is a circular, illumination-dominated view of vascular and neural retinal tissue. The features a network learns to discriminate cats from cars are not obviously the features that distinguish a microaneurysm from a healthy capillary, and transferring across that gap is an in-practice success rather than a guaranteed one. The dissertation treats the degree of transfer as an empirical matter bounded to its architectures and datasets, not as a property that can be assumed.

The recognition of this domain gap, together with the chronic scarcity of expert labels in medical imaging, has motivated a broad turn toward learning paradigms that depend less on large labeled corpora. Cheplygina et al. [45], surveying not-so-supervised learning in medical image analysis, organized this turn into semi-supervised, multiple-instance, and transfer-learning strategies for settings where labels are limited. As a general survey of medical imaging rather than a retinal-specific or experimental study, it is cited here for its framing of the problem — that label scarcity makes representation learning from less-supervised signals attractive — and not for any retinal result.

The most direct response to the domain gap is in-domain pretraining: learning representations from unlabeled fundus imagery itself, so that the initialization already encodes retinal structure — vascular topology, optic-disc and macular morphology, illumination and artifact statistics — before any diabetic-retinopathy label is seen. Self-supervised learning makes this feasible without annotation, by training the network on pretext objectives defined on the images alone, and the dissertation's corpus now contains the primary self-supervised methods of the SimCLR, MoCo, BYOL, and DINO families together with masked-image modelling, as well as direct evidence that in-domain self-supervised pretraining on a large retinal corpus yields generalizable retinal representations [46] (RETFound) and that in-domain medical pretraining can exceed ImageNet transfer [47] (MICLe). These sources establish the general credibility of in-domain self-supervised pretraining; what they do not supply is the specific configuration adopted here. RETFound's evidence is obtained with a

vision-transformer backbone, whereas this dissertation holds the convolutional backbone fixed and pretrains it CNN-natively on the four-channel pipeline tensor — a configuration none of the cited works evaluates. The in-domain self-supervised pretraining used in this dissertation is therefore presented as the candidate's own methodological design, specified in §3.3.2, whose efficacy is to be established empirically rather than imported from the literature.

This sets up the dissertation's two-regime design. Under the primary hypothesis, the two arms of the experiment are slaved to the preprocessing condition: the baseline configuration uses ImageNet-pretrained weights, while the full-pipeline configuration initializes the *same* convolutional backbone — ResNet-50 or EfficientNet-B3 — from ophthalmology-specific self-supervised pretraining on an unlabeled fundus corpus, using no diabetic-retinopathy labels. Keeping the architecture identical across arms is deliberate: an earlier version of this design considered a transformer-based retinal foundation model [46] (RETFound), but adopting it would have changed the architecture as well as the initialization, confounding the comparison with an architecture switch; the chosen CNN-native self-supervised initialization changes only the pretraining stage and preserves the backbone used in the baseline.

A direct and important consequence follows, and it is stated here as a matter of design honesty rather than discovered later. Because the full-pipeline arm differs from the baseline along *two* axes at once — the preprocessing pipeline and the pretraining source — any performance difference the experiment eventually reveals is the joint contribution of both factors, not of preprocessing in isolation. The dissertation accordingly does not, and may not, claim that the preprocessing pipeline alone produces the observed effect; permissible claims are restricted to the integrated configuration as a whole — that the integrated full-pipeline-plus-ophthalmology-pretraining configuration outperforms the baseline-plus-ImageNet configuration on a given metric by a given margin. This is a deliberate scoping decision: the central thesis concerns preprocessing and the CNN as a single integrated model, so the integrated configuration is the proper object of test, and the alternative of changing the architecture to isolate preprocessing would have introduced a worse confound. Decomposing the joint effect into separate preprocessing and pretraining contributions is left explicitly to future work outside this dissertation's scope.

With initialization and adaptation established as a determinant co-equal with architecture, and the composite nature of the experimental contrast fixed, two of the three methodological pillars of automated DR classification are in place: the architectures that learn, and the pretraining that initializes them. The third is the means by which a trained model's decisions can be inspected and held to account — the explainability methods examined in §1.3.3, which complete this survey of the methods the later chapters build upon.

§1.3.3 Explainability Methods in Medical Image Classification

The two methodological pillars established so far — the convolutional architectures of §1.3.1 and the pretraining regimes of §1.3.2 — produce a trained

model whose internal decision process is opaque. A deep CNN maps a fundus image to a five-class grade through millions of learned parameters, and the clinical setting in which such a model would operate places a premium on understanding *why* it produces a given output. This is the concern of explainability, the third methodological pillar surveyed here. Tjoa and Guan [48], surveying explainable artificial intelligence toward medical applications, organized the field's methods along a taxonomy distinguishing perceptive interpretability — explanations a human can see, such as highlighted image regions — from interpretability grounded in the mathematical structure of the model. Samek et al. [49] framed the complementary problem of *evaluating* such explanations, comparing relevance-propagation approaches favorably against simple sensitivity analysis and proposing perturbation-based assessment. These surveys motivate explainability without privileging a single method, and they make clear that "explanation" is a family of distinct techniques answering distinct questions.

For image classification, the most directly useful family is the one that localizes the spatial evidence behind a prediction. Zhou et al. [50] introduced class activation mapping (CAM), which exploits a global-average-pooling layer to project class-specific importance back onto the image, achieving weakly-supervised localization with a 37.1% top-5 localization error on the ILSVRC benchmark. CAM's requirement of a particular architectural component limited its applicability, and Selvaraju et al. [51] generalized it with Grad-CAM, which uses the gradients of a target class flowing into the final convolutional layer to weight that layer's feature maps, producing a class-discriminative localization map for essentially any CNN without architectural modification or retraining. Chattopadhyay et al. [52] subsequently refined the technique as Grad-CAM++, improving localization quality and the handling of multiple instances of a class within an image, reporting a lower average confidence drop (36.8% versus 46.6%) than Grad-CAM on ImageNet-class data. These figures are properties of the methods measured on natural-image benchmarks, not results on retinal lesions, and are reported as such.

A second family explains predictions without reference to spatial activation. Lundberg and Lee [53] proposed SHAP, a unified additive feature-attribution framework with theoretical guarantees of local accuracy, missingness, and consistency, and Ribeiro et al. [54] introduced LIME, which explains any classifier locally by fitting an interpretable surrogate model around a single prediction. These attribution methods are powerful and model-agnostic, but they answer a feature-importance question rather than producing the pixel-aligned spatial map that a retinal application most naturally compares against lesion locations. In this dissertation they are therefore retained as explainable-AI background rather than adopted as the operational tool.

The dissertation selects Grad-CAM from the activation-mapping lineage as its primary interpretability method, for a specific and bounded reason. Because Grad-CAM produces a spatial map over the image, that map can be compared directly against pixel-level lesion annotations — available in the IDRiD dataset — to quantify how much of a model's class-relevant activation falls on annotated lesions. This comparison is the basis of the explainability metrics introduced later: an attention–

lesion overlap (ALO) as the primary measure and an intersection-over-union (IoU) as a secondary one, evaluated in Experiment 4. Grad-CAM is preferred over CAM for its applicability to the dissertation's backbones without architectural change, and over the attribution methods for the natural spatial comparison its output affords; Grad-CAM++ and the attribution methods remain available as supporting or contrasting references.

It is essential to state precisely, and as a binding constraint, what such an overlap can and cannot mean. A Grad-CAM map indicates regions of high gradient-weighted activation in the final convolutional layer; it does not constitute clinical localization of pathology, and it is not a pixel-level diagnostic delineation of lesion boundaries. A region of strong activation is evidence of where the network's class signal concentrates, not a segmentation of disease. It follows that even a high attention–lesion overlap must not be read as a validation of the model's diagnostic reasoning: overlap is consistent with clinically plausible attention, but it does not establish that the prediction is *caused* by the lesion features in any mechanistic sense, nor that the activations trace lesion boundaries accurately. The tempting inference — that a model shown to "look at the lesions" has had its reasoning clinically validated — is precisely the inference this dissertation does not draw. Grad-CAM provides post-hoc interpretability, not a mechanistic explanation, and its use in Experiment 4 is framed throughout as explainability evidence rather than as clinical validation.

With this boundary fixed, the explainability pillar is in place alongside architecture and pretraining, and the survey of methods that the experimental chapters build upon is complete. The chapter has, to this point, assembled the problem (the disease, screening burden, image-quality, and device-shift conditions of §§1.1–1.2) and the method components proposed to address it (the architectures, pretraining strategies, and explainability tools of §1.3). What remains before the research problem can be formulated is to examine how these components have actually been assembled into working automated DR screening systems, and where those systems fall short. That critical analysis is the subject of §1.4.

§1.4 Critical Analysis of Existing Automated DR Screening Systems

The preceding sections assembled the problem of automated diabetic-retinopathy screening and the method components — architectures, pretraining, and explainability — used to address it. This section examines how those components have been assembled into working systems, and what a critical reading of that body of work reveals. The purpose is not to rank these systems or to position the present work as their competitor; it is to locate, beneath an undeniably impressive record of reported performance, a consistent methodological pattern that constitutes the gap this dissertation is designed to fill.

That the field has produced high-performing systems is not in dispute. The landmark demonstration that a deep convolutional network could detect referable diabetic retinopathy at expert-comparable levels reported areas under the ROC curve of 0.991 and 0.990 on two clinical validation sets for a binary referable-DR endpoint [13]; these figures are cited here only as a historical and contextual reference, and a

direct numerical comparison with the present work would be unsound given differences in task, architecture, pretraining, data partition, and reference standard. Beyond the research setting, autonomous systems have reached regulatory clearance: the pivotal trial of the IDx-DR system reported a sensitivity of 87.2%, a specificity of 90.7%, and an imageability rate of 96.1% across 900 patients at ten primary-care sites, supporting its authorization as an autonomous diagnostic device [55]. That trial, however, reported no preprocessing ablation and no comparison against a public benchmark, so it speaks to deployed clinical performance rather than to the contribution of any individual pipeline component.

The literature has also matured well beyond single-site internal validation. Ting et al. [56] validated a screening model achieving an AUC of 0.936 for referable DR on a primary cohort and reported performance across ten additional multiethnic datasets with AUCs ranging from 0.889 to 0.983, though the development data were private. Bellemo et al. [57] extended external validation to a sub-Saharan African population, reporting a referable-DR AUC of 0.973 (sensitivity 92.25%, specificity 89.04%) in a Zambian cohort, on a single-country dataset. Zhang et al. [58] conducted a multicentre external validation across four centres and 83,465 images, reporting an image-level AUROC of 0.9931 and per-patient AUROC of 0.9848 for referable DR with expert-agreement kappa values of 0.86–0.93, but on data drawn entirely from China and without reporting the preprocessing applied. Ruamviboonsuk et al. [59] went further still, evaluating a system prospectively within Thailand's national screening programme across nine sites and 7,651 patients, reporting 94.7% accuracy and 91.4% sensitivity for vision-threatening DR and a statistically significant advantage over the participating specialists ($p = 0.024$) — a high-quality prospective result, but again single-country and without transparency on architecture or preprocessing. Clinical-validation studies of commercial tools tell a similar story: the RetCAD system was reported at an AUC of 0.988 with 90.5% sensitivity and 97.1% specificity and a 96% workload reduction on a private Spanish cohort [60], and Baget-Bernaldiz et al. [61] reported an AUC of 0.988 on their own population and 0.968 on Messidor for a four-class ICDR endpoint. At the level of pooled evidence, a meta-analysis of ten deep-learning screening studies in primary care reported a pooled sensitivity of 87%, specificity of 90%, and a summary ROC AUC of 0.9543 [62], which both confirms the field's aggregate strength and, by pooling heterogeneous studies, cannot be read as independent confirmation of any single system.

[TAB-1.1: Survey of existing automated DR screening systems — lit-derived]

System / Study	Task (taxonomy)	Dataset & population	Key reported metric (in context)	Validation type	Key limitation
Gulshan et al. [13] #12	Binary referable DR	EyePAC S-1 (US) + Messido	AUC 0.991 / 0.990 (binary)	Dual external (retrospective)	Preprocessing deferred to supplement; single grader-pool

		r-2 (France), private dev	— <i>contextual reference only</i>		reference standard
IDx-DR — Abràmoff et al. [55] #77	Binary referable DR (autonomous)	900 pts, 10 US primary- care sites	Sens 87.2% / Spec 90.7% / imageabil- ity 96.1%	Prospective pivotal (FDA)	No preprocessing ablation; no public- benchmark comparison
Ting et al. [56] #11	Referable DR	SIDRP (private) + 10 multieth- nic sets	AUC 0.936; external AUC 0.889– 0.983	Multiethnic external	Private development data
Bellema et al. [57] #115	Referable DR	4,504 images, Zambia (Africa)	AUC 0.973; Sens 92.25% / Spec 89.04%	Cross- population external	Single-country cohort
Zhang et al. [58] #44	Referable DR	83,465 images, 4 centres, China	AUROC 0.9931 image / 0.9848 patient; κ 0.86–0.93	Multicentre external	China-only; preprocessing not reported
Ruamviboon- suk et al. [59] #45	Vision- threatenin- g DR	7,651 pts, 9 sites, Thailand	Acc 94.7% / Sens 91.4%; p = 0.024 vs specialist s	Prospective (national programme)	Single-country; architecture/preproc- essing opaque
RetCAD — Sánchez- Gutiérrez et al. [60] #03	Referable DR	Private (Ramón y Cajal, Spain)	AUC 0.988; Sens 90.5% / Spec 97.1%; 96% workload	Clinical validation	Private data; reporting venue not stated

			↓		
Baget-Bernaldiz et al. [61] #07	Four-class ICDR	Spanish private + Messidor	AUC 0.988 own pop / 0.968 Messidor (4-class)	External	Single-population development
Saxena et al. [44] #02	Binary DR	EyePACS dev → Messidor-1/-2	AUC 0.958 / 0.92 (binary)	Cross-dataset	Binary task; P1 preprocessing practice
Wewetzer et al. [62] #14	Referable DR (pooled)	10 primary-care DL studies	Pooled Sens 87% / Spec 90%; SROC AUC 0.9543	Meta-analysis	Pools heterogeneous studies (not independent)

Caption. Metrics are reported with their dataset, population, and class-taxonomy context and are **not** mutually comparable: tasks range from binary referable-DR to four-class ICDR; data are largely private and single-country; reference standards differ. The Gulshan et al. [13] figures appear as historical/contextual reference only, not as a benchmark for the present work.

The first conclusion to draw from this landscape is that its numbers cannot be lined up as a ranking. The endpoints differ — binary referable DR in most systems, four-class ICDR in others, vision-threatening DR in the prospective Thai study — and so do the datasets, populations, camera hardware, and reference standards. A higher reported AUC on a private single-country cohort says little about performance on a different distribution, and comparing such figures across studies would violate the metric-consistency discipline maintained throughout this dissertation. This non-comparability is itself part of the gap: the field lacks a controlled, common-protocol basis on which the contribution of any individual design choice can be read off.

A second conclusion concerns reproducibility. As noted in §1.2.2, an independent attempt to reproduce the landmark system on public data recovered substantially lower performance — an AUC of 0.853 on Messidor-2 against the originally reported 0.990 — when proprietary data and code were unavailable [12]. This is not an isolated caution: a PRISMA-based systematic review of DR deep learning from 2016 to 2023 identified overfitting, class imbalance, and weak early-stage detection as recurring, unresolved weaknesses across the field [63]; a broader technical-and-clinical review of deep learning in ophthalmology similarly flagged data quality, generalizability, and clinical-translation barriers as open problems [64]. The headline metrics of the landscape, in other words, are fragile under independent replication and rest on datasets whose imbalance and provenance are not always controlled.

The third and most consequential conclusion, for this dissertation's purposes, concerns preprocessing. Across these systems, image preprocessing is treated as ancillary data preparation rather than as a formalized model component, and it is correspondingly under-reported: the landmark study defers its preprocessing protocol to supplementary material; the multicentre and prospective deployments do not report their preprocessing at all; and the cross-dataset CNN studies describe normalization only in passing. This is the observable methodological practice that this dissertation identifies as paradigm P1 — preprocessing as auxiliary preparation not requiring methodological discussion — and it is named here as a description of what these papers do, not as an attributed claim by their authors that preprocessing is unimportant. The consequence is that, despite the field's volume and clinical success, the contribution of preprocessing has rarely been isolated, formalized, or evaluated as an integral part of the model.

The deployment landscape compounds these limitations in a different way. In prior work by the candidate, Yesmukhamedov et al. (2025) compared nine existing ophthalmic AI systems and documented that the deployed tools are typically narrow or constrained — IDx-DR limited to diabetic retinopathy, systems such as DeepMind's requiring advanced imaging equipment, and cloud-based screening tools dependent on internet connectivity — while relaying reported operating points such as 96% sensitivity and 93% specificity for referable DR from IDx-DR and EyeArt. That comparative analysis, whose system-architecture contribution is developed in Chapter 6, also projected deployment outcomes for a Kazakhstan rollout; those projections are third-party figures cited for context and are not findings of this research. Commercial deployed systems are, moreover, predominantly binary and opaque, and some high-performing models operate on other modalities entirely — for instance the OCTA-based classifier of Ryu et al. [36], outside this dissertation's fundus-photography scope, or the device-independent OCT referral system of De Fauw et al. [65], likewise a different modality but notable for decoupling its analysis from specific scanner hardware through an intermediate segmentation representation. Transparent, five-class, device-robust systems whose preprocessing is fully specified and evaluated are scarce.

Taken together, these conclusions locate the gap precisely, and it is a methodological gap rather than a performance deficit. The existing systems are, in many cases, highly accurate and in some cases prospectively and clinically validated; the present work makes no claim to outperform them, and no head-to-head comparison against any named system is undertaken. What the literature has not systematically done is formalize preprocessing as an integral model component, report it in full, and evaluate its contribution under controlled, multi-dataset, transparent conditions that also probe cross-dataset transfer and device-domain shift. That this is the gap — and not a need for yet another high single-dataset metric — is what gives the dissertation its specific shape. The next section formulates this gap into the explicit research problem and the objectives that follow from it.

§1.5 Formulation of the Research Problem

The preceding sections converge on a problem that can now be stated precisely. The clinical and technical context of §§1.1–1.2 established that diabetic-retinopathy screening must operate at population scale, on images whose quality and device of origin vary systematically, and against a target — the small, low-contrast lesions of early disease — that is most vulnerable to exactly those variations. The methodological survey of §1.3 and the critical analysis of §1.4 then established that, while the field has produced accurate and in some cases prospectively validated systems, it has done so while treating image preprocessing as ancillary data preparation: under-reported in the main text of landmark studies, omitted entirely from several deployment reports, and rarely isolated or formalized as a component of the model. The reproduction evidence [12] and the systematic-review findings of overfitting, class imbalance, and weak early-stage detection [63] show further that the headline performance of this literature is fragile and unevenly validated. The gap is therefore not a deficit of accuracy but a deficit of method: the contribution of preprocessing to automated DR classification has not been formalized and evaluated as an integral part of the model under controlled, transparent, multi-dataset conditions.

The research problem this dissertation addresses follows directly from that gap. It is the question of how image preprocessing can be formalized as an integral component of a CNN-based diabetic-retinopathy classification model — rather than treated as separable, optional data preparation — and how its contribution can be evaluated under controlled and transparent conditions that are representative of the resource-limited environments in which screening must actually be delivered. This problem has a conceptual dimension and an empirical one: conceptually, it requires reframing preprocessing from auxiliary preparation (the paradigm this dissertation labels P1) to an integral model component (the paradigm P2); empirically, it requires a study design in which that reframing can be tested rather than assumed.

The dissertation's response, stated here as its central thesis and objective rather than as a demonstrated result, is to develop and experimentally validate an integrated fundus-image-enhancement and CNN-classification framework for multi-stage diabetic-retinopathy diagnosis. The framework's preprocessing is specified as an ordered eight-stage pipeline — canonical orientation, optic-disc/fovea rotation normalization, field-of-view crop with isotropic resize, field-of-view mask generation as a fourth input channel, adaptive flat-field illumination correction, contrast enhancement by an upgraded CLAHE in LAB color space with a dual-constraint clip limit, training-time augmentation, and dataset-specific normalization — applied together with a CNN classifier and treated, for evaluation purposes, as a single model. This integration is the conceptual contribution: the pipeline is not a preface to the model but part of it, defining the feature space the classifier operates on.

Because the framework is the object of study, the research problem decomposes into a set of bounded, pre-registered hypotheses, each testing one facet of the integrated framework under matched conditions. The primary hypothesis (H-1) concerns integrated-pipeline dominance: whether the full pipeline, paired with

ophthalmology-specific self-supervised pretraining of the backbone, outperforms a baseline of minimal preprocessing with ImageNet pretraining on the primary EyePACS data. It is essential to the honesty of this design to record that the two arms differ along two axes at once — preprocessing and pretraining source — so that any observed effect is the joint contribution of the integrated configuration and may not be attributed to preprocessing in isolation. The remaining hypotheses probe distinct properties: component ablation and CLAHE/illumination parameter sensitivity (H-2), cross-dataset transferability to APTOS 2019 (H-4), explainability through attention–lesion overlap on IDRiD (H-5), robustness across imaging devices (H-6), and resistance to clinical degradation on IDRiD and Messidor-2 (H-7). Each hypothesis is bounded to its datasets, architectures, and tested ranges, and each is held to the explicit evidence criteria the later chapters apply rather than to an informal sense of improvement.

Two boundaries define what this problem formulation does and does not commit the dissertation to. First, the aim is formalization and controlled evaluation, not competition: the dissertation does not set out to outperform any named existing system, and undertakes no head-to-head comparison against one. Its contribution is the controlled, transparent test of preprocessing as an integral component — a methodological contribution that stands whatever the direction of the result, since the hypotheses are falsifiable and a null or contrary finding is to be reported as such, not silently revised. Second, the scope is bounded throughout to five-class DR grading on fundus photography, to the specified family of datasets, and to resource-limited computational conditions; no generalization beyond this architecture is claimed, and the system-architecture work that situates the framework in a deployment context is a design contribution rather than a validated clinical system.

With the research problem formulated, its conceptual response named, and its empirical structure previewed, the work of Chapter 1 is complete: it has motivated, scoped, and located the contribution. Section 1.C consolidates these threads and sets the transition to Chapter 2, where the theoretical foundations of the enhancement, classification, transfer-learning, and explainability components are developed in the detail the framework requires.

§1.C Conclusions to Chapter 1

This chapter has established the problem domain of automated diabetic-retinopathy screening and located within it the specific contribution the dissertation will make. It began with the clinical and epidemiological grounding: diabetic retinopathy is a progressive neurovascular disease whose severity is operationalized into the five-class ordinal scale (DR 0–4) that constitutes every downstream classifier's target, and whose treatable early grades rest on small, low-contrast lesions detectable only through systematic, population-scale screening. That screening must be delivered against a structural shortage of specialists and within resource-limited settings, which establishes the demand for an accessible, decision-support classifier and the computational envelope under which it must operate.

The chapter then characterized the technical conditions that make this demand difficult to meet. Fundus images acquired in real screening practice degrade along optical, photometric, geometric, and patient-related axes, and that degradation is target-correlated — it preferentially attacks the small early lesions on which screening most depends. The empirical literature confirms that image quality and data provenance are first-order determinants of measured classification performance, though not through any simple discard-the-bad-images mechanism, and that one substantial component of this variation is systematic rather than random: the imaging device imposes a structured domain shift across cameras and populations.

Against this problem, the chapter surveyed the method components that automated systems assemble — the convolutional architectures that perform classification, the transfer-learning and self-supervised pretraining strategies that initialize them, and the explainability methods that make their decisions inspectable — and then examined how those components have been combined into working systems. That critical analysis was the chapter's pivot. Existing systems are accurate and, in places, prospectively and clinically validated, yet the literature as a body is heterogeneous and non-comparable across tasks and populations, fragile under independent reproduction, and — most consequentially — treats image preprocessing as ancillary data preparation, under-reporting it and rarely evaluating it as an integral model component.

From this the dissertation's research problem was formulated: how to formalize preprocessing as an integral component of a CNN-based DR-classification model and evaluate its contribution under controlled, transparent, multi-dataset conditions representative of resource-limited deployment. The response is the conceptual reframing of preprocessing from auxiliary preparation (the paradigm labelled P1) to integral model component (P2), operationalized as an integrated eight-stage preprocessing and CNN framework and tested through a set of bounded, pre-registered, and falsifiable hypotheses spanning integrated-pipeline performance, component ablation, cross-dataset transfer, explainability, device robustness, and degradation resistance. The chapter has been explicit about the terms of this contribution: it is a methodological one — formalization and controlled evaluation — and not a claim to outperform any named existing system; the primary hypothesis varies preprocessing and pretraining jointly, so its eventual effect is attributable to the integrated configuration rather than to preprocessing alone; and every hypothesis remains falsifiable, to be reported as observed.

With the problem motivated, scoped to five-class DR grading on fundus photography within the specified family of datasets, and located as a methodological gap with a structured response, the groundwork of Chapter 1 is complete. Chapter 2 develops the theoretical foundations on which the framework rests — the mathematics of contrast enhancement and CLAHE, the operations and training of convolutional networks, the principles of transfer and in-domain self-supervised learning, and the formal basis of the explainability metrics — in the detail the integrated framework and its experimental evaluation require.

2 THEORETICAL FOUNDATIONS OF IMAGE PREPROCESSING AND DEEP LEARNING FOR FUNDUS IMAGE ANALYSIS

§2.1.1 Histogram Equalization and Adaptive Contrast Enhancement

Chapter 1 closed by locating the research problem as a methodological one: fundus image quality — and the contrast of the small, low-contrast lesions that define the screening-critical early grades in particular — conditions what an automated classifier can learn, and the integrated preprocessing–CNN framework adopted in §1.5 treats the enhancement of that quality as a model component rather than ancillary data preparation. This chapter supplies the theoretical foundations on which that framework rests, beginning with the contrast-enhancement lineage that culminates in the pipeline's contrast stage (Stage 5, per the operational pipeline definition OD-3). The lineage proceeds in three steps — global histogram equalization, adaptive histogram equalization, and contrast-limited (clipped) adaptive histogram equalization — and the purpose of this section is to establish why each step is necessary and what control parameter the final form exposes, without presuming that contrast enhancement is uniformly beneficial.

Histogram equalization is the elementary operation at the root of this lineage. It redistributes the pixel intensities of an image so that the resulting intensity histogram is approximately uniform, which is achieved by mapping each intensity through the (normalized) cumulative distribution function of the image's intensities; intensities that are densely populated are spread apart, expanding the dynamic range in which most of the image's information lies and so increasing global contrast. As a foundational image-enhancement technique it serves as the conceptual basis for the adaptive variants developed below. Its defining limitation, for fundus imagery specifically, follows directly from its globality: a single mapping is derived from one image-wide histogram, which for a circular fundus photograph is dominated by the large, near-uniform dark region outside and around the retinal field of view. The mapping that best equalizes that background-dominated histogram is not the mapping that best resolves a cluster of microaneurysms occupying a few pixels in one quadrant, because global histogram equalization has no mechanism to adapt its transformation to local image content.

Adaptive histogram equalization (AHE) addresses this locality deficit by computing a separate equalization mapping for each of many local regions, or tiles, into which the image is partitioned, so that the contrast transformation applied at a given location reflects the intensity statistics of that location's neighbourhood rather than of the whole image. This recovers exactly the local contrast that the global method discards. It does so, however, at a characteristic cost. In a region that is nearly homogeneous — low intensity variance, as in a smooth retinal background or the interior of a large vessel — the local histogram is concentrated in a narrow band, and equalizing it produces a transformation with a very steep slope over that band. A steep mapping amplifies small intensity differences indiscriminately, which means that whatever sensor and quantization noise is present in an otherwise featureless region is amplified along with any genuine signal. Pizer et al. [66], in the

foundational study of adaptive histogram equalization and its variants, described AHE as "an excellent contrast enhancement method for both natural images and medical and other initially nonvisual images" while noting plainly that "the basic method is slow, and under certain conditions the enhanced image has undesirable features" (p. 355); that study reports qualitative image comparisons on medical modalities rather than quantitative or retinal-specific evaluation, and its observations are taken here as the methodological synthesis it presents rather than as quantitatively validated results.

The resolution of the noise-amplification problem is the contribution for which the same study is the methodological origin. Pizer et al. [66] established that "limiting the slope of the mapping function is equivalent to clipping the height of the histogram" (p. 363): if the local histogram is truncated at a maximum count before the cumulative mapping is formed — and the truncated excess redistributed across the histogram — the maximum attainable slope of the transformation is bounded, and with it the maximum noise amplification in homogeneous regions. This clipped form of adaptive histogram equalization is the direct precursor of contrast-limited adaptive histogram equalization (CLAHE) — later given its canonical clip-and-interpolate algorithmic form by Zuiderveld [67] — and is the form adopted, in an upgraded parameterization, as the pipeline's contrast stage. The height at which the histogram is clipped — the clip limit — is therefore the governing control of the method: it sets the trade-off between contrast gain and noise suppression. Pizer et al. [66] themselves noted that appropriate clipping levels "vary across imaging modalities and acquisition conditions" (p. 367), which is the first appearance in this dissertation of a recurring caveat (DGL-5): clip-limit values, and CLAHE parameters generally, are optimized for particular image distributions and architectures and are not asserted to be portable across them. The precise dual-constraint definition of the clip limit used by the pipeline is derived in §2.1.2; the present section fixes only that the clip limit exists and what it controls.

That contrast enhancement materially affects fundus-image analysis is supported, though unevenly, by the empirical preprocessing literature. Hayati et al. [68] evaluated CLAHE against unenhanced inputs across four CNN architectures for binary diabetic-retinopathy classification on the APTOS 2019 dataset (224×224 inputs, accuracy as the sole metric, single training runs without confidence intervals — a limited-scope conference study), and reported that CLAHE raised the test accuracy of EfficientNet-B4 from 95.57% to 97.83% and improved VGG16 and InceptionV3 as well. The same study, however, reported that CLAHE *degraded* ResNet34 by 12.02 percentage points under the uniform hyperparameter configuration applied across all four models, an outcome the authors attributed to the absence of per-architecture tuning (pp. 62–64). This result is decisive for the framing of the dissertation's contrast stage: it demonstrates that the effect of a contrast-enhancement step is parameter- and architecture-dependent, so no claim that contrast enhancement uniformly or universally improves classification is admissible. It is precisely this contingency that justifies treating the contrast stage as a formalized, parameterized component to be characterized empirically (§4.3.2) rather than as a presumptively beneficial fixed step. The metrics cited are bound to APTOS 2019

binary classification at 224×224 and are not directly comparable to the dissertation's five-class DR 0–4 task on EyePACS.

Two further preprocessing studies illustrate the breadth of the contrast-enhancement literature while carrying limited evidentiary weight. Shaout and Han [69], in an arXiv preprint evaluated only through a ten-person subjective preference survey on the single small DRIVE vessel-segmentation dataset, reported that a fuzzy-logic contrast-enhancement step blended with CLAHE (applied with clip limit 2.0 and an 8×8 tile grid) was preferred over either method alone; the result indicates that contrast-enhancement design choices affect perceived retinal-image quality but, lacking quantitative metrics and peer review, supports no stronger conclusion. Chakka [70], in a study published in a secondary-school science journal and reporting training curves without numerical performance values, compared four elementary preprocessing operations for retinal disease classification on optical coherence tomography images — a different imaging modality from the fundus photography to which this dissertation is bounded — and is cited only to note that systematic preprocessing comparisons exist in the retinal-imaging literature. Neither source is relied upon as primary evidence.

The candidate's own prior work sits within this lineage as an evolutionary precursor of the contrast stage. In previously published results (Sapakova, Yesmukhamedov & Sapakov, 2025), an upgraded CLAHE filter — replacing the conventional fixed clip value with a controllable global-threshold formulation — was integrated with a fine-tuned ResNet-50 for five-class retinal-disease classification on the STARE database, and reported 100% accuracy, sensitivity, and specificity on that small, rotation-augmented dataset. Consistent with the scope boundaries of this dissertation, that perfect-performance figure is not transferable to the present experimental context (SB-1.2): it was obtained on a 5-disease task distinct from DR 0–4 grading, on roughly 150 source images augmented to a balanced set, without explicit overfitting analysis, and it is reported here strictly as the authors' prior result rather than as a general capability of the method. The dissertation extends that precursor in two specific ways developed in the chapters that follow: it replaces the single-threshold clip rule with the dual-constraint clip-limit formulation derived in §2.1.2, and it integrates the contrast stage into the unified pipeline (Stage 5, OD-3) whose parameters are characterized by the sensitivity analysis of Experiment 2 (§4.3.2).

(Figure 2.1)

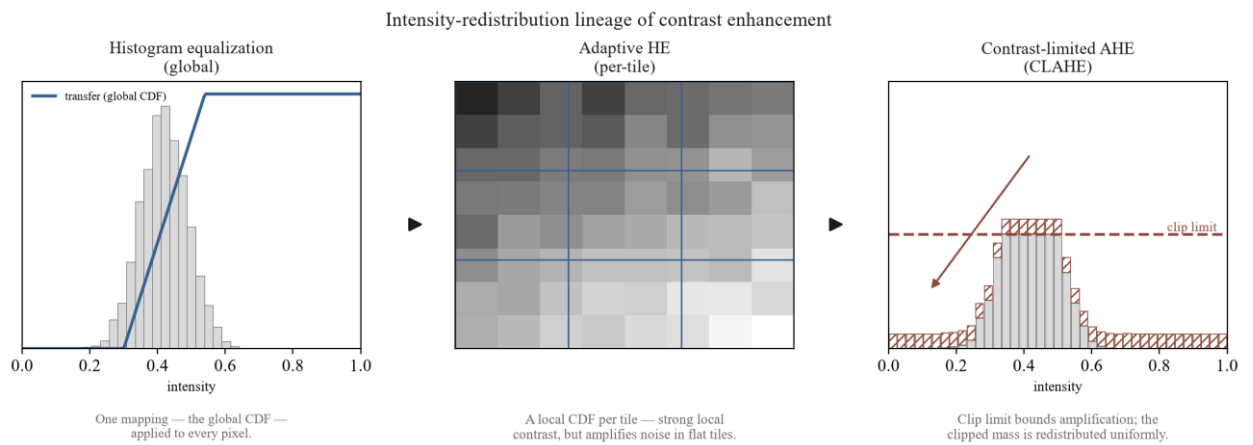


Figure 2.1 – Histogram equalization → adaptive equalization → contrast-limited (clipped) adaptive equalization – intensity-redistribution concept across the three stages of the lineage

In sum, the contrast-enhancement lineage establishes contrast-limited adaptive histogram equalization as the principled foundation for the pipeline's contrast stage: global equalization supplies the redistribution principle, adaptive equalization supplies the locality required to resolve focal lesions, and slope-limiting via histogram clipping supplies the noise control that makes local adaptation usable, with the clip limit as the single parameter governing the contrast-versus-noise trade-off. The empirical literature confirms that this stage is consequential but contingent, not universally beneficial. What remains is to formalize the clip limit itself — the dual-constraint definition that the pipeline adopts — which is the subject of §2.1.2.

§2.1.2 Formalization of CLAHE with Dual-Constraint Clip Limit

Section §2.1.1 fixed contrast-limited adaptive histogram equalization as the contrast foundation of the pipeline and isolated the clip limit — the height at which each tile's local histogram is truncated before its cumulative mapping is formed — as the single parameter governing the contrast-versus-noise trade-off, on the basis of the equivalence established by Pizer et al. [66] that "limiting the slope of the mapping function is equivalent to clipping the height of the histogram" (p. 363). What that section deliberately left open is the precise rule by which the clip limit is set. This section supplies that rule. It develops the formalization in three stages — the conventional contrast-limiting clip limit, the candidate's prior single-threshold upgrade, and the dual-constraint clip limit adopted by Stage 5 of the pipeline (OD-3) — and explains, for each, what the governing parameter controls and what failure mode the next refinement addresses.

Conventional contrast-limiting clip limit. In the standard formulation, the clip limit for a tile is set relative to the height the tile's histogram would have if its intensities were uniformly distributed. For a tile of area A pixels (the tile-area, the number of pixels in the contextual region) over L discrete gray levels, a uniform histogram assigns each level the average count A/L . The contrast-limiting clip limit truncates each histogram bin at a multiple of this average,

$$\text{CL}_{\text{conv}} = \beta \cdot \frac{A}{L}, \quad \text{where } \beta \geq 1, \tag{2.1}$$

where β is the clip factor: $\beta = 1$ clips the histogram to the uniform height (maximal contrast limiting), and larger β permits proportionally more contrast amplification before clipping engages. The clipped excess is redistributed across the histogram so that total tile mass is conserved. The same contrast-limiting principle appears in the candidate's prior work in the algebraically equivalent form $\text{CL} = \lceil L/T \rceil + \beta(\varphi - \lceil L/T \rceil)$, with φ the per-tile pixel population and T a threshold parameter (Sapakova, Yesmukhamedov & Sapakov, 2025, Eq. 1, p. 5). Equation (2.1) makes the role of the clip factor explicit: it scales the permitted bin height against the *uniform* per-bin count A/L , which is an accurate proxy for the average occupied-bin height only when the tile's intensities are spread across most of the L levels.

Single-threshold upgrade (prior own work). In previously published results, the candidate's prior work replaced the derived conventional clip with a single controllable global threshold, defining the clip limit directly as

$$\text{CL}_{T/80} = \frac{T}{80}, \tag{2.2}$$

where T is a global threshold value (Sapakova, Yesmukhamedov & Sapakov, 2025, Eq. 2, p. 5). This single-parameter form was reported to improve image distinctiveness "especially the tiny veins" (p. 5), raising an image-quality score from 33.23 under conventional CLAHE to 34.35 under the $T/80$ rule (Fig. 4), and was integrated with a fine-tuned ResNet-50 on the STARE database. Three constraints govern how that result enters the present dissertation. First, it is prior own work, cited as a previously published precursor rather than as independent evidence — and, because the two literature cards drawn from this study (#23 and #24) describe the same journal article, they constitute a single self-cited thread and are not treated as two independent confirmations (SIR-5). Second, the reported sensitivity figures rest on a sensitivity definition printed in that source as $\text{Sen} = \frac{\text{TP}}{(\text{TP} + \text{TN})}$ (Eq. 3, p. 9), which deviates from the standard $\text{Sen} = \frac{\text{TP}}{(\text{TP} + \text{FN})}$; this anomaly is noted here and the source's sensitivity values are not carried forward as comparable quantities (SIR-3). Third, the headline 100% accuracy, sensitivity, and specificity reported on the small, rotation-augmented STARE set is not transferable to the dissertation's experimental context (SB-1.2): the $T/80$ threshold was optimized for STARE's image distribution, and no parameter-level portability to EyePACS is assumed (DGL-5).

The single-threshold form is simpler than Equation (2.1) but carries a structural weakness that motivates the design. Because $T/80$ is a single scalar applied identically to every tile, it cannot respond to the wide variation in local intensity statistics across a fundus image. The conventional factor of Equation (2.1) does respond to the tile, but through the uniform proxy A/L , which is itself unreliable in precisely the regions a fundus image contains in abundance: large, near-uniform background tiles whose true histogram is concentrated in a narrow band of levels. In such a tile the occupied bins stand far above A/L , so a clip set as $\beta A/L$ truncates aggressively where contrast amplification is least wanted, or — for larger β — permits a steep mapping over the few occupied levels and amplifies background noise. A single per-bin factor and a single global threshold thus fail in

complementary ways: the former misjudges peaked local histograms, the latter ignores the local distribution entirely.

dual-constraint clip limit. The pipeline resolves this by constraining the clip limit with *both* terms simultaneously and taking the tighter of the two. For a tile of area A over $L = 256$ gray levels,

$$\text{CL}_{\text{DC}} = \min(\text{clip_factor} \cdot \frac{A}{256}, \text{global_threshold} \cdot A) \quad \text{\tag{2.3}}$$

where clip_factor and global_threshold are tunable hyperparameters (RESEARCH_ARCHITECTURE §3.2; OD-3 Stage 5). The first term is the conventional histogram-relative clip of Equation (2.1) with $L = 256$: it scales the permitted bin height against the uniform per-bin count and governs tiles whose intensities are well spread. The second term, $\text{global_threshold} \cdot A$, is a tile-relative absolute ceiling — a fixed fraction of the entire tile's pixel count, independent of how the intensities are distributed within the tile. Taking the minimum makes the constraints act as a logical conjunction: in a well-spread tile the histogram-relative term is the smaller and binds, delivering ordinary contrast limiting; in a strongly peaked tile, where the histogram-relative term would scale up with a degenerate distribution, the absolute ceiling is the smaller and binds, capping the attainable mapping slope regardless of how concentrated the local histogram is. The dual-constraint rule therefore bounds noise amplification under both regimes that the single-parameter forms handle poorly, which is the analytical rationale for adopting it as the contrast stage. Whether this rationale yields a measurable downstream advantage is a separate, empirical question, addressed by the parameter sweep of Experiment 2 (§4.3.2) and not asserted here.

The remaining specification of the stage follows the operational definition. The dual-constraint clip limit is applied to the L-channel of the LAB color space, over an 8×8 tile grid, with the clipped histogram excess redistributed and bilinear interpolation across tile boundaries as in the canonical CLAHE algorithm [67]; the operation is applied stochastically during training (with probability 0.8) and deterministically at inference (OD-3 Stage 5). The two hyperparameters clip_factor and global_threshold are left free at this stage by design: consistent with the parameter-portability caveat (DGL-5), no specific values are imported from the STARE-derived $T/80$ result or from any external study, and the values used by the pipeline are those selected by independent validation within the dissertation's own experimental framework. The dissertation's contribution at the level of formalization is therefore the dual-constraint rule of Equation (2.3) itself; the prior self-work contributed the single-threshold $T/80$ form of Equation (2.2), which the dual-constraint rule generalizes rather than reuses.

Table 2.1 contrasts the three formulations along the dimensions that matter for the design — the clip-limit definition, the parameters it exposes, its origin, and its portability.

TAB-2.1. CLAHE clip-limit formulations: conventional, single-threshold ($T/80$), and dual-constraint.

Formulation	Clip-limit definition	Control parameter	Origin	Portability /
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		(s)		limitati on
Convent ional contrast- limiting	$\text{CL} = \beta \cdot A/L$ — bin height clipped at a multiple of the uniform per-bin count (Eq. 2.1)	clip factor β (single)	Standard CLAHE; slope-limiting $\equiv$ clipping [66, p. 363]	Unifor m proxy A/L misjud ges peaked local histogr ams (large backgr ound tiles)
Single- threshol d (T/80)	$\text{CL} = T/80$ — clip set directly by one global threshold (Eq. 2.2)	global threshold T (single)	Prior own work on STARE (Sapakova, Yesmukhamedov & Sapakov, 2025)	STAR E- optimiz ed; not assume d portabl e to EyePA CS (DGL- 5); applied identic ally to every tile
dual- constrai nt	$\text{CL} = \min(\text{clip_factor} \cdot A/256, \text{global_threshold} \cdot A)$ — tighter of histogram-relative and tile-relative cap (Eq. 2.3)	clip_factor and global_thr eshold (two)	Dissertation design (OD-3 Stage 5; RESEARCH_ARCHITECTURE §3.2)	Parame ters tunable ; values validat ed in Exp 2 (§4.3.2) , not assume d portabl

				e (DGL-5); not claime d univers ally optimal (NC-17)
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The empirical preprocessing literature reinforces why this parameterization is treated as a quantity to be characterized rather than fixed by assumption. Hayati et al. [68], in evaluating CLAHE under a uniform configuration across four CNN architectures on APTOS 2019 (binary classification, accuracy-only, no confidence intervals), found the contrast step to help three architectures and degrade a fourth by 12.02 percentage points (pp. 62–64) — direct evidence that the downstream effect of a given clip configuration is contingent on architecture and tuning, and not comparable across differing datasets and tasks (SIR-3). The dual-constraint formulation does not escape this contingency; it exposes two parameters whose joint setting is precisely what Experiment 2 sweeps to identify the sensitivity profile required by H-2.

With the contrast stage now fully specified as a parameterized, dual-constraint operation on the LAB L-channel, the theoretical development turns to the complementary problem of suppressing noise and preserving structural edges — the spatial-filtering and noise-reduction foundations treated in §2.1.3.

§2.1.3 Spatial Filtering and Noise Reduction Methods

The formalization of the contrast stage in §2.1.2 fixed the clip limit as the mechanism that bounds noise amplification during contrast enhancement, but it did not address the complementary low-level operation of noise reduction itself. Contrast enhancement and noise reduction stand in tension: amplifying local contrast tends to amplify noise, while suppressing noise tends to blur the structures contrast enhancement is meant to reveal. This tension is acute in fundus imaging for a reason already established in Chapter 1 — the diagnostically decisive features of the early, screening-critical grades, principally microaneurysms and thin vessels, are themselves small and low in contrast, so any operation that smooths the image risks erasing precisely the signal a grading model depends on. The theoretical foundations of edge- and structure-preserving filtering, which formalize how smoothing can be made content-adaptive rather than indiscriminate, are therefore relevant background to the pipeline's low-level design, and are the subject of this section.

A linear, domain-only smoothing filter — one that averages each pixel with its spatial neighbours irrespective of their values — reduces noise but blurs edges and washes out low-contrast detail, because it cannot distinguish a genuine intensity

boundary from a noise fluctuation. Bilateral filtering, introduced by Tomasi and Manduchi [71], resolves this by making the averaging weights depend on photometric similarity as well as spatial proximity. In their formulation the filtered value is

$$h(x) = k^{-1}(x) \int f(\xi) c(\xi - x) s(\text{big}(f(\xi), f(x))) d\xi,$$

where $c(\xi - x)$ measures geometric closeness in the spatial domain and $s(f(\xi), f(x))$ measures photometric similarity in the intensity range, so that, in the authors' words, "only perceptually similar colors are averaged together, and only perceptually important edges are preserved" (p. 2). Edges are preserved because pixels across an intensity boundary, being photometrically dissimilar, receive negligible weight and so do not bleed into one another. The bilateral filter is a non-iterative, local operation; it is presented through analytical derivation and qualitative image examples rather than quantitative benchmarking, and the authors note that automatic selection of its parameters remains an open problem and that the method has no medical-imaging evaluation — so it is cited here for the principle it introduces, not as evidence of any downstream diagnostic gain.

Non-local means denoising, documented by Buades, Coll, and Morel [72], generalizes the notion of similarity from the local neighbourhood to the whole image. Its premise is that "the most similar pixels to a given pixel have no reason to be close at all" (p. 209): rather than averaging spatially adjacent pixels, the method averages pixels whose surrounding patches resemble the patch around the target, with resemblance "evaluated by comparing a whole window around each pixel, and not just the color" (p. 209). By exploiting the redundancy of repeated structures across an image — the same edge or texture recurring in many locations — non-local means is reported to "remove the noise while keeping the fine structures and details" (p. 211). Like the bilateral filter, it is an algorithm-description paper presenting parameter recommendations and a qualitative example without quantitative performance metrics or medical-imaging validation, and it is cited on the same restricted basis.

Considered together, bilateral filtering and non-local means are two solutions to a single problem: how to suppress noise without destroying the structures that carry information. Both replace the indiscriminate averaging of linear smoothing with a similarity-weighted averaging — bilateral filtering by photometric similarity within a spatial window, non-local means by patch self-similarity across the image — and both thereby formalize the requirement that a smoothing operation in a diagnostic pipeline be content-adaptive. This requirement is the durable contribution of the spatial-filtering literature to the present work; the specific algorithms, being qualitatively evaluated on non-medical example images, do not themselves establish anything about retinal classification.

The pipeline does not adopt bilateral filtering or non-local means as an explicit stage, and the reason is consistent with the trade-off just described. Inserting a dedicated denoiser ahead of classification carries the risk it is designed to mitigate — that aggressive smoothing erases the low-contrast microaneurysms that distinguish the early grades — and that risk is least acceptable precisely where screening value is highest. The pipeline instead manages noise at two points already specified: the contrast stage limits rather than amplifies noise through the dual-constraint clip limit

(Stage 5, §2.1.2), which caps the attainable mapping slope in homogeneous regions where noise would otherwise be magnified, and uneven illumination is corrected by the adaptive flat-field stage (Stage 4, OD-3). The edge- and structure-preserving principle that bilateral filtering and non-local means formalize is therefore honored as a design constraint — operate adaptively, conserve fine structure — rather than implemented as a separate component. This is a design rationale, not a claim of superiority over a denoising-augmented pipeline.

With the low-level image-operation foundations now in place — contrast enhancement in §2.1.1–§2.1.2 and the noise-reduction principle here — the theoretical development turns from the operations that prepare the image to the operations that consume it: the convolution, pooling, and feature-extraction mechanics of the convolutional neural network, treated in §2.2.1.

§2.2.1 Convolution, Pooling, and Feature Extraction Operations

The §2.1 sections established how the preprocessing stages prepare a fundus image — enhancing contrast under a clip-limited adaptive scheme and correcting illumination while conserving the small, low-contrast structures that define the early DR grades. The theoretical development now turns to the model component that consumes that prepared image: the convolutional neural network (CNN), whose feature-extraction operations transform pixel intensities into the representation a classifier acts upon. This section establishes the mechanics of that transformation — convolution, nonlinearity, and pooling, composed into a learned feature hierarchy — and the connectivity innovations that make deep hierarchies trainable, before drawing out why, in the fundus-lesion setting, the design of feature extraction cannot be separated from the preprocessing that precedes it.

The elementary operation of a CNN is convolution: a small bank of learnable filters is convolved across the spatial extent of the input, each filter computing a weighted sum over a local receptive field to produce a feature map. Two properties distinguish this from the dense (fully connected) connectivity of a classical neural network. First, the weights of a filter are shared across all spatial positions, so a feature detector learned in one location is applied everywhere, conferring translation equivariance — a shift in the input produces a corresponding shift in the feature map. Second, because each output unit depends only on a local receptive field rather than the entire input, the parameter count is drastically lower than a dense layer over the same input, which is what makes training on large images tractable. The viability of this design at scale was established by Krizhevsky, Sutskever, and Hinton [19], whose network of "five convolutional layers ... and three fully-connected layers" with 60 million parameters substantially outperformed prior hand-engineered feature pipelines on ImageNet (reporting top-5 test error of 17.0%); the result is a natural-image benchmark, reported here without confidence intervals or external validation, and is cited for the architectural mechanism it demonstrated rather than for any medical-imaging conclusion.

Two further operations complete the elementary feature-extraction block. A pointwise nonlinearity follows each convolution; the rectified linear unit (ReLU),

which Krizhevsky et al. [19] showed lets deep CNNs "train several times faster than their equivalents with tanh units" (Section 3.1), is the standard choice and is the activation used throughout the dissertation's architectures. Pooling — typically max-pooling over small windows with stride — then downsamples each feature map spatially. Pooling serves two purposes: it confers a degree of local translation invariance, so that the presence of a feature matters more than its exact position, and it enlarges the effective receptive field of subsequent layers, allowing them to integrate over wider spatial context. Stacking the convolution → nonlinearity → pooling block repeatedly produces a hierarchy of representations in which early layers respond to low-level structure such as edges and textures and deeper layers compose these into progressively more abstract patterns, as depicted schematically in FIG-2.2.

(Figure 2.2)

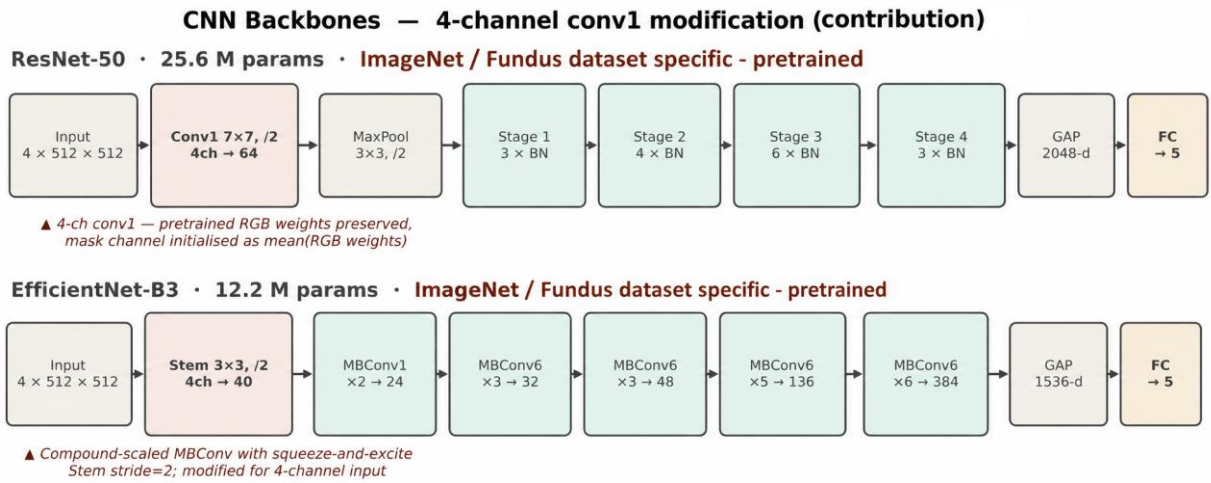


Figure 2.2 – CNN feature-hierarchy and convolution–pooling schematic

The expressive power of this hierarchy grows with depth, but depth is not freely obtained. He, Zhang, Ren, and Sun [23] characterized a degradation problem whereby, beyond a point, "with the network depth increasing, accuracy gets saturated and then degrades rapidly" — an optimization failure rather than overfitting, since deeper plain networks exhibited higher training error. Their resolution was residual learning: reformulating each block to learn a residual function with reference to its input, realized through identity shortcut connections that, in their words, "add neither extra parameter nor computational complexity" (p. 2). This eased the optimization of very deep networks and enabled the training of models exceeding one hundred layers, with the 50-layer bottleneck variant (ResNet-50) reaching a single-model top-5 ImageNet error of 5.25%. ResNet-50 is one of the two backbone architectures evaluated in this dissertation; it is adopted as an established, widely used design rather than as a claimed optimum, and no exhaustive search over the architecture space is undertaken (SB-3.1). These figures, again, are natural-image benchmarks without statistical testing or medical validation.

A parallel connectivity lineage offers an alternative response to the same depth-trainability problem. Huang, Liu, van der Maaten, and Weinberger [24] introduced the densely connected network, in which each layer "receives the feature-maps of all preceding layers as input through concatenation," a design the authors

reported to "alleviate the vanishing-gradient problem, strengthen feature propagation, encourage feature reuse, and substantially reduce the number of parameters" (Abstract). Residual and dense connectivity are not ranked here against one another; they are presented as the two principal mechanisms by which the feature hierarchy is kept trainable at depth, and they establish the architectural vocabulary — shortcut connections, feature reuse, parameter efficiency — on which the dissertation's backbone choices in §3.2 draw. Like the preceding architectures, DenseNet was evaluated only on computer-vision benchmarks, with no external or medical validation.

These foundations carry a specific consequence for fundus imaging that bears directly on the dissertation's central thesis. The pooling operations that give a CNN its invariance and receptive-field growth do so by discarding spatial resolution, and the features most vulnerable to that loss are precisely the smallest — the microaneurysms and fine vascular changes that distinguish the early, screening-critical grades. Wan et al. [33], proposing a lesion-segmentation network (EAD-Net) for diabetic retinopathy, designed it to use only a single max-pooling stage specifically to "preserve small-lesion features," supplementing it with dual attention and dilated convolutions; even so, the network's microaneurysm segmentation was weak (sensitivity 17.32% on the authors' local dataset; an IDRiD challenge AUPR of 0.2408), a result the authors themselves flag and which is reported here with the caveats that the study used small datasets, reported no confidence intervals, and conducted no cross-dataset transfer. The lesson for the present work is analytical rather than architectural: increasing architectural sophistication does not by itself resolve the detection of small, low-contrast lesions, which is exactly the burden the preprocessing stages are designed to share. This is one concrete expression of the dissertation's framing of the model as preprocessing plus CNN — the two components addressing the small-lesion problem jointly — and it is consistent with the constraint (CFC-2.8) that no eventual performance effect will be attributed to preprocessing or architecture in isolation.

Feature extraction, as developed here, yields the representation on which the network's classifier head ultimately operates. How that representation is trained — and in particular how training is shaped by the severe class imbalance characteristic of diabetic-retinopathy datasets — is governed by the loss function and optimization scheme, which §2.2.2 takes up next.

§2.2.2 Loss Functions and Optimization for Imbalanced Medical Datasets

The convolutional feature extractor of §2.2.1 produces a representation that a classifier head maps to class scores; how that mapping is trained is governed by the loss function, and in diabetic-retinopathy classification the choice of loss is dominated by a single property of the data — severe class imbalance. The five-class DR grading taxonomy fixed in §1.1.1 is acutely skewed in practice: in the candidate's prior experiments on the APTOS 2019 dataset, the no-disease class accounted for 73.5% of the training set while the severe and proliferative classes (DR 3 and DR 4)

accounted for only 2.5% and 2.0% respectively (previously published results; Sapakova et al., 2025; Yesmukhamedov et al., 2025, which report the same experiment and are treated here as a single prior-work thread). Pratt et al. [27] likewise observed that classes 3 and 4 together "came from ... fewer than three percent" of their large Kaggle DR training set, requiring deliberate changes "to ensure it could still learn the features of these images" (p. 201). This section establishes why the standard objective is inadequate under such skew and what imbalance-aware alternatives the dissertation adopts.

The default objective for multi-class classification is softmax cross-entropy, which for a single example penalizes the negative log-probability the model assigns to the true class. Summed over a training batch, the gradient of this loss is an average over examples, and when one class supplies the overwhelming majority of examples that average is dominated by the majority class. The practical consequence under the DR distribution is that the network can minimize the mean loss while effectively ignoring the rare severe grades, producing high overall accuracy that conceals poor minority-class recall — and, compounded by limited data, overfitting: the candidate's prior work reported a training F1-score of 0.87 against a test F1-score of 0.62 on the frozen-layer model, attributed in part to "class imbalance and limited representation of severe cases" (previously published results). Cross-entropy, in short, optimizes a quantity misaligned with the clinical objective of detecting the uncommon but sight-threatening cases.

Cost-sensitive learning corrects this misalignment at the level of the loss rather than the data. The per-class contribution to the loss is rescaled by a weight — most simply the inverse of the class frequency, more generally a misclassification cost — so that an error on a rare class incurs proportionally larger penalty and thus a larger gradient. More refined weighting schemes exist: Cui et al. [73] re-weight by the "effective number of samples" rather than raw inverse frequency, tempering the over-weighting of extremely rare classes that pure inverse-frequency weighting can induce, though on long-tailed natural-image benchmarks rather than on DR data. Pratt et al. [27] implemented an early form of class weighting through dynamic per-batch class weights adjusted relative to the proportion of no-DR images in each batch. In the most systematic synthesis of the approach, Araf, Idri, and Chairi [74], reviewing 173 studies of cost-sensitive learning for imbalanced medical data, reported that cost-sensitive learning "preserves the data distribution while ensuring computational efficiency" relative to resampling and is particularly suited to medical settings in which false negatives are clinically severe. That review is a vote-counting synthesis without meta-analysis and is not specific to diabetic retinopathy, and it found that "only two studies were classified as validation research" (p. 20) — so cost-sensitive learning is adopted here as a reasoned, distribution-preserving design choice rather than as an empirically settled superiority over resampling.

The loss function adopted by the training protocol is focal loss, introduced by Lin et al. [75], which extends weighted cross-entropy with a multiplicative modulating factor that decreases as the model's confidence in the correct class increases. The effect is to down-weight the contribution of examples the network already classifies easily — predominantly the abundant no-DR images — and to

concentrate the gradient on hard and minority examples, combining the class-frequency weighting of cost-sensitive learning with an additional per-example difficulty weighting. The dissertation employs focal loss with a focusing parameter $\gamma = 2$ and a class-balancing weight α set to inverse class frequency (a configuration specified in the methodology and depicted in the focal-loss schematic of §3.3.4). Lin et al. [75] introduced and evaluated focal loss on the COCO dense-object-detection benchmark rather than on retinal or medical imagery, so it is adopted here on the strength of its established loss-design rationale, not as a result transferred from a DR study; its full formalization, including the modulating-factor expression and parameter rationale, is deferred to §3.3.4, where it is specified as a methodological component of the system.

The same imbalance that shapes the loss also dictates how performance must be measured, because the majority class that distorts the objective equally distorts the most familiar metric. Overall accuracy is inflated by the dominant no-DR class — a model predicting "no DR" for every image would score near the majority proportion — and is therefore demoted, in this dissertation's evidence hierarchy, below metrics that account for the distribution. The primary measures are the weighted F1-score, ROC-AUC, and Cohen's κ with quadratic weights; the last is especially appropriate to the ordinal structure established in §1.1.1, since its quadratic weighting "ensured that errors between adjacent classes ... are penalized less than errors between distant classes," which the candidate's prior work noted has "clinical significance, as errors in transitions between mild stages are less critical than missing severe cases" (previously published results). This alignment of metric to the ordinal, unequal-cost structure of DR is the basis of the sub-claim (carried into §3.4 and Chapter 4) that imbalance governs both loss and evaluation design.

That class imbalance is a field-wide rather than idiosyncratic concern is corroborated by survey and benchmark evidence, with the caveats their study designs require. Senapati et al. [63], in a PRISMA-based systematic review of diabetic-retinopathy artificial intelligence, identified class imbalance and overfitting among the field's recurring methodological gaps; as a field-mapping survey it provides no primary benchmarking. Arora et al. [31] reported an EfficientNetB0 model reaching an average accuracy of 0.8653 on a Kaggle DR set undersampled to mitigate imbalance, but with a maximum training accuracy of 97.11% indicating a substantial overfitting gap and no external validation, a result of limited epistemic weight. These EfficientNetB0 findings, like the candidate's own EfficientNetB0 experiments, are bounded to that architecture: consistent with the architecture-generalization prohibition, no conclusion about the broader class of efficient CNN architectures is drawn from them without explicit comparative experiments.

In sum, the severe imbalance of DR data renders unweighted cross-entropy inadequate, motivating cost-sensitive weighting and the focal-loss modulation the protocol adopts, and dictating evaluation by weighted F1, ROC-AUC, and quadratically weighted κ rather than accuracy. Imbalance and limited data also exacerbate overfitting — visible already in the train-versus-test gaps reported above — which is controlled not by the loss but by the regularization techniques (dropout, batch normalization, and data augmentation) that §2.2.3 takes up next.

§2.2.3 Regularization Techniques: Dropout, Batch Normalization, and Data Augmentation

The loss design of §2.2.2 corrects the objective toward the rare severe grades, but it does not address the second consequence of imbalanced, limited DR data already evident there: overfitting, the divergence between strong training performance and weaker held-out performance — illustrated by the candidate's prior result of a training F1-score of 0.87 against a test F1-score of 0.62. Regularization is the family of techniques that narrows that gap. This section establishes the regularizers relevant to the dissertation's architectures, which operate at three levels — the weights of the network, the control of the training process, and the data itself — and shows why data augmentation, acting at the third level, is integrated directly into the pipeline.

At the level of the network's weights, dropout is the canonical regularizer. During training it randomly deactivates a fraction of units, which prevents units from co-adapting into fragile, jointly-tuned detectors and forces the network toward more redundant, robust representations. Its importance was established alongside the first large-scale CNN — Krizhevsky, Sutskever, and Hinton [19] reported plainly that "without dropout, our network exhibits substantial overfitting" (§4.2) — and formalized shortly afterward by Srivastava et al. [76], who characterized dropout as an efficient approximation to averaging over exponentially many thinned networks. Both are natural-image results cited here for the mechanism rather than for any medical-imaging conclusion. The technique carries directly into diabetic-retinopathy CNNs — Pratt et al. [27] applied dropout at rate 0.5 after the final convolutional block and the first fully connected layer, and the candidate's prior enhanced architecture used a dropout rate of 0.4.

Batch normalization is a complementary weight-level technique. Introduced by Ioffe and Szegedy [77], it normalizes the activations of a layer across each training batch, which stabilizes and accelerates optimization and confers a mild regularizing effect through the noise introduced by batch statistics — an effect that weakens as the batch shrinks, a caveat relevant to this dissertation's batch size of sixteen and revisited in §3.2. In this dissertation's terms it is an architectural component present in the enhanced network and absent from the shallow baseline, a distinction that matters for the factorial comparisons of Chapter 4; Pratt et al. [27] likewise applied batch normalization after convolutional layers. Neither dropout nor batch normalization is claimed to prevent overfitting universally; each is a contributing control whose effect is empirical and architecture-dependent.

A second level of regularization controls the training process rather than the network. Early stopping halts training once a validation metric ceases to improve, preventing the model from continuing to fit the training set past the point of best generalization; the candidate's prior work stopped training when the validation Cohen's κ "did not improve for k epochs, preventing overfitting" (previously published results; the two prior papers report the same experiment and are treated as a single prior-work thread). Learning-rate scheduling — reducing the learning rate when a monitored metric plateaus — serves a related purpose, and the same prior

work paired ReduceLROnPlateau with early stopping. These training-control measures reduced but did not eliminate the train–test gap, and that work explicitly noted that further improvement would require additional data and class balancing — a candid indication that regularization mitigates rather than resolves overfitting under limited, imbalanced data.

The third level acts on the data. Data augmentation synthesizes additional training examples by applying label-preserving transformations to existing images, expanding the effective dataset and enforcing invariance to the transformed factors. The practice again traces to Krizhevsky et al. [19], who combined random crops and horizontal reflections with a principal-component-based perturbation of RGB intensities; the design space is far broader — surveyed by Shorten and Khoshgoftaar [78] and extended by methods such as mixup [79] and RandAugment [80], all natural-image studies — though the pipeline adopts only the geometric and photometric subset specified in §3.1.3. Augmentation is standard in DR CNNs: Pratt et al. [27] applied real-time per-epoch augmentation (rotations, flips, and shifts), and the candidate's prior work applied rotation, scaling, shifting, brightness adjustment, and skew. It is also distinctive among regularizers in serving a second function directly relevant to §2.2.2: because minority-class images can be augmented more aggressively than majority-class images, augmentation also rebalances the effective class distribution — consistent with a systematic imbalance study reporting oversampling as an effective mitigation without an attendant overfitting penalty [81] — and the candidate's prior work applied it explicitly "to address dataset imbalance and improve variability." This dual role — regularizer and imbalance-mitigation mechanism — is why augmentation is not treated as an external training trick but integrated into the pipeline as Stage 6 (OD-3): a train-only stage combining a unified affine transform with principal-component color jitter and brightness/contrast variation, applied before the final normalization (the augmentation renders and the color-jitter basis are documented in §3.1.3).

With regularization established, the CNN-training fundamentals of §2.2 are complete: §2.2.1 fixed how the network extracts features, §2.2.2 how it is trained under class imbalance, and this section how that training is regularized against overfitting. What remains, before the methodology can specify the full system, is how the network is *initialized* — the transfer-learning and feature-transferability questions that determine what the network knows before fine-tuning begins, which §2.3 takes up next.

§2.3.1 Feature Transferability Across Visual Domains

The regularization techniques of §2.2.3 complete the account of how a convolutional network is trained, but they presuppose a starting point — the weights from which training begins. In practice these weights are rarely random: a network is initialized from those of a model already trained on a large source dataset, and only then adapted to the target task — the paradigm that Pan and Yang [82] systematized as transfer learning, classifying its settings by what differs between source and target domain and task and cautioning that a poor source–target match can yield negative

transfer. The effectiveness of this transfer is not automatic, however, and the theory of feature transferability — how much of what a network has learned on one domain carries usefully to another — directly conditions the dissertation's pretraining design. This section establishes that theory and uses it to motivate the two pretraining regimes the experimental framework compares.

The foundational empirical result is that transferability is layer-dependent. Yosinski, Clune, Bengio, and Lipson [83] measured, layer by layer, how well features learned on one task transfer to another and found that "layers one and two transfer almost perfectly" between tasks while higher layers progressively specialize to the source task (§4.1). This follows the now-standard understanding that early convolutional layers learn generic structure — edges, colors, simple textures — common to most visual domains, whereas deeper layers compose these into representations tuned to the specific source categories. Crucially, the same study established that transferability "decreases as the distance between the base task and target task increases" (Abstract), and that the result is reported on ImageNet-derived splits only, without confidence intervals or statistical testing and with no medical-imaging data — so it grounds a general principle rather than a retinal-specific quantity. A practically important corollary is that transferred features, once fine-tuned, can "generalize better than those trained directly on the target dataset" (§4.1): transfer is not merely a convenience but can improve the final model.

The magnitude and predictability of transfer were quantified at architecture scale by Kornblith, Shlens, and Le [25], who evaluated sixteen CNN architectures across twelve natural-image datasets and found that ImageNet accuracy strongly predicts transfer accuracy — a correlation of $r = 0.99$ for fixed penultimate-layer features and $r = 0.96$ for full fine-tuning — with fine-tuning outperforming fixed-feature transfer in 179 of 192 dataset–model combinations and accelerating convergence roughly seventeen-fold. These findings, also confined to natural-image classification with no medical or clinical validation, establish that better source representations transfer better and that fine-tuning is generally the stronger adaptation mode. The same study, however, reported that transfer benefits diminish as the target diverges from ImageNet labels and as the target dataset grows — a caveat that bears directly on medical imaging.

The retinal-fundus domain is precisely such a divergent target. Fundus photographs share little category structure with ImageNet's natural objects, so the natural-image→fundus shift is a large domain distance of the kind Yosinski's principle predicts will most degrade the transfer of the higher, task-specific layers — the layers that carry the most discriminative information. This is the setting Cheplygina, de Bruijne, and Pluim [45] survey under the heading of learning with limited supervision, observing that "the lack of labeled data is a major obstacle" in medical image analysis and that transfer learning is one of the principal responses to it; as a survey, it organizes the problem without contributing original retinal benchmarks. The most cited demonstration that ImageNet features transfer across a large domain gap into medical imaging is Esteva et al. [37], who fine-tuned an ImageNet-pretrained Inception- network to dermatologist-level skin-cancer classification — a cross-domain transfer success, though on dermatology rather than

retinal fundus images and therefore off this dissertation's task. That transfer to the DR task specifically is nonetheless useful is shown by Arrieta et al. [84], whose semi-supervised approach reached an AUC of 0.94 on EyePACS and 0.89 on Messidor-2 using only 2% of labels — evidence, on a binary referable-DR framing, that transfer can be highly effective from few labels, while remaining a single study rather than a general guarantee.

These results jointly motivate the dissertation's pretraining design without licensing any claim that one regime is superior in advance. Under the binding transfer-learning principle (DGL-6), the H-1 experiment compares two initialization regimes for the same CNN backbones: a baseline arm initialized from ImageNet — a cross-domain transfer of the kind whose limits this section has just described — and an integrated arm initialized from ophthalmology-specific self-supervised pretraining on unlabeled fundus imagery, which reduces the domain gap by learning representations in the target domain itself. The degree to which either regime transfers to the dissertation's specific datasets is not theoretically guaranteed and is evaluated empirically within the experimental framework only; and because the integrated arm differs from the baseline along both the preprocessing and the pretraining axis, no eventual performance difference will be attributable to pretraining (or to preprocessing) in isolation. The theory of in-domain self-supervised pretraining that the regime rests on is developed in §2.3.3.

Transferability, then, is real but bounded — strongest in the lower layers, decaying with domain distance, and most uncertain precisely for a target as far from natural images as the retinal fundus. What this section has left open is the practical question its findings raise: given transferred weights, how should they be adapted — held frozen as fixed features, or progressively unfrozen and fine-tuned? That choice of adaptation strategy is the subject of §2.3.2.

§2.3.2 Frozen-Layer versus Progressive Fine-Tuning Strategies

Section §2.3.1 established that transferred features are useful but bounded, and closed on the question its findings raise: given a network initialized from transferred weights, how should those weights be adapted to the target task? The answer is a choice of adaptation strategy, occupying a spectrum between two poles. At one pole is the frozen-base, or fixed-feature, strategy: the pretrained backbone is held frozen and used purely as a feature extractor, while only a new classification head is trained on the target data. At the other is progressive fine-tuning: the head is trained first, after which the upper layers of the backbone are unfrozen and adapted, allowing the transferred representation itself to specialize to the target domain. These are distinct constructs and are kept terminologically distinct throughout this dissertation — "feature extraction" and "frozen-base strategy" denote the frozen pole, "progressive" or "two-stage fine-tuning" the adapted pole.

Transfer-learning theory favors fine-tuning, particularly under domain shift. Yosinski et al. [83] reported that "transferring features and then fine-tuning them results in networks that generalize better than those trained directly on the target dataset" (§4.1), and identified an optimization penalty — co-adaptation of

neighbouring units — that a frozen split can incur but that fine-tuning relieves. At architecture scale, Kornblith et al. [85] found fine-tuning superior to fixed-feature transfer in 179 of 192 dataset–model combinations on natural-image tasks. Both results are reported on natural-image classification without medical validation, so they ground an expectation rather than a retinal-specific quantity; but that expectation is strongest precisely where the target diverges most from the source, which §2.3.1 identified as the natural-image→fundus case.

The candidate's prior diabetic-retinopathy experiments provide direct, if bounded, confirmation on the target task itself. In previously published results — reported across two papers that describe the same experiment and are treated here as a single prior-work thread — an EfficientNetB0 model adapted to five-class DR classification on APTOS 2019 was evaluated under both strategies. The frozen-base configuration reached a test F1-score of 0.62; progressive fine-tuning of the upper layers raised it to 0.74, a 12-percentage-point gain, with macro-averaged F1 improving from 0.72 to 0.77 and weighted-averaged F1 from 0.74 to 0.81, and a narrowing of the train-to-test gap that indicated reduced overfitting. The authors concluded that "unfreezing and fine-tuning the top layers leads to more balanced and robust generalization on unseen data, whereas fully freezing the layers limits the model's ability to generalize." Two caveats bound this evidence: overall accuracy was unchanged at 0.80 under both strategies (the gains appear in the imbalance-aware metrics, consistent with §2.2.2), and the result is specific to the EfficientNetB0 architecture and APTOS data and is not generalized to the broader class of efficient CNN architectures without explicit comparative experiments.

On the convergent strength of this theory and prior evidence, the design adopts a two-stage progressive fine-tuning protocol — training the classification head first, then unfreezing and adapting the upper backbone layers — whose specification is given in §3.3.3. It is important to locate this choice correctly within the experimental design: the two-stage protocol is a shared training method applied uniformly across the H-1 configurations, not an independent variable of the H-1 comparison (the earlier hypothesis that compared training methods was dropped). It therefore does not itself contribute to, and cannot be credited with, any performance difference observed between the baseline and integrated arms.

A distinct family of adaptation methods addresses domain shift at the level of the learned representation rather than through layer freezing. Ganin et al. [86] introduced domain-adversarial training, in which a gradient-reversal layer drives the feature extractor toward representations that are "discriminative for the main learning task and ... indiscriminate with respect to the shift between domains" — features predictive of the label but uninformative about which domain an input came from. This is a conceptually different adaptation strategy, demonstrated on digit and object benchmarks rather than medical imaging, and it is relevant to the dissertation's device-domain-shift evaluation (§4.7) as a point of comparison; it is not, however, the adaptation mode chosen for the H-1 backbones, which use straightforward progressive fine-tuning.

With the adaptation strategy settled, the transfer-learning foundations that the experimental framework actively exercises — feature extraction, training under

imbalance, regularization, and the transfer and adaptation of pretrained weights — are in place. The in-domain self-supervised pretraining that the integrated arm relies upon is treated separately in §2.3.3. Chapter 2 now turns to the single theoretical strand of the dissertation that is not exercised by the classification experiments but stands as a simulation-based contribution in its own right: the coupled thermal-optical model of fundus tissue response, developed in §2.4.1.

§2.3.3 In-Domain Self-Supervised Pretraining for Retinal Imaging

The transfer-learning strategies formalized in §2.3.1 and §2.3.2 took as given a backbone whose weights are inherited from supervised pretraining on natural images, and asked only how those weights should be adapted — frozen or progressively fine-tuned — to the fundus domain. This section develops the alternative source of initialization that the integrated arm of the dissertation's first hypothesis adopts: ophthalmology-specific self-supervised pretraining of the convolutional backbone on an unlabelled retinal corpus, producing retina-aware weights before any diabetic-retinopathy label is seen. The thesis of the section is that such in-domain initialization is theoretically motivated to be more label-efficient and more clinically generalizable than natural-image initialization — but this is a motivation, not a guarantee, and consistent with DGL-6 it is evaluated empirically within the dissertation's framework rather than assumed. It must also be stated at the outset that, under the composite independent variable of H-1 (CFC-2.8), this pretraining source is varied jointly with the preprocessing pipeline; it is therefore reported only as one component of the integrated configuration and is never advanced as an independently attributable factor.

The motivation begins with the data economics of medical imaging. Cheplygina et al. [45] characterized the lack of labelled data as a frequent obstacle in medical image analysis and surveyed the weak-supervision paradigms — semi-supervised, multiple-instance, and transfer learning — that reduce dependence on full annotation. Self-supervised learning extends this logic by deriving the supervisory signal from the unlabelled images themselves: a pretext objective is solved on large unlabelled corpora to learn general representations, which are then fine-tuned on a small labelled set for the target task. Shurrab and Duwairi [87], surveying self-supervised learning across medical imaging, organized the field into predictive (pretext-task), contrastive, and generative families, and argued that such methods reduce the dependence on expensive medical annotations — the precise constraint under which population-scale diabetic-retinopathy screening operates.

The design space these surveys map is populated by a set of method primaries developed largely on natural images. Contrastive approaches learn representations by drawing augmented views of the same image together while pushing distinct images apart, instantiated with a normalized temperature-scaled objective in SimCLR [88] and with a momentum encoder and a memory queue in MoCo [89]. Negative-free approaches dispense with explicit negative pairs: BYOL [90] bootstraps a target network, while SimSiam [91] shows that a stop-gradient operation alone prevents representational collapse. DINO [92] frames pretraining as self-distillation.

Generative masked modelling, finally, is exemplified by the masked autoencoder of He et al. [93], which masks a high proportion of image patches — on the order of 75% — and reconstructs the missing pixels, yielding a nontrivial and efficient label-free objective. These results are reported on natural-image benchmarks (principally ImageNet), so they establish the available design space rather than retinal performance, and their figures are not commensurable with the fundus task (SIR-3).

The decisive question for this dissertation is whether pretraining *in domain* — on retinal images — improves on natural-image pretraining. Two independent lines of evidence bear on it. Azizi et al. [47], working in dermatology and chest radiography, reported that self-supervised pretraining on domain images, with multi-instance positive pairs drawn from multiple images per patient, advanced classification over a supervised ImageNet baseline by 6.7% top-1 accuracy in dermatology and 1.1% mean AUC in chest radiography, with improved robustness to distribution shift; because that study is dermatological and radiographic rather than retinal, its transfer to fundus imaging is an extrapolation and is cited here only as cross-domain method-level support (SIR-2). The keystone retinal evidence is RETFound [46], a foundation model pretrained by masked autoencoding on approximately 1.6 million unlabelled retinal images and then adapted to disease-detection tasks with explicit labels; the authors reported that it consistently outperformed comparison models — including the contrastive baselines SimCLR, DINO and MoCo- — in diagnosis and prognosis of sight-threatening eye diseases, with the largest gains under limited labelled data. These two sources are kept as separate lines and not pooled as joint confirmation (SIR-5).

This evidence carries an architectural qualification that the dissertation must make explicit. The keystone in-domain demonstrations — RETFound and the masked autoencoder it builds on — are Vision-Transformer models, whereas the backbones adopted in this work are convolutional networks (ResNet-50 and EfficientNet-B3, §3.2.1). The in-domain principle therefore transfers to the dissertation's setting most directly through the convolution-compatible members of the design space: the contrastive and negative-free family (SimCLR, MoCo, BYOL, SimSiam) and the multi-instance contrastive scheme of Azizi et al. [47], all of which operate on ResNet backbones. RETFound establishes that retinal in-domain pretraining is beneficial in principle; the convolution-native realization of that principle is selected empirically. Several inherited limitations bound the strength of this motivation: RETFound is retrospective, the device and population provenance of its pretraining corpus is not exhaustively controlled, and — most relevant here — it includes no preprocessing ablation, so it cannot adjudicate whether pretraining or preprocessing drives the observed advantage (SIR-2).

That last limitation is exactly why the dissertation does not treat pretraining source as a rival to preprocessing but as its partner. RETFound foregrounds the pretraining source as the principal performance lever; the integrated arm of H-1 accordingly combines ophthalmology-specific self-supervised initialization with the preprocessing pipeline and, per CFC-2.8, reports their joint effect rather than apportioning credit between them. Nothing in this section is a clinical-validation claim: the retina-aware initialization is a representation-learning choice whose

diagnostic value is established, if at all, only by the controlled experiments of Chapter 4. With its theoretical basis fixed here, the operational pretraining procedure — corpus, objective, and the convolution-native instantiation — is specified in §3.3.2, and its empirical contribution is evaluated under H-1.

§2.4.1 Coupled Thermal-Optical Model of Fundus Tissue Response

The foundations developed through §2.3 — preprocessing operations, convolutional feature extraction, training under class imbalance, regularization, and transfer learning — are all exercised directly by the diagnostic-classification experiments of Chapters 4 and 5. This section marks a deliberate change of epistemic register. It presents a coupled thermal-optical model of laser photocoagulation on fundus tissue: a *theoretical*, simulation-based contribution that stands apart from the classification system, addresses the *treatment* rather than the *diagnosis* of diabetic retinopathy, and — this must be stated at the outset — is a qualitative model that has not been validated against experimental or clinical data and makes no clinical-grade claim. It is presented here as previously published own work (the modeling study reported in the Herald of KazUTB, 2024), re-expressed in independent notation, and its standing is that of a secondary theoretical contribution; it does not bear on hypotheses H-1 through H-7 or on any claim about diagnostic performance.

The clinical motivation for the model lies in laser photocoagulation, a standard therapy for diabetic retinopathy in which focused laser energy induces protein coagulation to arrest the proliferation of abnormal retinal vessels. Effective therapy requires delivering enough energy to coagulate the target while sparing surrounding healthy tissue, and this balance depends on how laser energy is absorbed and how the resulting heat distributes across the heterogeneous, layered structure of the fundus. The modeling problem, accordingly, is to predict the spatiotemporal temperature field produced in tissue by a laser pulse of given power, spot size, and duration — a problem that couples an optical sub-model (how much energy is deposited where) to a thermal sub-model (how that energy diffuses as heat).

The optical sub-model begins with the spatial profile of the incident beam. The laser is taken to have a Gaussian transverse intensity distribution,

$$I_0(r) = \frac{P}{\pi a^2} \exp\left(-\frac{r^2}{a^2}\right),$$

where P is the laser power, a the beam spot radius, and r the radial distance from the beam axis (after the formulation of the prior modeling study, Eq. 7). As the beam penetrates tissue, its intensity is attenuated with depth according to the Beer–Lambert law,

$$I(r, z) = I_0(r) \exp\left(-\int_0^z \beta(r, \xi) d\xi\right),$$

where z is depth and $\beta(r, \xi)$ is the spatially varying absorption coefficient of the tissue (Eq. 2). The integral in the exponent accumulates absorption along the path, so that more superficial tissue, encountered first, intercepts a larger share of the incident energy than deeper tissue.

The coupling from optics to heat is made through an energy-balance relation. Over an elementary volume ΔV , the energy absorbed from the beam in a time

interval Δt raises the local temperature in proportion to the tissue's volumetric heat capacity C_o ,

$$\Delta V \cdot C_o(x,y,z) \cdot \Delta T(x,y,z) = W \cdot \Delta t,$$

where W is the absorbed power in the volume (Eq. 3). Combining this with the attenuated intensity yields the temperature rise deposited by absorption,

$$\Delta T(x,y,z) = \frac{I_0 \cdot \exp(-\int_0^z \beta(x,y,z) dx)}{\beta(x,y,z) \cdot \Delta t \cdot C_o(x,y,z) \cdot \sigma(x,y,z)},$$

in which the local absorption coefficient β acts as the source term converting transmitted intensity into deposited heat (Eq. 4). For a slowly varying absorption coefficient over a thin depth increment, the fractional absorption simplifies to $1 - \exp(-\int_z^{z+\Delta z} \beta(x,y,z) dx) \approx \beta(r,z) \cdot \Delta z$ (Eq. 5), which renders the source term tractable for discretization.

This absorbed-energy temperature rise furnishes the initial condition for the thermal sub-model proper, the heat-conduction equation with spatially varying properties,

$$C_o \cdot \sigma(x,y,z,T) \cdot \frac{\partial T}{\partial t} = \nabla \cdot (k(x,y,z,T) \cdot \nabla T),$$

where k is the thermal conductivity, allowed to vary with position and temperature to represent tissue inhomogeneity, and the initial field combines the deposited rise ΔT with the baseline tissue temperature (Eq. 8). The prior study solved this equation numerically by an explicit finite-difference scheme implemented in Python with the NumPy and SciPy libraries, with boundary conditions placed far enough from the beam axis that thermal energy was assumed not to reach them.

The reported simulation results are qualitative. Across the modeled tissue layers, surface tissue (cornea) was found to heat rapidly owing to high near-surface energy absorption, while deeper layers (choroid and retina) heated slowly and reached a comparatively stable temperature, with the temperature field evolving over an interval of roughly zero to 4.5 seconds of exposure. No specific numerical temperature values, damage thresholds, or error margins were reported; the findings are presented through heat maps and temperature-versus-depth and temperature-versus-time graphs rather than tabulated quantities. The prior study stated that its results "confirm the effectiveness of using laser therapy" and that the approach could "lead to improved clinical outcomes"; consistent with the rule against extrapolating modeling results, these are recorded here as statements made by that source, not as validated findings of this dissertation.

The epistemic limits of the model must be made explicit, because they determine how it may be used. The model is qualitative and has not been validated against experimental or clinical measurements, so it does not constitute an experimentally validated clinical model and supports no clinical-grade claim. Its construction rests on simplifying assumptions that bound its scope: tissue optical and thermal properties are treated as static during exposure, neglecting the property changes that coagulation itself induces; each fundus layer is modeled as homogeneous despite the acknowledged inhomogeneity of real tissue; the Gaussian beam profile is held fixed, omitting scattering and beam distortion; and, notably, the heat-conduction equation contains no blood-perfusion term — the Pennes bioheat

contribution that would represent convective heat removal by retinal and choroidal blood flow is absent, although perfusion is physiologically significant in this vascular tissue. The evidence base is also thin: the model is supported by a single prior publication co-authored by the candidate, with no external corroboration. The conceptual structure of the coupled model — the optical attenuation feeding the thermal diffusion across layers — is summarized in FIG-2.4.

(Figure 2.4)

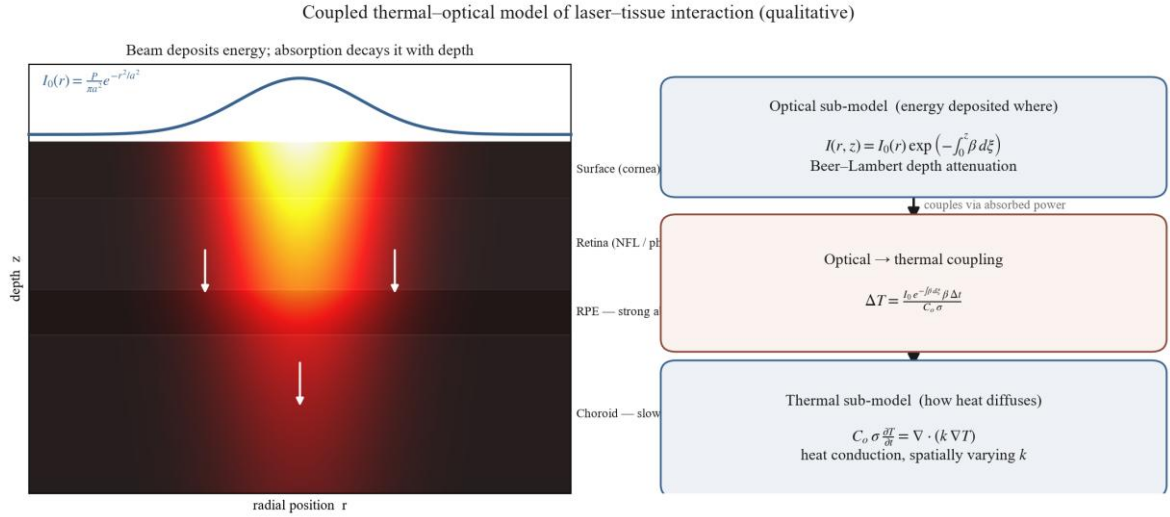


Figure 2.4 – Coupled thermal-optical model of laser-tissue interaction – Gaussian-beam intensity, Beer-Lambert depth attenuation, and heat conduction across fundus layers

Within the dissertation's argument, this model occupies a clearly delimited place. It is a secondary, theoretical contribution concerning the *treatment*-side physics of laser photocoagulation, offered as a basis for reasoning about how laser parameters shape the thermal field, and it is reflected in the statement of the dissertation's theoretical significance rather than among its empirical, classification-oriented results. It is epistemically independent of the integrated preprocessing-CNN diagnostic system that is the dissertation's central concern. With this theoretical strand set in its proper bounds, the chapter returns to the diagnostic system and to the interpretability of its predictions, beginning with the formalization of class activation mapping and Grad-CAM in §2.5.1.

§2.5.1 CAM / Grad-CAM Theory and Formalization

Having set the simulation-only thermal model of §2.4.1 in its proper, bounded place, the chapter returns to the diagnostic system and to a property it must possess to be clinically usable: interpretability. A convolutional classifier outputs a grade, but a screening decision-support tool is far more trustworthy if it can indicate *where* in the image its evidence for that grade lies. This section formalizes the family of methods the dissertation uses for that purpose — class activation mapping and its gradient-based generalizations — and fixes, from the outset, the strict limit on what these methods establish: a class activation map indicates regions of gradient-weighted

activation in the network's final convolutional layer; it is an interpretability visualization, not a pixel-level clinical delineation of lesion boundaries (NC-14).

The lineage begins with class activation mapping (CAM), introduced by Zhou et al. [50]. Their insight was architectural: if the final convolutional feature maps are reduced by global average pooling immediately before the classification layer, then the learned classification weight w_k^c connecting feature map k to class c can be projected back onto that feature map's spatial activations $f_k(x,y)$ to form a class activation map,

$$M_c(x,y) = \sum_k w_k^c f_k(x,y),$$

which, in the authors' words, "directly indicates the importance of activations at each spatial location" for the class. Zhou et al. [50] demonstrated that global average pooling "enables CNNs to retain localization ability despite training only with image-level labels," achieving weakly supervised localization on ImageNet. Two constraints attend the method: it requires the network to be built with a global-average-pooling penultimate stage (and thus often retrained), and its weakly supervised localization, while notable, remained substantially worse than fully supervised localization — a natural-image benchmark result with no medical-imaging evaluation.

Grad-CAM, introduced by Selvaraju et al. [51], removes the architectural constraint by replacing the classification weights with gradients. For a target class c and a convolutional feature map A^k , the importance of that map is computed as the global average of the gradient of the class score y^c with respect to the map's activations,

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k},$$

and the localization map is formed as a weighted combination of the feature maps passed through a rectifier,

$$L_{\text{Grad-CAM}}^c = \text{ReLU}(\sum_k \alpha_k^c A^k)$$

The rectifier retains only the features that exert a positive influence on the class score, suppressing regions that argue against it. Because the weights are obtained by backpropagation rather than from a specific pooling structure, Grad-CAM applies to essentially any convolutional architecture without modification or retraining, and it reduces to CAM as a special case for GAP networks.

Grad-CAM++ [52] refines the weighting further. Rather than averaging the class-score gradients uniformly across spatial positions, it derives pixel-wise gradient weights from higher-order derivative terms, which the authors report yields "better object localization as well as explaining occurrences of multiple object instances in a single image." On an ImageNet faithfulness evaluation it reduced the average drop in class confidence under explanation-masking from 46.56% (Grad-CAM) to 36.84%. These figures are reported here as the source's benchmark comparison on natural images — the study is an arXiv preprint with a small human-evaluation cohort (thirteen participants) and no medical or external validation — and not as a general ranking the dissertation asserts; the dissertation's choice among Grad-CAM variants is a methodological decision recorded in Chapter 3, not a superiority claim made here.

The schematic relationship among these formulations — the projection of class evidence onto final-layer activations, gradient-weighted in the Grad-CAM generalization — is summarized in FIG-2.3.

(Figure 2.3)

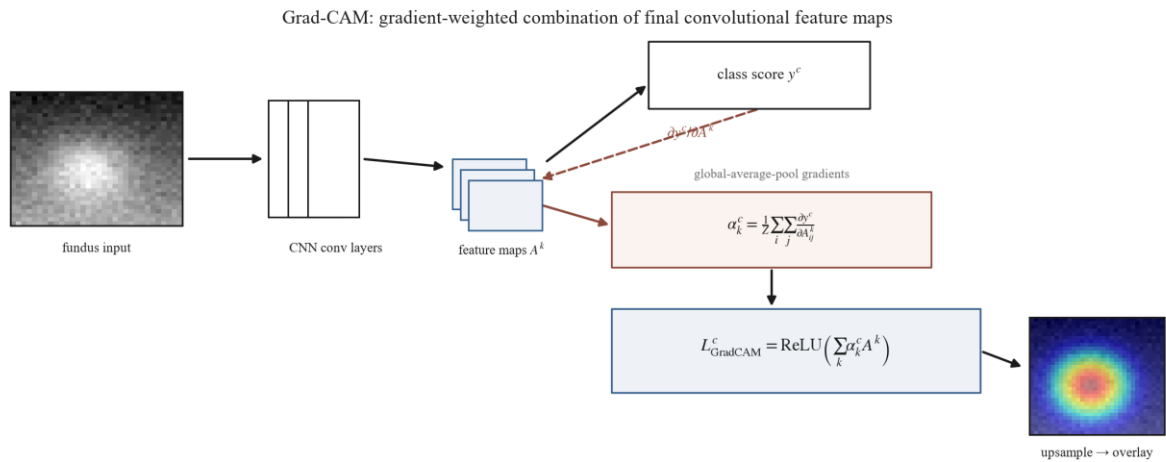


Figure 2.3 – Grad-CAM mathematical formulation – gradient-weighted combination of final convolutional feature maps with ReLU

The decisive point for the dissertation's use of these methods is the non-claim that bounds them. A Grad-CAM map is computed at the spatial resolution of the final convolutional layer — coarse relative to the input image — and is upsampled for visualization; it highlights where gradient-weighted activation for the predicted class concentrates. It does not segment lesions, and it does not represent a pixel-level diagnostic determination of where pathology is. Consequently, when the dissertation later compares Grad-CAM activations against pixel-level lesion masks, a high degree of overlap is to be read as evidence that the model's attention is directed toward clinically relevant structures, not as a demonstration that the model has clinically localized pathology or reasoned as a clinician would. This distinction (NC-14) is binding on every subsequent use of these maps and is the reason the overlap analysis of §2.5.3 is framed as an interpretability measure rather than a detection benchmark.

With the localization map formally defined, two questions remain. The first is how such a map is to be *interpreted* — what it does and does not license a reader to conclude — which §2.5.2 addresses. The second is how it is to be *quantified* against ground-truth lesion annotations, through the Attention–Lesion Overlap and Intersection-over-Union metrics, which §2.5.3 formalizes ahead of their experimental use in Chapter 4.

§2.5.2 Attention Map Interpretation

The formalization in §2.5.1 produced a localization map; this section addresses what such a map may legitimately be taken to mean. The question is not incidental, because the value of an attention map in a screening context depends entirely on the strength of the conclusions a reader is entitled to draw from it, and overreading is a real risk when a heat map is overlaid persuasively on a retinal image. The interpretation defended here is deliberately conservative: a Grad-CAM map indicates

where the network's evidence for the predicted class is concentrated, and nothing stronger.

What the map legitimately shows is class-discriminative activation. By construction it weights the final convolutional features by their influence on a specific class score, so a region of high response is one whose activation, if removed, would most reduce the model's confidence in the predicted grade. This is a genuine and useful signal: it distinguishes the regions the model relied upon from those it effectively ignored, and it is class-specific rather than a generic saliency. It is on this basis that an attention map can support a judgment of *plausibility* — whether the model's evidence falls on retinal structures a clinician would regard as relevant — which is the role it plays in the qualitative analysis of Chapter 4.

Three properties bound that judgment. The first is resolution. The map is computed at the spatial resolution of the final convolutional layer, which is coarse relative to the input image, and is then upsampled for display; its apparent boundaries are therefore approximate. The smallest diagnostically decisive lesions — the microaneurysms that define the early grades — may be correctly attended without being precisely delimited, so the absence of a sharply lesion-shaped activation is not evidence that the lesion was missed, nor is a diffuse activation evidence of imprecise reasoning. The second is method and layer dependence. The map depends on which layer's activations are used and on how the gradients are weighted; that the weighting scheme materially changes the result is shown by Grad-CAM++ [52], whose pixel-wise reweighting was reported to improve localization and the handling of "multiple occurrences of the same class" relative to Grad-CAM. An attention map is thus a property of a model-and-method pairing, not an absolute readout of the model alone, and comparisons are meaningful only when the method is held fixed.

The third and most consequential property is the gap between activation and pathology, which is the substance of the dissertation's binding non-claim on this subject (NC-14). A region of concentrated class-discriminative activation is not a determination that pathology is located there. A network may attend to context that co-occurs with disease rather than to the lesion itself, or to structures whose appearance is correlated with grade for non-causal reasons; weakly supervised localization of this kind, as Zhou et al. [50] reported for class activation mapping, remains substantially less accurate than fully supervised localization even on natural images. Consequently, when an attention map overlaps a clinician-annotated lesion, that overlap is suggestive evidence that the model's attention is directed toward clinically relevant structures — but it is not proof that the model has localized pathology or reasoned diagnostically, and the converse high-overlap reading must not be made.

These interpretive limits do not diminish the usefulness of attention maps; they fix the register in which they are useful. An attention map answers the question "where did the model's class evidence concentrate, and is that plausible?" — a question of interpretability and trust — and it does not answer the question "where is the pathology?", which belongs to clinical determination. Holding the maps to the first question is what keeps their use honest. Turning the relationship between attention and annotated lesions into a quantity — so that "plausibility" can be

measured rather than merely inspected — requires explicit overlap metrics, and those metrics, the Attention–Lesion Overlap and Intersection-over-Union, are formalized in §2.5.3, where they too are framed as interpretability measures rather than as a detection benchmark.

§2.5.3 ALO and IoU as Quantitative Explainability Metrics

Section §2.5.2 confined the interpretation of an attention map to a judgment of plausibility — whether the model's class evidence falls on clinically relevant structures — and noted that turning that judgment into a number requires an explicit overlap measure. This section formalizes the two such measures the dissertation uses: the Attention–Lesion Overlap (ALO), its primary explainability metric, and the Intersection-over-Union (IoU), retained as a secondary measure. Both compare the spatial region of a Grad-CAM activation, thresholded to a binary mask, against a pixel-level lesion annotation, and both are interpretability measures rather than determinations of clinical localization (NC-14).

The primary metric is defined relative to the lesion. The Attention–Lesion Overlap is the fraction of the annotated lesion area that the attention region covers,

$$\mathrm{ALO} = \frac{\mathrm{area}(\text{Grad}\text{-}\text{CAM} \cap \text{lesion mask})}{\mathrm{area}(\text{lesion mask})},$$

so that ALO approaches one when the model's class-discriminative activation falls over the whole of an annotated lesion and approaches zero when it falls elsewhere. ALO is asymmetric and lesion-referenced by design: it asks the single interpretability question of interest — does the model attend to the lesion? — and is the primary explainability metric under the dissertation's explainability hypothesis (H-5).

The secondary metric is symmetric. The Intersection-over-Union normalizes the intersection by the *union* of the two regions,

$$\mathrm{IoU} = \frac{\mathrm{area}(\text{Grad}\text{-}\text{CAM} \cap \text{lesion mask})}{\mathrm{area}(\text{Grad}\text{-}\text{CAM} \cup \text{lesion mask})},$$

penalizing both lesion area the attention fails to cover and attention area that spills beyond the lesion. IoU originates in object detection, where Everingham et al. [94] established, for the PASCAL VOC benchmark, the convention that a detected bounding box is counted correct when its overlap with the ground-truth box exceeds one half — $\mathrm{area}(B_p \cap B_{gt}) / \mathrm{area}(B_p \cup B_{gt}) > 0.5$ — and where Rezatofghi et al. [95] later generalized IoU so that it "provides a gradient in all possible cases, including non-overlapping situations," noting IoU's own weakness that for non-overlapping shapes "the IoU value will be zero and will not reflect how far the two shapes are." Both are arXiv/benchmark object-detection references with no medical-imaging evaluation, and they are cited here only for the overlap definition, not for any retinal result.

The choice of ALO rather than IoU as the primary metric follows directly from the nature of the attention maps established in §2.5.1–§2.5.2. A Grad-CAM map is computed at the coarse resolution of the final convolutional layer and is class-

discriminative, so by construction its thresholded region tends to extend beyond the precise boundary of any single lesion. A symmetric metric such as IoU treats that overflow as error, charging the union denominator for activation that lies outside the lesion mask; it therefore systematically understates genuine attention–lesion correspondence whenever the attention map is — as it inherently is — coarser than the lesion. ALO, by referencing only the lesion area, measures lesion *coverage* and is robust to this coarseness, which is why it is primary. IoU is retained as a stricter, secondary check: where ALO confirms that attention covers the lesion, IoU additionally reports how concentrated that attention is on the lesion rather than its surroundings, but a low IoU accompanying a high ALO is interpreted as a property of the explanation method's resolution, not as a localization failure of the model.

Crucially, neither metric is a detection benchmark, and the borrowing of IoU from object detection must not be allowed to import a detection interpretation. In object detection an IoU threshold adjudicates whether an object was *found*; here, IoU and ALO quantify the correspondence between a coarse interpretability heat map and an annotation, and a high value indicates that the model's attention is directed toward clinically relevant structures, not that the model has localized or detected pathology. This is the metric-level expression of the binding non-claim (NC-14) carried through §2.5: explainability overlap is an interpretability signal, and the IoU criterion is used as a borrowed overlap measure rather than as the detection threshold it originally defined.

These definitions complete the explainability formalization. The comparison the metrics are built for — whether the preprocessing pipeline raises attention–lesion overlap relative to the baseline, the condition $\mathrm{ALO}_{\text{preproc}} > \mathrm{ALO}_{\text{baseline}}$ posed by H-5 — is an empirical question reserved for Experiment 4 (§4.5), where it is evaluated quantitatively against the IDRiD lesion masks and illustrated qualitatively on the clinical dataset; no such result is asserted here. With contrast enhancement, CNN training, transfer learning, the thermal model, and explainability now formalized, one theoretical strand remains: the image-quality metrics by which the preprocessing pipeline's effect on the image itself — independent of any classifier — is evaluated, which §2.6 develops.

§2.6 Image Quality Metrics for Preprocessing Evaluation: CNR, VVI, Entropy, SSIM

The explainability metrics of §2.5.3 measure a property of the trained classifier — where its attention falls. This final theoretical section turns to metrics of a different kind, which measure a property of the *image* after preprocessing, independently of any classifier. The motivation is intrinsic to the dissertation's framing of preprocessing as a model component: if the preprocessing pipeline is an integral part of the model rather than incidental data preparation, its effect ought to be observable directly on the image it produces, and not only inferred from downstream accuracy. There is also a methodological reason to want a classifier-independent view. Under the integrated-pipeline hypothesis the baseline and full-pipeline arms differ along more than one axis, so a change in classification performance cannot be

attributed to preprocessing in isolation (a constraint developed fully in the experimental chapters); an image-level metric, by contrast, isolates what the preprocessing stages actually do to the image. The four metrics defined here serve that purpose, and they are reported as supplementary evidence — informative about pipeline behavior, but not by themselves able to establish or refute the primary diagnostic hypotheses.

The first metric, the contrast-to-noise ratio (CNR), quantifies how well diagnostically relevant structures stand out from their background relative to the level of image noise — operationally, the difference in mean intensity between vessel and background regions normalized by the background noise. CNR targets exactly the property the illumination and contrast stages are designed to improve: the adaptive flat-field correction (Stage 4) removes the low-frequency illumination gradients that depress local contrast, and the contrast-limited adaptive histogram equalization of Stage 5 amplifies local contrast under a noise-bounding clip limit, so CNR is expected to rise after these stages. That expectation is a design hypothesis to be tested in Experiment 2, not a result asserted here.

The second metric, the vessel visibility index (VVI), addresses the same structures from the standpoint of detectability rather than signal ratio: it measures how clearly the retinal vasculature can be discerned in the processed image. The retinal vessels are both the most contrast-sensitive structures in the fundus and among the most diagnostically informative, and they are the structures most directly affected by the flat-field and contrast stages, so VVI is expected, like CNR, to improve under preprocessing. CNR and VVI together characterize the *enhancement* side of the pipeline's effect.

The third metric, image entropy, measures the information content of the processed image from its intensity distribution. As the dissertation's glossary records, entropy is "expected to increase after contrast enhancement, reflecting greater utilization of the intensity range": a low-contrast image confines most of its pixels to a narrow band of intensities, whereas contrast enhancement spreads them across a wider range, raising the entropy of the intensity histogram. Entropy thus provides a distribution-level summary of the redistribution effect that §2.1.1 described mechanistically for histogram equalization.

The fourth metric performs a different and necessary function. Whereas CNR, VVI, and entropy reward enhancement, the structural similarity index (SSIM), introduced by Wang et al. [96] as a structural-similarity measure for image-quality assessment, measures how much of the original image's structure is preserved in the processed image. SSIM is included as a guard: contrast enhancement can be carried too far, introducing artifacts or distorting anatomical structure, and a metric that only rewards contrast would not detect such damage. A high SSIM between the processed and original image confirms that the enhancement has preserved structural information rather than fabricated or destroyed it, so that the gains reported by the other three metrics reflect genuine enhancement rather than destructive over-processing. The four metrics are therefore complementary: three measure how much the pipeline improves visibility, and the fourth verifies that it does so without structural damage.

The evidentiary standing and the sourcing of these metrics must both be stated plainly. In the dissertation's evidence hierarchy they are supplementary: they describe what preprocessing does to the image and support interpretation of the pipeline's behavior, but they cannot independently establish or refute the primary diagnostic hypotheses, which rest on the classification metrics of §2.2.2. Their literature grounding is also thin and general rather than diabetic-retinopathy-specific: SSIM derives from a general image-quality benchmark [96] with no retinal application in its source, and the contrast-to-noise ratio and vessel visibility index are standard signal- and structure-level constructs without a dedicated primary source in the dissertation's corpus, defined operationally for this work in the methodology (TAB-3.3, §3.4.1). This thin, general-purpose sourcing is acknowledged rather than concealed: the metrics are used as established image-analysis tools, not presented as DR-validated measures. Fundus-specific image-quality assessment is nonetheless an established research area in its own right — Zago et al. [9] trained a retinal-image-quality classifier that generalized across databases (DRIMDB and ELSA-Brasil) — which confirms that image quality is a measurable, retinal-domain property, even though that study assesses overall image gradability rather than supplying a primary source for the particular CNR and VVI constructs used here. A related but separate family — the model-calibration metrics, the Expected Calibration Error and Brier Score formalized by Guo et al. [97] — is sometimes grouped with quality measures but is distinct: those quantify the reliability of predicted probabilities, a property of the classifier, and belong with the diagnostic metrics rather than with the image-quality metrics defined here.

The conceptual relationship among the four image-quality metrics — three enhancement measures balanced against one structural-preservation guard — is summarized in FIG-2.5.

(Figure 2.5)

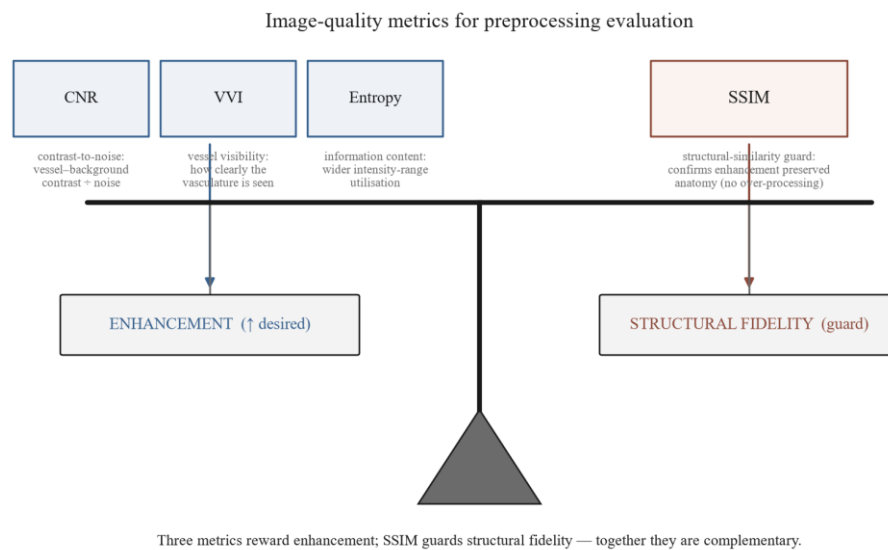


Figure 2.5 – Image-quality metrics for preprocessing evaluation – CNR, VVI, and entropy as enhancement measures balanced by SSIM as a structural-preservation guard

These metrics complete the theoretical apparatus the dissertation requires. Their formal definitions are consolidated in TAB-3.3 (§3.4.1) and they are applied to the pipeline's stages, including the flat-field σ sweep, in Experiment 2 (§4.3.3), where the expectations stated above are tested rather than assumed. With contrast enhancement, spatial filtering, convolutional feature extraction, loss and regularization, transfer learning, the thermal-optical model, explainability, and now preprocessing-evaluation metrics all formalized, the theoretical foundations of the dissertation are established. Chapter 2's conclusions, in §2.C, synthesize these foundations and distinguish the experimentally-grounded strands from the single theoretical, simulation-only one.

§2.C Conclusions to Chapter 2

Chapter 2 has supplied the theoretical foundations on which the remainder of the dissertation rests. Where Chapter 1 located the research problem and committed to an integrated preprocessing–CNN framework as the response, this chapter has given each component of that framework its mathematical and theoretical basis, so that the methodology of Chapter 3 and the experiments of Chapters 4 and 5 can be specified and interpreted on solid ground rather than asserted.

The greater part of the chapter develops foundations that the empirical chapters exercise directly. The contrast-enhancement lineage of §2.1 — from histogram equalization through adaptive equalization to the contrast-limited, dual-constraint formulation of CLAHE, together with the edge-preserving noise-control principle — grounds the preprocessing stages of the pipeline. The convolutional-network theory of §2.2 — convolution, pooling, and the feature hierarchy; the loss functions that train it under severe class imbalance; and the regularization that controls the resulting overfitting — grounds the classifier and its training. The transfer-learning theory of §2.3 — the layer-dependent, domain-bounded transferability of features, the choice between frozen-base and progressive fine-tuning, and the in-domain self-supervised pretraining (§2.3.3) that initializes the integrated arm — grounds the initialization and adaptation of the network's weights. The explainability formalization of §2.5 — the CAM and Grad-CAM construction, the bounded interpretation of attention maps, and the Attention–Lesion Overlap and Intersection-over-Union metrics — grounds the interpretability analysis. And the image-quality metrics of §2.6 — contrast-to-noise ratio, vessel visibility index, entropy, and structural similarity — provide a classifier-independent means of evaluating the preprocessing component. Each of these strands is foundational only: the empirical questions they frame — the CLAHE sensitivity profile, the integrated-pipeline comparison, cross-dataset transfer, attention–lesion overlap, and image-quality change — are reserved for Chapters 4 and 5 and are not anticipated here.

One strand stands apart in epistemic status. The coupled thermal-optical model of fundus tissue in §2.4 is a theoretical, simulation-only contribution concerning the laser-photocoagulation *treatment* of diabetic retinopathy rather than its diagnostic classification. It is qualitative, supported by a single prior publication, and — as emphasized at its presentation — has not been validated against experimental or

clinical data; it is a secondary contribution, epistemically independent of the integrated diagnostic system that is the dissertation's central concern, and it is carried forward only as such.

No theoretical strand remains outstanding. The in-domain self-supervised pretraining for retinal imaging (§2.3.3), which underlies the integrated arm's initialization, was the one strand earlier deferred for want of dedicated primary sources; those sources were acquired in the v6.1.0 corpus expansion, and the section has since been written alongside the rest of §2.3. It grounds the ophthalmology-specific initialization of the integrated arm as a representation-learning choice whose efficacy is established empirically in Chapter 4 rather than assumed here. With §2.3.3 in place, Chapter 2 is content-complete: every foundation the experimental chapters require has its theoretical basis established.

With these foundations established, the dissertation is positioned to specify its methodology. Chapter 3 builds the preprocessing pipeline on the contrast and noise-control theory of §2.1, the ResNet-50 and EfficientNet-B3 architectures on the convolutional theory of §2.2, the pretraining and two-stage fine-tuning protocol on the transfer-learning theory of §2.3, and the multi-metric evaluation framework on the explainability and image-quality theory of §2.5 and §2.6. The epistemic disciplines that have governed this chapter — that Grad-CAM is an interpretability tool and not clinical localization, that preprocessing parameters and transfer are not assumed portable but evaluated, that the integrated pipeline's effect is not attributable to any single factor, and that the primary metrics are the imbalance-aware ones — are carried into that methodology unchanged.

3 METHODOLOGY OF INTEGRATED PREPROCESSING-CNN PIPELINE DESIGN

§3.1.1 Pipeline Stage Specification: 8-Stage System

Chapter 2 supplied the theoretical foundations of every component of the diagnostic framework committed to in Chapter 1 — the contrast and noise-reduction theory behind the enhancement stages, the convolutional and optimization theory behind the classifier, the transfer-learning theory behind its initialization, and the explainability and image-quality theory behind its evaluation. This chapter specifies the methodology those foundations support, beginning with the preprocessing pipeline. The organizing commitment of the specification is the proposition, stated in the central thesis of this work, that the diagnostic model is a two-stage system, `model = preprocessing + CNN`, in which the preprocessing stage is an integral component of the model rather than ancillary data preparation: it defines the feature space available to the convolutional network and therefore co-determines what the network can learn (CENTRAL_THESIS; IT-1). The decision to formalize preprocessing as part of the model is, in the first instance, a conceptual one, motivated by the integrated preprocessing-CNN paradigm (paradigm P2) that this

dissertation instantiates; the eight ordered stages set out below are the engineering realization of that commitment.

This conceptual decision is most clearly seen against the alternative methodological practice it departs from. In the end-to-end paradigm (paradigm P1), of which Gulshan et al. [13] is taken in this dissertation as the canonical representative, preprocessing is treated as ancillary data preparation: the published methods defer preprocessing details to the supplementary material and locate the methodological emphasis in architecture, data scale, and training protocol [13]. This is a description of observable methodological practice, not an attribution of any theoretical claim about preprocessing to those authors (SIR-9). The present work makes the opposite methodological choice: it specifies the pipeline with the same rigor it later applies to the network, places the pipeline under controlled experimental contrast (Experiment 1, §4.2), and decomposes it into ablatable stages (Experiment 2, §4.3). Whether each stage earns its place in that pipeline is an empirical question reserved for those experiments; the task of the present section is to define the construct precisely enough that the question can be asked. The pipeline specified here is the canonical eight-stage configuration of OD-3.

The pipeline is a deterministic, order-dependent transformation: each stage consumes the output of the one before it, and the sequence cannot be reordered without changing the result. Stages 0 through 5 and Stage 7 are always applied, at both training and inference time; Stage 6 (augmentation) is applied at training time only. The end-to-end sequence is shown in Figure 3.1 and, in its relation to the downstream classifier, in Figure 3.14. The eight stages are specified below in execution order.

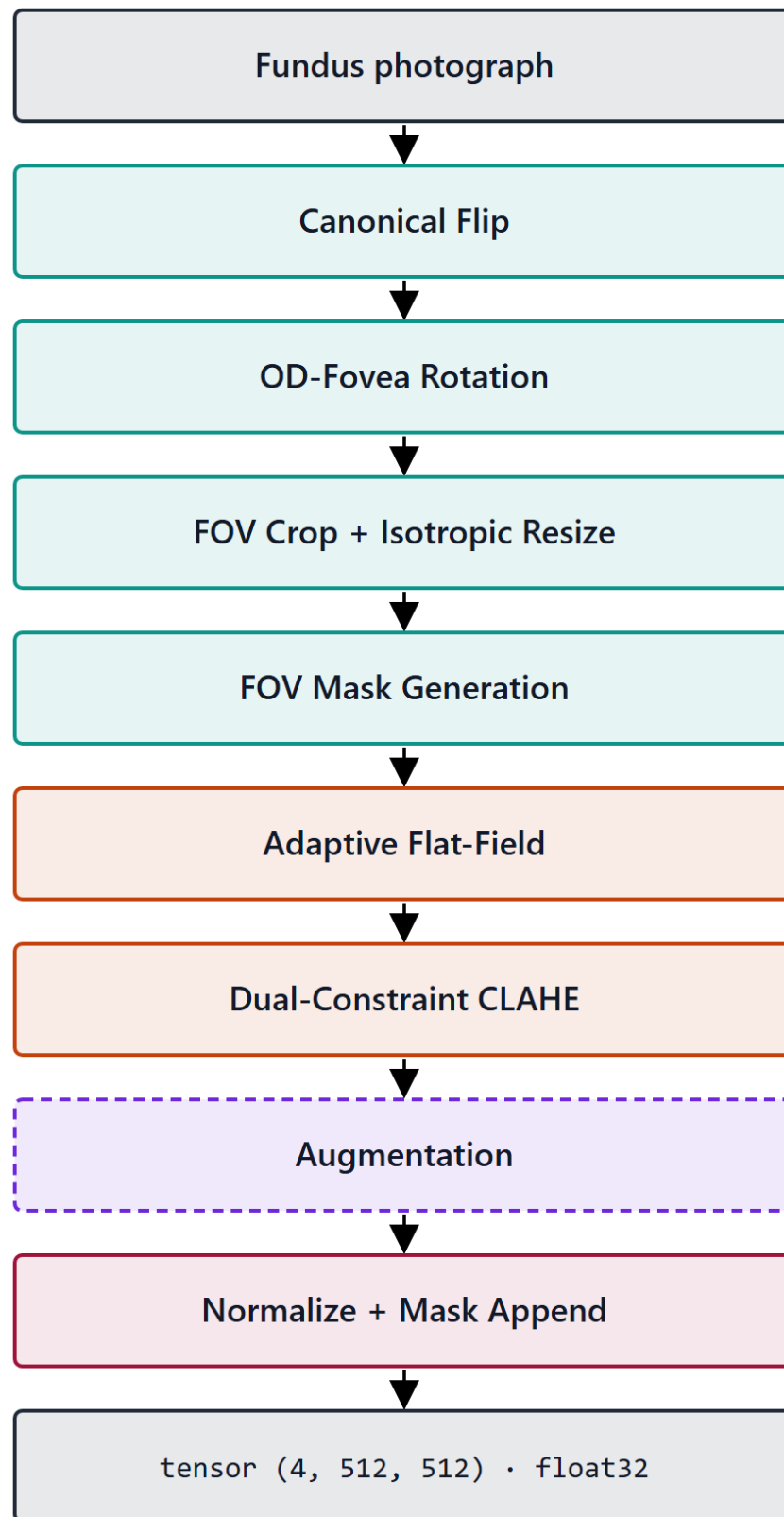


Figure 3.1 – 8-stage preprocessing pipeline overview (vertical)

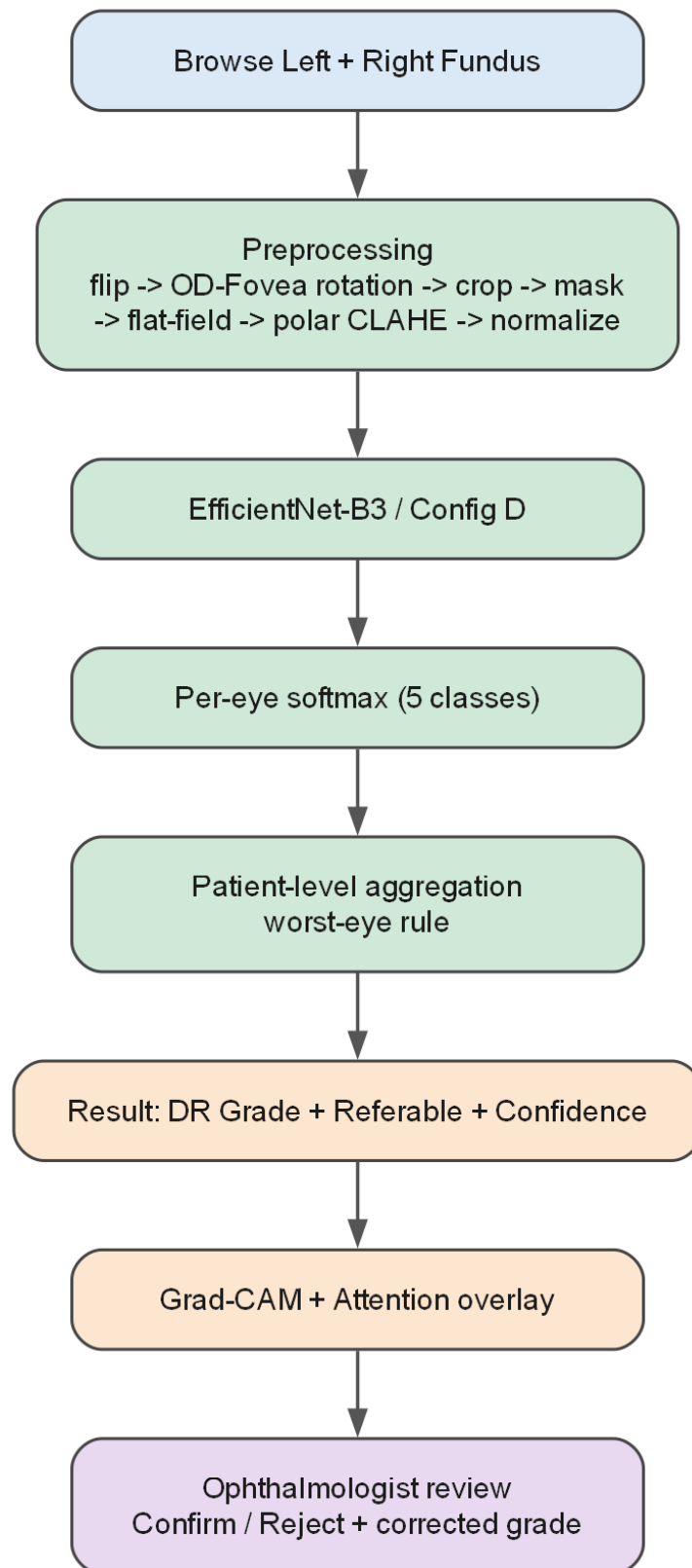


Figure 3.14 – End-to-end model flowchart, model = preprocessing + CNN

Stage 0 — Canonical flip. Fundus images of the left and right eye are approximate mirror images of one another: in the right eye the optic disc lies temporally to the right of the macula, and in the left eye to its left. Stage 0 removes this laterality degree of freedom by horizontally flipping every left-eye image to the right-eye canonical orientation (optic disc right, macula left); right-eye images pass through unchanged. Eye laterality is taken from image metadata where available and

from heuristic detection otherwise. The stage is always on. Its rationale is to spend the network's capacity on disease-relevant variation rather than on a known, removable geometric symmetry, and to give the subsequent rotation-normalization stage a consistent starting orientation (Figure 3.2).

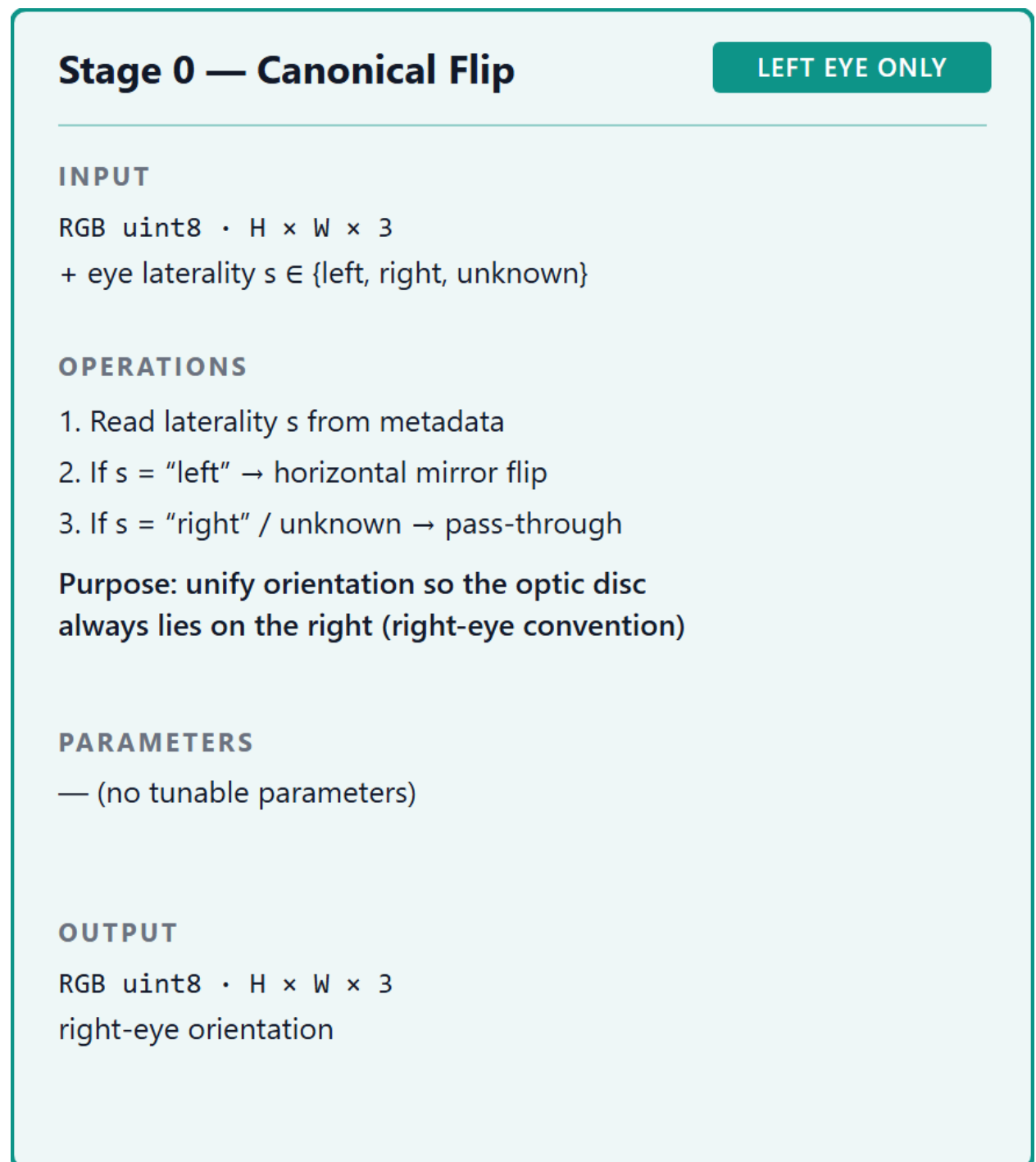


Figure 3.2 – Stage 0 – canonical flip (L→R)

Stage 1 — Optic-disc-fovea rotation normalization. Stage 1 standardizes in-plane rotation by detecting two anatomical landmarks and aligning the axis between them to the horizontal. The optic disc is localized as the brightest coherent region of the fundus, and the fovea as the darkest region subject to a distance prior relative to the optic disc; the image is then rotated so that the optic-disc-to-fovea axis is horizontal. The stage is always on, but it carries an explicit low-confidence fallback:

when landmark detection is judged unreliable the rotation is skipped, and a fallback rotation dispersion of $\sigma = 13.0^\circ$ is used. The augmentation rotation dispersion of Stage 6 is, by design, adaptive per image, derived from the uncertainty of this detection rather than fixed (OD-3 Stage 1).

The reliability of this classical-vision detection step was measured directly, rather than assumed, on the IDRiD dataset (516 images), for which independent reference coordinates are available; the overlay montage is reproduced in Figure 3.10 and the quantitative results in the validation record (RES-VAL). The following analysis of those results is the dissertation's own. Optic-disc localization was moderate: the detector placed the disc within one optic-disc radius of the reference in 67.3% of training-split images and 61.2% of test-split images, and within half a radius in 34.9% and 32.0% respectively, with a median displacement of roughly 237–260 pixels. Fovea localization was, by contrast, unreliable: the detector placed the fovea within one optic-disc radius of the reference in 0.0% of training-split images and 0.97% of test-split images, with a median displacement on the order of five optic-disc radii. A further finding is that the detector's internal `confident` flag was uninformative on this set — it reported confidence for 100% of images and therefore could not discriminate the successful localizations from the failed ones. These measurements are bounded to IDRiD (a single dataset, Kowa camera, 516 images) and are not claimed to generalize (DGL-1). They are, however, the empirical justification for the conservative design of Stage 1: because reliable fovea localization cannot be assumed, the stage is built to degrade gracefully — preferring the skip-and-fallback path and an uncertainty-adaptive augmentation dispersion over a confident but possibly mis-aligned rotation (Figure 3.3).

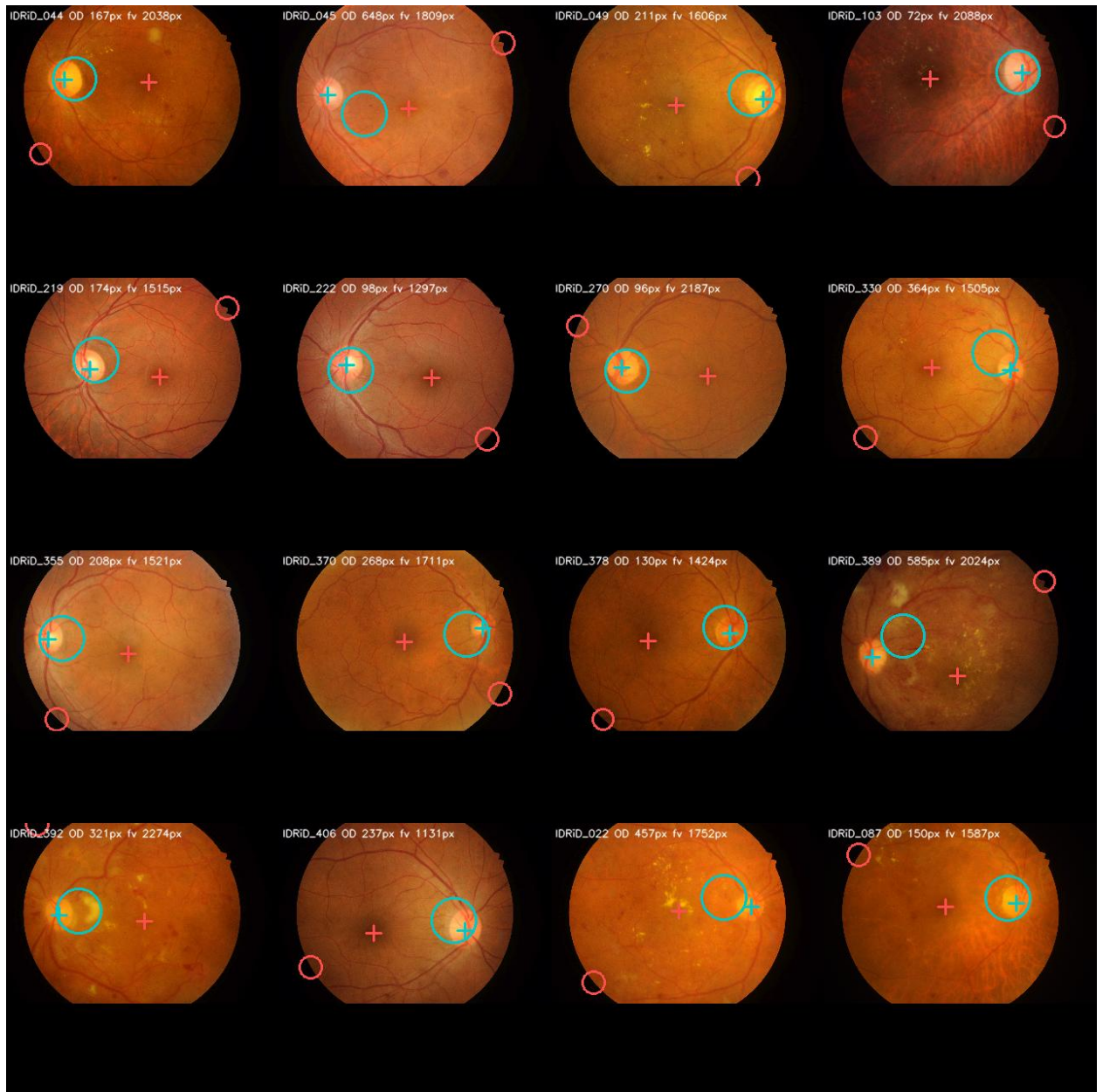


Figure 3.10 – OD/fovea detector validation montage (IDRiD, 516 images)

Stage 1 — OD-Fovea Rotation

ALWAYS ON

INPUT

RGB uint8 · H × W × 3
right-eye-oriented from Stage 0

OPERATIONS

1. Detect OD centre (green-channel bright-region percentile threshold)
2. Detect fovea centre (darkest macular region within search window)
3. Compute $\theta = \angle(\text{OD} \rightarrow \text{Fovea}, \text{x-axis})$
4. Rotate image by $-\theta$

PARAMETERS

OD: top-percentile (green-channel)
Fovea: search window relative to OD

OUTPUT

RGB uint8 · H × W × 3
retinal axis aligned to horizontal

Figure 3.3 – Stage 1 – OD-fovea rotation normalization

Stage 2 — Field-of-view crop and isotropic resize. Fundus images arrive at heterogeneous resolutions and aspect ratios, with the circular field of view (FOV) framed differently by each camera. Stage 2 detects the image foreground — the illuminated fundus disc — using a foreground-detection procedure rather than a Hough-circle fit, crops to the FOV region, and rescales it isotropically to a 512×512 canvas with centered zero-padding. Isotropy is the operative property: scaling both axes by the same factor preserves the circular geometry of the fundus and the true aspect ratio of lesions, so that a microaneurysm is not stretched into an ellipse by an anisotropic resize. The 512×512 resolution and the centered-padding convention are

fixed for every dataset, giving the network a geometrically consistent input frame. The stage is always on (Figure 3.4).

Stage 2 — FOV Crop + Isotropic Resize

ALWAYS ON

INPUT

RGB uint8 · H × W × 3

OPERATIONS

1. Detect FOV bounding box (PIL foreground / brightness threshold)
2. Crop with margin m around (cx, cy, R)
3. Isotropic scale: $2R \rightarrow 512$ px
4. Centre zero-pad to 512×512

Contrast with baseline: baseline uses stretch-resize, distorting fundus geometry.

PARAMETERS

target = 512×512 · preserve aspect ratio

OUTPUT

RGB uint8 · $512 \times 512 \times 3$

circular FOV preserved + zero-padding map

Figure 3.4 – Stage 2 – FOV crop + isotropic resize

Stage 3 — Field-of-view mask generation. The centered zero-padding introduced by Stage 2 creates a border of non-fundus pixels whose value carries no retinal information. Stage 3 makes this distinction explicit by generating a binary mask — 1.0 where the pixel is real fundus data, 0.0 where it is padding — and appending it to the image as a fourth input channel. The mask is produced alongside the resized image in Stage 2 and is always on. Carrying the FOV as an explicit channel, rather than leaving the network to infer the fundus boundary from pixel intensities, is one of the named design features of the pipeline

(RESEARCH_ARCHITECTURE §10): it lets the downstream flat-field correction (Stage 4) and normalization (Stage 7) operate on genuine retinal pixels only, and gives the network an unambiguous signal of where valid data ends (Figure 3.5).

Stage 3 — FOV Mask Generation

SIDE BRANCH

INPUT

padding map / FOV geometry from Stage 2

OPERATIONS

1. $M[i, j] = 1.0$ if pixel inside FOV
 $M[i, j] = 0.0$ if pixel is zero-padding
2. Optional Gaussian smoothing of mask boundary (soften hard edge)

Routed in parallel to Stages 4–6, then
appended as 4th channel at Stage 7.

PARAMETERS

mask values $\in \{0.0, 1.0\}$

OUTPUT

float32 mask · 512 × 512

$M \in \{0.0, 1.0\}$

Figure 3.5 – Stage 3 – FOV mask (4th channel)

Stage 4 — Flat-field correction. Fundus images exhibit slowly varying illumination gradients — vignetting and uneven flash coverage — that are unrelated to pathology but compete with the low-contrast microvascular signal the classifier must read. Stage 4 corrects this by subtracting a heavily blurred estimate of the local background and re-centering the intensity range,

$$\text{corrected} = \text{image} - \text{GaussianBlur}(\text{image}, \sigma) + 128$$

with the blur scale set adaptively to the image geometry as $\sigma = 0.07 \times D$, where D is the FOV diameter in pixels derived from the Stage-3 mask. The correction is applied inside the FOV mask only, so that the zero-padding border is not drawn into the estimate. Tying σ to the per-image FOV diameter, rather than fixing it in absolute pixels, is the stage's defining choice: it keeps the spatial scale of the background estimate constant relative to the retina across images cropped and resized from different source resolutions. The stage is always on; the sensitivity of performance to the σ factor across the range $0.05 \cdot D$ – $0.10 \cdot D$ is itself characterized empirically in Experiment 2 (§4.3.3) and is not asserted here (Figure 3.6).

Stage 4 — Adaptive Flat-Field

ALWAYS ON

INPUT

RGB uint8 · 512 × 512 × 3

OPERATIONS

1. Estimate background illumination

$B = \text{GaussianBlur}(I, \sigma)$

2. Subtract: $I' = I - B + 128$

3. Clip to valid range [0, 255]

σ is adaptive to FOV geometry — replaces the fixed $\sigma = 45$ from V4.

PARAMETERS

$\sigma = 0.07 \cdot D$ (D = FOV diameter)

sweep range in Exp 2: $\sigma \in [0.05, 0.10] \cdot D$

OUTPUT

RGB uint8 · 512 × 512 × 3

uniform illumination, vignetting removed

Figure 3.6 – Stage 4 – adaptive flat-field correction

Stage 5 — Contrast-limited adaptive histogram equalization (CLAHE).

Stage 5 enhances local contrast using the dual-constraint formulation developed theoretically in §2.1.2. The clip limit for each tile is set as the tighter of two constraints,

$$\text{CL} = \min\left(\text{clip_factor} \cdot \frac{\text{tile_area}}{256}, \text{global_threshold} \cdot \text{tile_area}\right),$$

over an 8×8 tile grid applied to the L-channel of the LAB color space, with the operation applied stochastically during training (probability 0.8) and deterministically at inference (OD-3 Stage 5). Operating on the LAB luminance channel enhances contrast without shifting hue; the stochastic application at train time turns the enhancement into a source of controlled variability. The full justification of the dual-constraint rule and of its two hyperparameters is the subject of §3.1.2; what matters here is that the parameters are not imported from any external study — the parameters validated on STARE in the candidate's prior work are not assumed to transfer to EyePACS (DGL-5) — but are fixed by validation within the dissertation's own framework. The stage is always on (Figure 3.7).

Stage 5 — Dual-Constraint CLAHE

STOCHASTIC

INPUT

RGB uint8 · 512 × 512 × 3

OPERATIONS

1. RGB → LAB colour space
2. Apply CLAHE to L-channel only with dual-constraint clip limit:
$$CL_tile = \min(c \cdot area / 256 , t \cdot area)$$
3. At train: apply with $p = 0.8$ (stochastic)
At inference: always apply
4. LAB → RGB

PARAMETERS

clip_factor $c = 2.0$ · threshold $t = 0.01$
tile_grid = 8 × 8

OUTPUT

RGB uint8 · 512 × 512 × 3
enhanced local contrast (microvasculature)

Figure 3.7 – Stage 5 – dual-constraint CLAHE (with polar variant and vessel maps)

Stage 6 — Augmentation. Stage 6 is the only stage applied at training time only; at inference it is bypassed entirely, so that the production transformation is deterministic. During training it applies a unified affine transform — rotation, with the dispersion adaptive from the Stage-1 detection uncertainty; zoom in the range [0.9, 1.1]; and optional shear and stretch — together with brightness and contrast jitter and a principal-component (PCA) color jitter. The role of augmentation here is dual: it enlarges the effective training distribution to counter overfitting, and, applied selectively, it is one of the levers available against the severe class imbalance of EyePACS (SB-2.1). The full augmentation specification, including the PCA color

basis actually estimated from the EyePACS training set, is given in §3.1.3 (Figure 3.8).

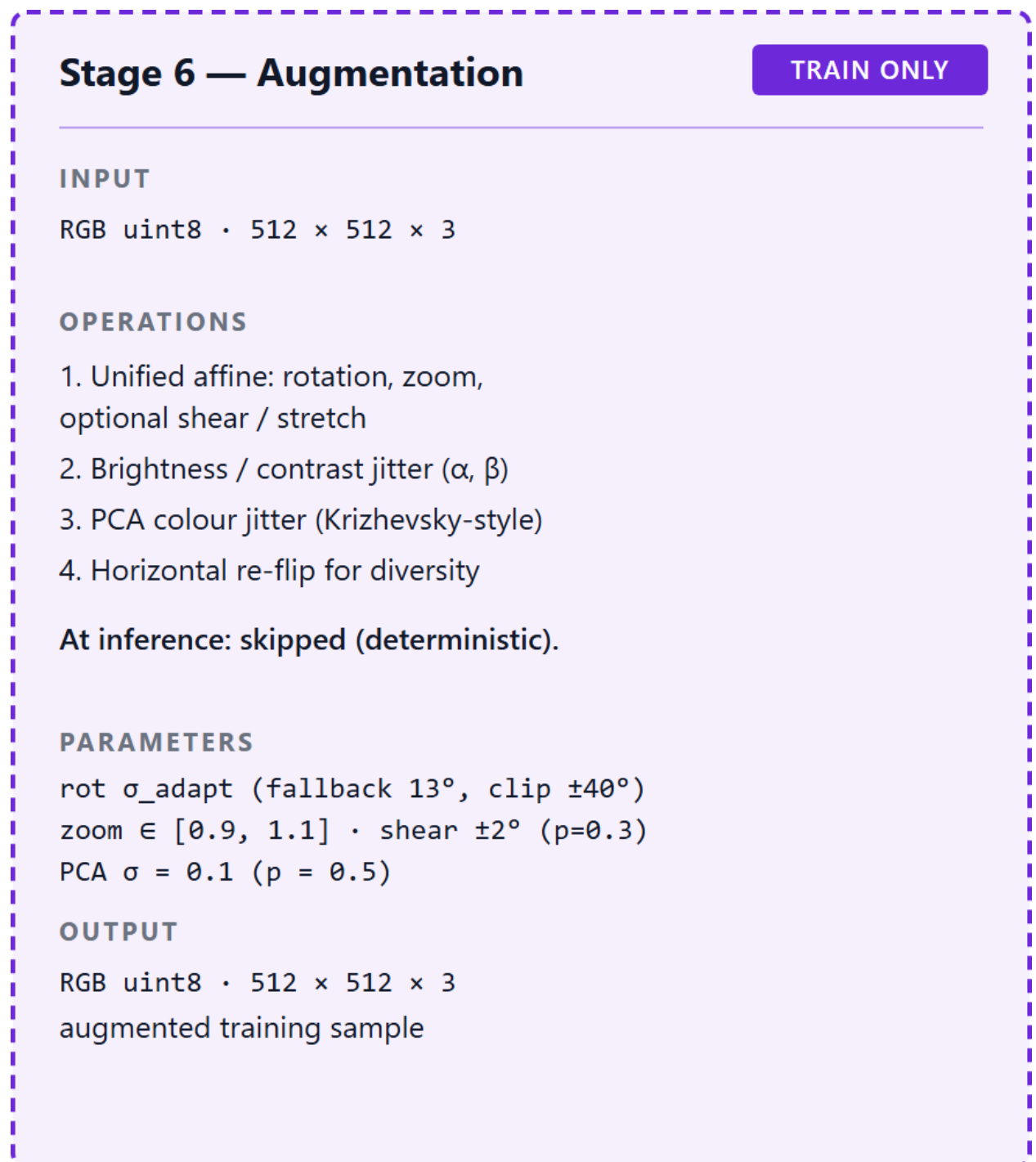


Figure 3.8 – Stage 6 – augmentation (rotation/scale/shear/PCA-color/brightness)

Stage 7 — Dataset-specific normalization. The final stage converts the image to a tensor and standardizes its channel statistics. It first maps the HWC uint8 image to a CHW float32 tensor in $[0, 1]$, then applies channel-wise normalization ($x - \text{mean}$) / std, and appends the FOV mask as the fourth channel, yielding an output tensor of shape (4, 512, 512). The normalization statistics are dataset-specific rather than inherited from ImageNet: the per-channel mean and standard deviation are computed on the EyePACS training set, after Stages 0–4 have been applied, using only pixels where the FOV mask equals 1.0 — so that the zero-padding does not bias

the statistics. These statistics were computed as a real artifact of this work (RES-NORM): over roughly 1.25 billion in-mask pixels from 5,000 EyePACS images, the per-channel means were approximately [0.506, 0.505, 0.504] and the standard deviations approximately [0.090, 0.074, 0.058] on the [0, 1] scale. The pronounced asymmetry of the standard deviations across the colour channels — the red channel varying roughly 1.6 times as much as the blue — is precisely the dataset-specific structure that an ImageNet normalization would mismatch, and is the rationale for computing the statistics on the target distribution. The stage is always on and always last (Figure 3.9).

Stage 7 — Normalize + Mask Append

ALWAYS ON

INPUT

RGB uint8 · 512 × 512 × 3
+ FOV mask M (from Stage 3, float32 512²)

OPERATIONS

1. uint8 → float32, scale to [0, 1]
2. Per-channel normalize
$$x'_c = (x_c - \mu_c) / \sigma_c$$
3. Append M as 4th channel

Baseline uses ImageNet (μ , σ) on 3 channels.
Pipeline uses dataset-specific stats on 4ch.

PARAMETERS

(μ_c , σ_c) computed on training set
after deterministic Stages 0–4

OUTPUT

float32 tensor · (4, 512, 512)
CNN-ready

Figure 3.9 – Stage 7 – dataset-specific normalization

Two operating states of the pipeline are defined for the experimental design (OD-3). In its **ACTIVE** state — the full pipeline — all eight stages are applied, Stage 6 at training time only, and the output is the 4-channel (RGB + FOV mask) tensor described above. In its **ABSENT** state — the experimental baseline — none of the eight stages is applied; the image is instead stretch-resized to 512×512 and normalized with the fixed ImageNet statistics (mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225]), yielding a 3-channel tensor with no FOV mask. This ACTIVE-versus-ABSENT contrast is the operational construct that Experiment 1 manipulates as the preprocessing axis of its factorial design (§4.2); it is an internal construct defined by OD-3, terminologically distinct from the P1 representative discussed above (SB-1.12). Between the two extremes, Experiment 2 defines a seven-level ablation ladder — baseline, then cumulatively adding flip, rotation, isotropic-resize-plus-mask, flat-field, CLAHE, and finally the full pipeline — to quantify each stage's marginal contribution (§4.3). That the full eight-stage configuration is the one specified here does not entail that it is the universally optimal preprocessing configuration; the marginal value of each stage, and the possibility that some stage is redundant or harmful for a given architecture, is exactly what the ablation is designed to test, and is not asserted in advance (NC-17).

The pipeline so specified is the dissertation's own design. It develops a lineage of the candidate's previously published results — earlier work in which conventional preprocessing raised validation accuracy from 71% to 86% with a four-layer convolutional network (Sapakova, Yesmukhamedov & Sapakov, 2025; reported with ROC-AUC 0.9638), and in which an upgraded CLAHE variant was studied on the STARE database (whose headline figures are not transferable to the present context, SB-1.2; same article, treated as a single self-cited thread, SIR-5) — into a formalized, eight-stage, four-channel pipeline. The wider literature motivates the standardization objective without establishing it: cross-dataset studies report performance variation between fundus sources processed as an exogenous step [44], benchmark studies report that image quality bears on grading reliability [15, 8], and enhancement studies report that contrast operations such as CLAHE can help or, for at least one architecture, harm classification, depending on the dataset and tuning [68, 69] — context that is consistent across differing datasets and tasks only at the level of motivation, not of transferable numbers (SIR-3). The objective the pipeline is designed to serve — reducing variability across imaging devices and acquisition conditions while preserving diagnostically relevant features, and doing so without introducing destructive structural artifacts [96] — is a design objective stated here, not a demonstrated result; its evaluation belongs to Chapter 4. With the pipeline fixed as a precise construct, §3.1.2 returns to Stage 5 to complete the formalization of the dual-constraint CLAHE, §3.1.3 details the Stage-6 augmentation and its estimated PCA color basis, and §3.1.4 specifies the external-image ingestion protocol that precedes Stage 0 for clinical data.

§3.1.2 Modified CLAHE Algorithm with Dual-Constraint Clip Limit

Stage 5 of the pipeline, introduced in §3.1.1 as the contrast-enhancement stage, realizes as a concrete image operation the dual-constraint clip limit formalized theoretically in §2.1.2. The theoretical section established the rule and its rationale: the clip limit for each tile is set as the tighter of a histogram-relative term and a tile-relative absolute ceiling,

$$\text{CL} = \min\left(\text{clip_factor} \cdot \frac{\text{tile_area}}{256}, \text{global_threshold} \cdot \text{tile_area}\right), \text{ cf. Eq. 2.3}$$

so that ordinary contrast limiting governs well-spread local histograms while the absolute ceiling caps the attainable mapping slope in strongly peaked ones (§2.1.2). The present section does not re-derive this rule; it specifies how the rule is applied as Stage 5 — the colour space it operates in, the tiling and interpolation it uses, the manner of its application at training versus inference time, and how its two free hyperparameters are fixed.

Colour-space realization. The operation is applied to the luminance channel of the LAB colour space rather than to the RGB channels directly. The image is converted from RGB to LAB, the dual-constraint contrast-limited adaptive histogram equalization is applied to the L (lightness) channel alone, and the result is converted back to RGB. Confining the equalization to the luminance channel enhances local contrast — the visibility of low-amplitude microvascular structure against the retinal background — without altering the a and b chrominance channels, so that hue is preserved and the characteristic colour of lesions and vasculature is not distorted by the contrast operation. This is the engineering reason for the LAB choice: contrast and colour are separated, and only contrast is manipulated.

Tiling, redistribution, and interpolation. The L-channel is partitioned into an 8×8 grid of contextual tiles (OD-3 Stage 5). Within each tile a local histogram is formed and clipped at the dual-constraint limit of Equation (cf. 2.3); the excess mass removed by clipping is redistributed across the histogram so that the total tile mass is conserved, and the clipped histogram's cumulative distribution defines that tile's intensity mapping. To avoid the block artifacts that independent per-tile mappings would produce at tile boundaries, the per-pixel mapping is computed by bilinear interpolation between the mappings of the neighbouring tile centres — the standard CLAHE reconstruction. The 8×8 grid fixes the spatial scale of the contextual region relative to the 512×512 frame established by Stage 2, so that the contextual region has a consistent size across images.

Stochastic application. The stage is applied stochastically at training time, with probability 0.8, and deterministically at inference time (OD-3 Stage 5). The asymmetry is deliberate. At inference the production transformation must be reproducible, so the enhancement is always applied; at training, applying it on a randomly selected 80% of inputs turns the contrast operation into an additional, controlled source of input variability, coordinated with the augmentation role of Stage 6 (§3.1.3). The training distribution therefore contains both enhanced and unenhanced versions of the data, which discourages the network from depending on a

single fixed contrast profile while still presenting the enhanced profile as the dominant and the inference-time condition.

Hyperparameters and how they are fixed. Equation (cf. 2.3) exposes two free hyperparameters — the clip factor and the global threshold — and the realization deliberately leaves them free. Their values are not imported from any external study or from the candidate's prior work: the single-threshold $T/80$ form validated on the STARE database in previously published results (Sapakova, Yesmukhamedov & Sapakov, 2025 — cited as a single self-authored thread, the two literature cards drawn from it describing the same article, SIR-4/SIR-5) was optimized for a different image distribution, and its headline figures, including a reported 100% accuracy on that small dataset, are not transferable to the present experimental context (SB-1.2). Consistent with the parameter-portability constraint (DGL-5), the clip factor and global threshold used in the pipeline are determined by the parameter sweep of Experiment 2 (§4.3.2), which characterizes the downstream sensitivity profile required by hypothesis H-2; no specific values are asserted here. That this empirical determination is necessary rather than optional is reinforced by the external evidence that the effect of a contrast operation is contingent on architecture and tuning — Hayati et al. [68], evaluating CLAHE under a single fixed configuration across four networks on APTOS 2019 (binary task, accuracy-only, no confidence intervals), found it to help three architectures and degrade a fourth by 12.02 percentage points. The dual-constraint formulation is therefore presented as a tunable design whose parameters are fixed by validation, not as a configuration claimed optimal in advance (NC-17).

Spatially-adaptive refinement (FIG-3.7). The implementation additionally explored a spatially-adaptive refinement of the stage, in which the tiling adapts to retinal geometry and local vessel density rather than to a fixed Cartesian grid; the variant, together with the intermediate vessel-detection and vessel-density maps it uses, is shown in Figure 3.7. A consequential design detail of this variant is that it pivots its adaptation on the centroid of the FOV mask (Stage 3) rather than on a detected fovea — a choice directly justified by the Stage-1 validation reported in §3.1.1, where fovea localization on IDRiD proved unreliable (within one optic-disc radius of the reference in under 1% of images); pivoting on the geometrically robust FOV centroid avoids propagating the detector's fovea-localization failures into the contrast stage. The canonical Stage-5 specification of OD-3, and the configuration carried forward in this methodology, remains the dual-constraint CLAHE on the LAB L-channel over an 8×8 tile grid as specified above; the adoption status of the spatially-adaptive variant as a pipeline default is an implementation matter not yet reflected in the operational definition and is flagged for governance reconciliation [VERIFY: OD-3 Stage 5 vs shipped polar-CLAHE default].

With the contrast stage specified as a parameterized, luminance-channel, stochastically-applied operation whose parameters are fixed by Experiment 2, the pipeline specification turns to Stage 6, the train-time augmentation with which Stage 5's stochasticity is coordinated, in §3.1.3.

§3.1.3 Augmentation Strategy for Class Imbalance Mitigation

Stage 6 is the only stage of the pipeline applied at training time alone. Where §3.1.2 specified the stochastic application of Stage-5 CLAHE — a transformation present at both training and inference but randomized during training — Stage 6 is bypassed entirely at inference, so that the production transformation a deployed model applies to a new image is fully deterministic. The reason for this asymmetry is methodological hygiene: augmentation deliberately fabricates plausible variants of the training data, and admitting such fabricated variants into validation or test partitions would inflate apparent performance. Augmentation is therefore confined to the training path, and the cross-validation protocol (§3.4.2) further guarantees that no augmented image appears in any evaluation partition (RESEARCH_ARCHITECTURE §9.1). The stage composes three families of perturbation — geometric, photometric, and chromatic — applied in that order to the uint8 image before the final normalization of Stage 7 (Figure 3.8).

Geometric perturbation (unified affine). The geometric family is a single unified affine transform combining rotation, zoom in the range $[0.9, 1.1]$, and optional shear and stretch (OD-3 Stage 6). Its purpose is to present the network with the range of plausible viewpoint and framing variation that arises across fundus acquisitions without altering diagnostic content. The rotation component is distinctive: its dispersion is not a fixed constant but is adaptive per image, derived from the uncertainty of the optic-disc–fovea detection performed in Stage 1 (§3.1.1). This couples augmentation to the reliability of canonical orientation — an image whose orientation could be normalized confidently is rotated within a tighter band, while one for which Stage 1 fell back to its low-confidence path (fallback dispersion $\sigma = 13.0^\circ$) is allowed a wider rotational spread — so that the augmentation does not assert a precision of alignment that the detector did not achieve. Because the affine transform scales the two image axes together, it preserves the circular fundus geometry and the true aspect ratio of lesions, consistent with the isotropy established by Stage 2.

Photometric perturbation. The photometric family applies brightness and contrast jitter. Together with the stochastic application of Stage-5 CLAHE (§3.1.2), this exposes the network during training to a range of contrast and luminance profiles rather than to a single fixed one, with the intent of discouraging dependence on a particular illumination or contrast level — a robustness objective rather than a demonstrated outcome.

Chromatic perturbation (PCA colour jitter). The chromatic family is a principal-component colour jitter in the style introduced for ImageNet classification by Krizhevsky et al. [19], in which a random multiple of the image's colour principal components, scaled by their eigenvalues, is added to every pixel: the perturbation added to a pixel is $[\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3], [\alpha_1 \lambda_1, \alpha_2 \lambda_2, \alpha_3 \lambda_3]^{\text{top}}$, where $\mathbf{p}_i$ and λ_i are the eigenvectors and eigenvalues of the RGB pixel covariance and each α_i is drawn from $\mathcal{N}(0, \sigma)$ with $\sigma = 0.1$ (OD-3 Stage 6). The

method is borrowed from Krizhevsky et al. [19]; the colour basis, however, is not — it was estimated directly on the EyePACS training set as a real artifact of this work (RES-PCA). The estimated eigenvalues are markedly anisotropic — approximately 20.4, 128.1, and 2374.0 — so that a single dominant chromatic axis accounts for roughly eighteen times the colour variance of the next and over a hundred times that of the least. The practical consequence is that the colour jitter perturbs predominantly along the direction in which EyePACS fundus colour actually varies most, rather than along arbitrary RGB directions; it is a dataset-specific, principled colour augmentation rather than uniform channel noise (DGL-1 bounds the basis to EyePACS).

Dual role and class imbalance. Stage 6 serves two purposes simultaneously. As regularization, it enlarges the effective training distribution and reduces overfitting, the mechanism analyzed theoretically in §2.2.3. As a response to data imbalance, it is one of two design levers the framework uses against the severe class skew of the EyePACS grading distribution (SB-2.1): augmentation can be emphasized for under-represented severity grades to enlarge their effective sample, increasing the diversity of minority-class examples the network sees per epoch (SC-1.4). It is, however, only one lever. The principal mechanism for imbalance is the inverse-frequency-weighted focal loss specified in §3.3.4; augmentation complements it by acting on the input distribution rather than on the loss surface. Neither lever is claimed here to resolve the imbalance, and the marginal contribution of the augmentation level — like that of every other stage — is a question for the Experiment 2 ablation (§4.3), not a property asserted in advance (NC-17).

With Stage 6 specified, the train-time pipeline is complete; Stage 7 (specified in §3.1.1) finalizes the four-channel tensor, and the eight-stage specification is closed. The remaining preprocessing concern is the handling of clinical images that arrive outside the controlled conditions the eight stages assume — heterogeneous, sometimes degraded, sometimes mislabeled — which §3.1.4 addresses as the external-image ingestion protocol that precedes Stage 0 for such data.

§3.1.4 External Image Ingestion Protocol

The eight-stage pipeline specified in §3.1.1–§3.1.3 assumes a well-formed input at Stage 0: a single-eye colour fundus photograph in which the field of view is recoverable, the laterality is known, and the diabetic-retinopathy grade is a valid label in the range 0–4. Publicly distributed research datasets such as EyePACS largely satisfy this assumption by construction. Clinical images exported from a hospital information system do not always satisfy it. The following protocol, which conditions such images for entry into Stage 0, is a design contribution of this dissertation: it has no antecedent in the literature corpus assembled for this work, and the analysis below is therefore the dissertation's own rather than an attribution to any source. Its purpose is to make explicit the input contract of the model — to state which clinical images the model can accept and how out-of-contract images are handled — so that the clinical-validation experiments (Experiments 4, 5, and 7) operate on a defined population rather than on an ad hoc selection.

Specifying this contract is itself a methodological act under the integrated-pipeline (P2) view adopted in this chapter [UNSOURCED CLAIM: that the input contract is a methodological boundary rather than mere engineering plumbing is the dissertation's own analytical position, not a sourced claim]. Because preprocessing is part of the model, the rule that decides which clinical images enter preprocessing is part of the model's boundary, not an external convenience. The protocol comprises four components, each defined by the downstream precondition it protects.

Format and container normalization. Clinical exports arrive in heterogeneous file formats, bit depths, and colour encodings, sometimes with embedded viewer annotations or burned-in identifiers. The first component normalizes every input to the canonical representation the pipeline expects — an 8-bit RGB raster — converting colour spaces and bit depths and rejecting containers that cannot be decoded to a single fundus raster. This establishes the uint8 RGB input contract on which Stage 0 and all subsequent stages depend.

Field-of-view and quality gating. Stage 2 of the pipeline recovers the field of view by foreground detection and crops to it; this presupposes that a coherent illuminated fundus disc is present. The second component gates on that precondition: images in which a field of view cannot be recovered — because the frame is blank, grossly over- or under-exposed, or dominated by artifact — are flagged and withheld rather than passed silently into a pipeline whose geometric and illumination stages would then operate on meaningless content. This gating is operationally a quality filter, consistent with the operational definition of image quality as the capacity to support downstream feature detection (OD-1), but it is applied here as an admissibility check at ingestion rather than as a graded quality score.

Laterality and metadata reconciliation. Stage 0 (canonical flip) requires the eye laterality so that left-eye images can be reoriented to the right-eye canonical frame. The third component supplies and reconciles this signal: where the export metadata records laterality it is used directly, and where it is absent or inconsistent a heuristic determination is substituted, with the same conservative posture adopted for landmark detection in Stage 1 — a low-confidence determination is recorded as such rather than asserted. Metadata fields needed for patient-level cross-validation grouping (§3.4.2) are reconciled at the same point, since patient identity must be preserved to prevent leakage even when image filenames are inconsistent.

Label and grade sanity. The fourth component validates the supplied label against the five-class DR taxonomy (DR 0–4): grades outside the range, or images with missing grades, are flagged for adjudication rather than coerced into a class. This protects the integrity of the evaluation partitions, whose metrics presuppose valid reference labels.

The validity of this protocol is bounded. It was designed against, and is validated only for, the specific Kazakh medical-center clinical data used in this work (60 images, RESEARCH_ARCHITECTURE §2.1.8); generalization of the ingestion protocol to clinical data from other sources, with different export conventions and degradation profiles, would require independent validation and is not claimed here (NC-15). Because that clinical data is held under institutional agreement and is not publicly available, the reproducibility of any result dependent on it is structurally

limited (SB-2.2). The protocol does not perform, and makes no claim about, regulatory-grade de-identification or compliance; the handling of patient-identifying information is treated as a system-design concern in §6.4.1, where it is presented as a design specification rather than a certified compliance status (SB-4.2). The protocol is, in sum, a defined and bounded input contract for the model — not a validated general-purpose clinical-image cleaner (CFC-2.4).

With the ingestion protocol specified, the formalization of the preprocessing pipeline (§3.1) is complete: from the entry conditions for clinical data, through the eight ordered stages, to the four-channel tensor delivered to the network. Section 3.2 turns to the second component of the two-stage model — the convolutional architectures that consume that tensor.

§3.2.1 ResNet-50 and EfficientNet-B3 as Primary Experimental Architectures

Section 3.1 specified the preprocessing component of the two-stage model and the four-channel tensor it delivers (or the three-channel tensor of the baseline condition). This section specifies the second component — the convolutional network that consumes that tensor — by fixing the two primary backbone architectures used throughout the experimental design. The two are chosen from distinct architecture families, and the rationale for using two rather than one is methodological rather than a search for the single best network: it is what allows any effect observed under the preprocessing contrast of Experiment 1 to be tested for replication across architecture families (Figure 3.14). The relationship of the two backbones to the input tensor is shown in Figure 3.11.

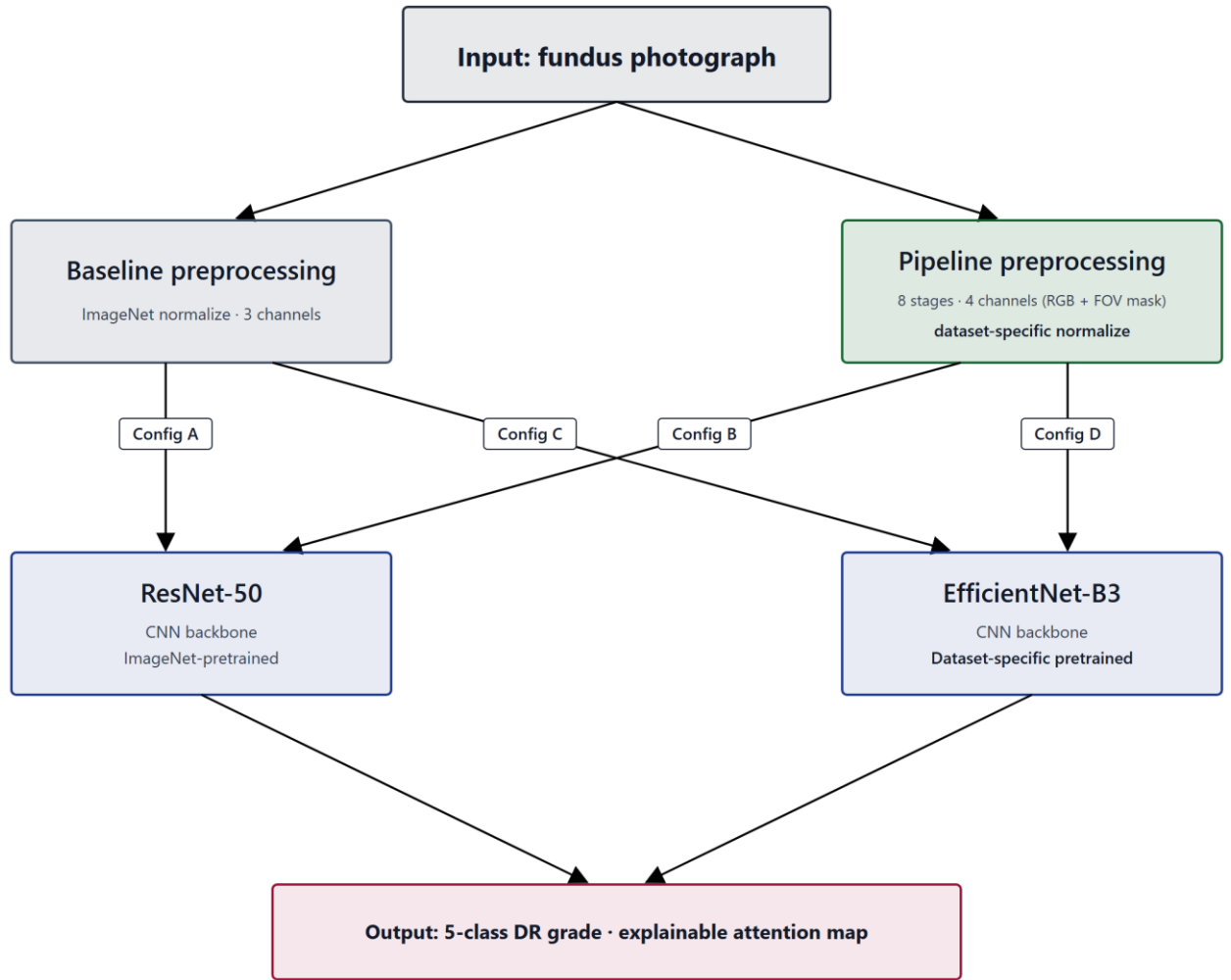


Figure 3.11 – ResNet-50 / EfficientNet-B3 backbone architecture

ResNet-50. The first backbone is ResNet-50, the 50-layer instance of the residual-learning architecture introduced by He et al. [23]. Its defining element is the residual connection — an identity shortcut that lets each block learn a residual function relative to its input rather than an unreferenced mapping — which makes very deep networks trainable by easing gradient propagation. ResNet-50 represents the residual-connection family of convolutional networks in this dissertation's design.

EfficientNet-B3. The second backbone is EfficientNet-B3, a member of the EfficientNet family introduced by Tan and Le [25], whose defining element is compound scaling: depth, width, and input resolution are scaled together by a single compound coefficient rather than independently. EfficientNet-B3 represents the compound-scaling family. The two backbones therefore embody two genuinely different design principles for building a deep convolutional feature extractor — residual depth on one hand, balanced compound scaling on the other — which is the property that makes them a useful pair.

Why two architecture families. The experimental contrast of Experiment 1 manipulates the preprocessing condition (baseline versus full pipeline) within each backbone: ResNet-50 appears in both the baseline configuration (A) and the integrated configuration (B), and EfficientNet-B3 appears in both the baseline configuration (C) and the integrated configuration (D) (RESEARCH_ARCHITECTURE §5.1). Reading the preprocessing contrast within

each fixed backbone, and then asking whether the two backbones agree, is exactly the cross-architecture replication required by the sufficient-validation criterion (EH-4): an effect confirmed for only one architecture would not meet it. A single backbone could not provide this evidence; the pair is the minimum design that can. This is a design rationale for the two-architecture structure, not a claim about either architecture's standing — neither ResNet-50 nor EfficientNet-B3 is asserted to be the globally optimal network for diabetic-retinopathy classification, and no exhaustive search over the architecture space is performed or claimed (SB-3.1, NC-6).

Why convolutional, and not a transformer. The wider landscape includes vision transformers [39], later EfficientNet variants [26], and CNN–transformer hybrids such as the EfficientNet–Swin combination evaluated on APTOS 2019 by Xu et al. [28] — cited here only to situate the choice, with no head-to-head comparison implied (SIR-3). The decision to keep both backbones convolutional is deliberate and tied to the causal logic of the study; it is also well-precedented — the landmark automated-DR systems were themselves built on convolutional backbones such as Inception- [22] — cited here only to situate the choice, not as a comparator (SIR-3). Because the integrated arm of Experiment 1 is initialized from ophthalmology-specific self-supervised pretraining rather than from ImageNet (§3.3.2), keeping the architecture identical across the baseline and integrated arms ensures that the preprocessing-plus-pretraining contrast is not confounded by an architecture change. A transformer backbone would have changed both the architecture and the initialization at once; a CNN-native self-supervised initialization changes only the initialization, preserving the convolutional design across arms (INVARIANTS v6.0.0 §X). The CNN-versus-transformer question is therefore outside the scope of this dissertation's experimental claims.

Input-channel handling. The two backbones must accept inputs of different channel count across the experimental conditions: three channels in the baseline condition (RGB) and four in the full condition (RGB plus the FOV mask of Stage 3). The first convolutional layer is adapted accordingly. For weights inherited from a three-channel pretraining source, the pretrained weights are copied to the first three input channels and the fourth (mask) channel is initialized from their per-channel mean (the protocol in `experiments/src/models/resnet.py`); because the integrated arm's self-supervised pretraining is performed in-house, however, the encoder can alternatively be pretrained directly on the four-channel tensor, which removes the channel-count mismatch entirely (AOQ-2 simplified, INVARIANTS v6.0.0 §X; RESEARCH_ARCHITECTURE §4.2bis). Either way the FOV-mask stage (Stage 3) is preserved, and the four-channel input contract of OD-3 is maintained.

Training configuration and hardware. Both backbones are trained under the single standardized configuration specified in TAB-3.1 and detailed in §3.4.2 and §4.1.3 (Adam, batch size 16, 512×512 input, focal loss, 5-fold patient-level cross-validation, fixed seed). Both backbones rely intrinsically on batch normalization [77], whose stabilizing and mildly regularizing effect derives from batch statistics and therefore weakens at the small batch size of sixteen imposed by the memory budget — a caveat carried from §2.2.3. One configuration detail is architecture-specific:

mixed-precision (fp16) training is enabled for ResNet-50 but disabled for EfficientNet, for which half-precision produced numerical overflow during training. This is a hardware- and implementation-bound setting, and like all efficiency-related conditions in this work it is bounded to the documented hardware (DGL-2); it does not generalize to other compute contexts without re-evaluation.

Two scope conditions close the specification. First, the two primary backbones supersede the smaller networks of the candidate's earlier work — including the four-layer convolutional classifier of previously published results (Sapakova, Yesmukhamedov & Sapakov, 2025), which the present design extends by moving to established, deeper backbones under a controlled factorial. Second, the EfficientNetB0 results reported in the candidate's transfer-learning self-publications are specific to that single network and are not generalized here to the class of efficient CNN architectures, of which EfficientNet-B3 is a distinct, larger member (SIR-7). The manner in which these backbones are initialized — ImageNet for the baseline arm, ophthalmology-specific self-supervised pretraining for the integrated arm, with the degree of transfer treated as an empirical question rather than a guarantee (DGL-6) — is the subject of §3.3. Before that, §3.2.2 documents the historical reference architectures that bound the architectural-complexity definition (OD-2).

§3.2.2 Historical Reference Architectures (Reference Only)

The two primary backbones specified in §3.2.1 are not the only convolutional architectures relevant to this dissertation's design; they sit on a complexity scale whose endpoints are defined by reference to two simpler historical architectures, and they succeed a line of the candidate's earlier networks. This section documents those reference architectures for two purposes — to ground the operational definition of architectural complexity (OD-2) and to record the prior empirical lineage — while making explicit that they are reference points only, not the experimental backbones of the present study.

The operational definition of architectural complexity is anchored at two reference points (OD-2). The **low-complexity reference** is a compact convolutional classifier comprising two convolutional blocks with 32 to 64 filters, no batch normalization, no dropout, and a sigmoid output. The **high-complexity reference** is a deeper classifier comprising four convolutional blocks with 32 to 256 filters, batch normalization, a dropout rate of 0.4, and a softmax output over the five DR classes; this corresponds to the four-layer convolutional network reported in the candidate's previously published preprocessing study (Sapakova, Yesmukhamedov & Sapakov, 2025). Together these two architectures fix the scale along which the complexity of any network in the study is read; architectures lying outside these bounds are not evaluated within this dissertation (OD-2). The two reference points also illustrate concretely the regularization components whose theory was developed in §2.2.3 — batch normalization and dropout are present in the high-complexity reference and absent in the low-complexity one — and so connect the complexity definition to the regularization discussion rather than leaving it abstract.

The prior empirical lineage is the candidate's transfer-learning work on EfficientNetB0. Those previously published results evaluated a frozen-feature configuration against a progressively fine-tuned one on EfficientNetB0 and reported the fine-tuned configuration to outperform the frozen one (the two self-publications reporting this experiment describe the same experimental thread and are treated here as a single citation, not as independent confirmation, per SIR-4 and SIR-5). Two constraints govern how this record enters the present chapter. First, those results were obtained with EfficientNetB0 specifically and are not generalized to the class of efficient CNN architectures; EfficientNet-B3, the §3.2.1 backbone, is a distinct and larger member of the EfficientNet family, and no transfer of the B0 findings to it is asserted (SIR-7). Second, the fine-tuning behaviour observed in that record motivates the two-stage fine-tuning protocol specified in §3.3.3, but it is cited as prior own work to be extended, not as a settled property of the present backbones (SIR-4).

These architectures are documented as references, not adopted as experimental backbones. The networks that carry the experiments of this dissertation are the ResNet-50 and EfficientNet-B3 pair of §3.2.1; the low- and high-complexity references serve to bound the complexity definition, and the EfficientNetB0 networks serve as the prior empirical record the study builds on. No claim of optimality is made for any of these architectures, and no exhaustive comparison across the architecture space is undertaken (SB-3.1, NC-6). With the architectural references documented, §3.3 turns to how the primary backbones are adapted for five-class classification, initialized by pretraining, and fine-tuned.

§3.3.1 Architecture Adaptation for Five-Class DR Classification

The two primary backbones specified in §3.2 are pretrained networks whose final classification layer is sized to their pretraining task, not to diabetic-retinopathy grading. Adapting them to the dissertation's task is the first step of §3.3, and it is deliberately separated here from two distinct decisions treated in the following sections: the source of the pretrained weights (§3.3.2) and the schedule by which the adapted network is fine-tuned (§3.3.3). The present section concerns only the structural adaptation — how the network's output is reshaped for the five-class problem.

The adaptation replaces the pretraining-task classification head of each backbone with a new head producing a five-way softmax distribution over the diabetic-retinopathy grades DR 0 through DR 4 (the five-class taxonomy of IT-1), while retaining the convolutional feature extractor that the pretraining produced. The softmax formulation and its associated loss are those developed in §2.2.1–§2.2.2; what is fixed here is the output cardinality and structure — five mutually exclusive ordinal grades — rather than the binary referable/non-referable formulation used in some of the screening literature. This five-class softmax structure follows the high-complexity reference of OD-2 and the candidate's prior five-class network (Sapakova, Yesmukhamedov & Sapakov, 2025), cited as prior own work and as a single thread with its companion card (SIR-4, SIR-5).

The methodologically important property of the adaptation is that it is identical across every configuration of the experimental design. The same five-way softmax head, replacing the same pretraining head, is applied to both ResNet-50 and EfficientNet-B3, and within each backbone to both the baseline arm and the integrated arm. Only the two axes under study — the preprocessing condition and, slaved to it, the pretraining source — vary across the four configurations; the output structure does not. This is what keeps the Experiment 1 factorial a clean contrast: because the head is held constant, any performance difference observed across configurations cannot be attributed to a difference in output structure, and the contrast is referable to the preprocessing-and-pretraining axes it is designed to test. Holding the adaptation identical across the two architectures is also what makes cross-architecture replication meaningful: the sufficient-validation criterion (EH-4) asks whether the same direction of effect appears for both backbones, and that question is well posed only if the two backbones are adapted to the task in the same way.

The candidate's prior transfer-learning work on EfficientNetB0 — which compared a frozen-feature configuration against a progressively fine-tuned one (reported across two self-publications describing a single experimental thread, SIR-4/SIR-5) — informs the fine-tuning schedule specified in §3.3.3 rather than the head adaptation specified here, and its EfficientNetB0 findings are not generalized to the present backbones or to the class of efficient CNN architectures (SIR-7). The adaptation is, finally, the structural mechanism by which transfer from the pretraining domain to the DR task is attempted; whether that transfer in fact yields useful representations is not guaranteed by the adaptation and is evaluated empirically within the experimental framework, not assumed (DGL-6). With the output structure fixed in common across all configurations, §3.3.2 specifies the initialization that distinguishes the integrated arm — ophthalmology-specific self-supervised pretraining of the same backbone — and §3.3.3 the fine-tuning schedule applied after adaptation.

§3.3.2 Ophthalmology-Specific Self-Supervised Pretraining of the CNN Backbone (integrated Arm)

Section 3.3.1 fixed the structural adaptation — the five-class softmax head — in common across all four configurations of Experiment 1. What distinguishes the configurations is not the head but the *initialization* of the backbone before fine-tuning, and this section specifies that initialization for the integrated arm. In the baseline arm (configurations A and C) the backbone is initialized from ImageNet weights, the standard cross-domain transfer from natural images. In the integrated arm (configurations B and D) the backbone is instead initialized from ophthalmology-specific self-supervised pretraining on an unlabeled retinal fundus corpus, performed on the same ResNet-50 or EfficientNet-B3 architecture used in the baseline arm (RESEARCH_ARCHITECTURE §4.2bis; DGL-6).

The pretraining protocol. The integrated-arm initialization is produced by a CNN-compatible domain-adaptive self-supervised learning protocol drawn from the DINO [92], BYOL [90], SimCLR [88], and MoCo [89] family of methods, with the

specific protocol selected empirically and recorded once chosen. No diabetic-retinopathy labels are used at any point during this pretraining; the objective is purely representational — to learn, from unlabeled fundus images, features that capture the structure of retinal imagery: vascular topology, optic-disc and macular morphology, retinal texture, illumination variability, and imaging artifacts (RESEARCH_ARCHITECTURE §4.2bis). The resulting weights then initialize the same adapted network that the baseline arm initializes from ImageNet, after which both arms are fine-tuned identically on the labeled DR task (§3.3.3). Because the self-supervised pretraining is performed in-house rather than loaded from an external checkpoint, the encoder can be pretrained directly on the four-channel tensor (RGB plus FOV mask), which removes the input-channel mismatch that a three-channel external checkpoint would create; the copy-and-mean-initialization of the fourth channel described in §3.2.1 remains available as a fallback (AOQ-2 simplified, INVARIANTS v6.0.0 §X).

Why an in-domain, CNN-native initialization. The design rationale for this choice is to obtain an in-domain initialization without confounding the experiment with an architecture change. A retinal foundation model such as RETFound is published as a ViT-Large vision transformer; initializing the integrated arm from it would change both the architecture and the initialization relative to the baseline arm, so that any observed difference would conflate the preprocessing contribution with an architecture difference. A CNN-native self-supervised initialization changes only the initialization stage and preserves the convolutional backbone across both arms (the reasoning recorded in INVARIANTS v6.0.0 §X, which resolves the earlier open question on backbone architecture in favour of CNN-compatible SSL, restores the 2×2 architecture symmetry across arms, and renders the external foundation-model license question moot). The in-domain initialization is thus the dissertation's own methodological response to a confound, not an adoption of an existing foundation model.

The composite independent variable. The consequence of this design for the interpretation of Experiment 1 is binding and must be stated before any result is reported. The integrated arm differs from the baseline arm along two axes simultaneously — the preprocessing axis (the full pipeline versus stretch-resize with ImageNet normalization) and the pretraining axis (ophthalmology-specific self-supervised initialization versus ImageNet initialization). The independent variable of Experiment 1 is therefore the composite factor (*preprocessing* $\times$ *pretraining source*), and any performance difference between the arms reflects the joint contribution of both axes. It follows that the H-1 effect may not be attributed to the preprocessing pipeline alone, nor to the pretraining source alone: claims of the form "the preprocessing pipeline produces the observed improvement" or "preprocessing dominates" are not admissible (CFC-2.8). The only admissible form of the H-1 claim is at the level of the integrated configuration — that the integrated preprocessing-plus-ophthalmology-SSL configuration outperforms the baseline-plus-ImageNet configuration on a given metric by a given margin. Decomposition of the composite effect into separate preprocessing and pretraining contributions would require an

additional factorial that is explicitly outside the scope of this dissertation and is named as future work.

Literature status and honesty conditions. Two limitations of this specification are stated plainly. First, the self-supervised methods on which the integrated-arm initialization draws are now grounded in the assembled corpus. Beyond the general survey of self-supervised, multiple-instance, and transfer learning under limited labels in medical imaging [45], the corpus now contains the primary methods named above together with the masked-autoencoder objective [93] and SimSiam [91], a dedicated survey of self-supervised learning in medical imaging [87], and — most directly — evidence that in-domain self-supervised pretraining on retinal fundus images yields generalizable retinal representations [46] (RETFound) and that in-domain medical self-supervised pretraining can exceed ImageNet transfer [47] (MICLe). These sources establish the general credibility of the direction; what they do not establish is the specific configuration this dissertation adopts. RETFound's in-domain evidence is obtained with a ViT-Large backbone and MICLe's with non-fundus medical modalities, whereas the integrated arm holds the convolutional backbone fixed and pretrains it CNN-natively on the four-channel tensor — a configuration none of these sources evaluates [UNSOURCED CLAIM: the expectation that CNN-native in-domain self-supervised pretraining on the four-channel tensor yields a fundus-relevant initialization competitive with the ImageNet baseline is, at this stage, the dissertation's own methodological hypothesis; it is consistent with the in-domain SSL evidence of RETFound and MICLe but is not directly established by any retinal-SSL primary, which evaluate ViT or non-fundus modalities]. The in-domain self-supervised initialization is, accordingly, presented as a candidate methodological contribution whose efficacy is to be established empirically, not as a result inherited from the literature. Second, this section specifies a design, not an outcome: the ophthalmology-self-supervised arm (configurations B and D) is a specification to be executed, and no self-supervised pretraining results are reported here or claimed elsewhere in this dissertation in advance of that execution.

With the integrated-arm initialization specified and bounded, §3.3.3 specifies the two-stage fine-tuning schedule by which the adapted, initialized network — under either initialization — is trained on the labeled DR task, and §3.3.4 specifies the loss it is trained against.

§3.3.3 Two-Stage Fine-Tuning Protocol Design

Once the backbone has been adapted to the five-class task (§3.3.1) and initialized — from ImageNet in the baseline arm or from ophthalmology-specific self-supervised pretraining in the integrated arm (§3.3.2) — it is trained on the labeled DR task by a two-stage fine-tuning schedule. The schedule is the same for every configuration; it is the procedure by which the adapted, initialized network is optimized, distinct from the initialization that precedes it and from the loss it minimizes (§3.3.4).

The two stages address a tension inherent in fine-tuning a pretrained network with a freshly initialized head. In the first stage the backbone is frozen and only the

new five-class classification head is trained. With the pretrained feature extractor held fixed, the head learns to map established features to the DR grades without the large, poorly-conditioned gradients of an untrained head propagating back into and disrupting the pretrained weights. In the second stage the upper layers of the backbone are progressively unfrozen and trained jointly with the head, allowing the higher-level features — which are the most task-specific and therefore the most likely to benefit from adaptation — to be tuned to the characteristics of fundus imagery and the DR grading task, while the lower, more generic layers are adapted more conservatively or left fixed. This frozen-then-progressive ordering is the standard progressive fine-tuning strategy whose theory was set out in §2.3.2.

The schedule is grounded in the candidate's prior transfer-learning work. Those previously published results compared a frozen-feature configuration against a progressively fine-tuned one on EfficientNetB0 and reported the progressively fine-tuned configuration to be the stronger of the two — a weighted F1-score of approximately 0.74 fine-tuned against approximately 0.62 frozen, and a precision of roughly 75% against 65% (reported across two self-publications that describe a single experimental thread and are cited here as one, per SIR-4 and SIR-5). Two qualifications govern the use of this record. First, it was obtained with EfficientNetB0 and is not generalized to the present backbones or to the class of efficient CNN architectures (SIR-7); it motivates the choice of a progressive schedule rather than establishing its outcome for ResNet-50 or EfficientNet-B3. Second, the comparison of frozen against progressive fine-tuning is, in this dissertation, a training-method decision and not a hypothesis under test: the earlier hypothesis concerning fine-tuning strategy was dropped in an earlier governance revision, so no such hypothesis is evaluated or reinstated here, and the schedule is simply fixed as the method by which all configurations are trained.

Fixing the schedule uniformly serves the same purpose as fixing the head uniformly (§3.3.1). The two-stage protocol is applied identically to ResNet-50 and EfficientNet-B3 and to both the baseline and the integrated arm, so that the training procedure is not a source of difference across the four configurations and the Experiment 1 contrast remains attributable to the preprocessing-and-pretraining axes under study, with cross-architecture agreement (EH-4) read under a common schedule. As with the adaptation, the schedule is the mechanism by which transfer is attempted; the degree to which fine-tuning yields effective task adaptation under either initialization is not guaranteed and is evaluated empirically (DGL-6). With the training schedule fixed, §3.3.4 specifies the loss function the schedule minimizes — a weighted formulation chosen for the imbalanced, ordinal structure of the DR grading task.

§3.3.4 Weighted Loss Function Formulation for Ordinal Class Structure

The two-stage schedule of §3.3.3 minimizes a loss function that must be chosen for the structure of the diabetic-retinopathy grading task: a five-class problem whose grades are ordinal and whose training distribution, on EyePACS, is severely

imbalanced (SB-2.1). Under an unweighted cross-entropy objective, the gradient signal would be dominated by the majority no-DR (DR 0) class, and the network could minimize the average loss while performing poorly on the rare severe grades that are clinically the most consequential. The loss is therefore specified to counter this directly, and it constitutes the principal of the two design levers the framework uses against class imbalance — the second being the Stage-6 augmentation of §3.1.3.

The training objective is the focal loss [75] with an inverse-frequency class weighting (RESEARCH_ARCHITECTURE §4.0):

$$\text{FL}(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t),$$

where p_t is the model's estimated probability for the true class of an example, γ is the focusing parameter (set to $\gamma = 2$), and α_t is a per-class weight. Two mechanisms act together in this formulation. The focal modulating factor $(1 - p_t)^\gamma$ down-weights examples that are already classified confidently and correctly (for which p_t is near 1, so the factor is near 0), redirecting the training emphasis toward hard and misclassified examples — the schematic of this re-weighting is shown in Figure 3.12. The class weight α_t is set to the inverse frequency of each class, so that rare grades contribute to the loss in proportion to their scarcity rather than being swamped by the common ones; these weights are derived from the EyePACS class distribution documented in §4.1.2. The focal modulation thus addresses the easy/hard imbalance of individual examples, while the inverse-frequency weighting addresses the between-class frequency imbalance, and the two are complementary.

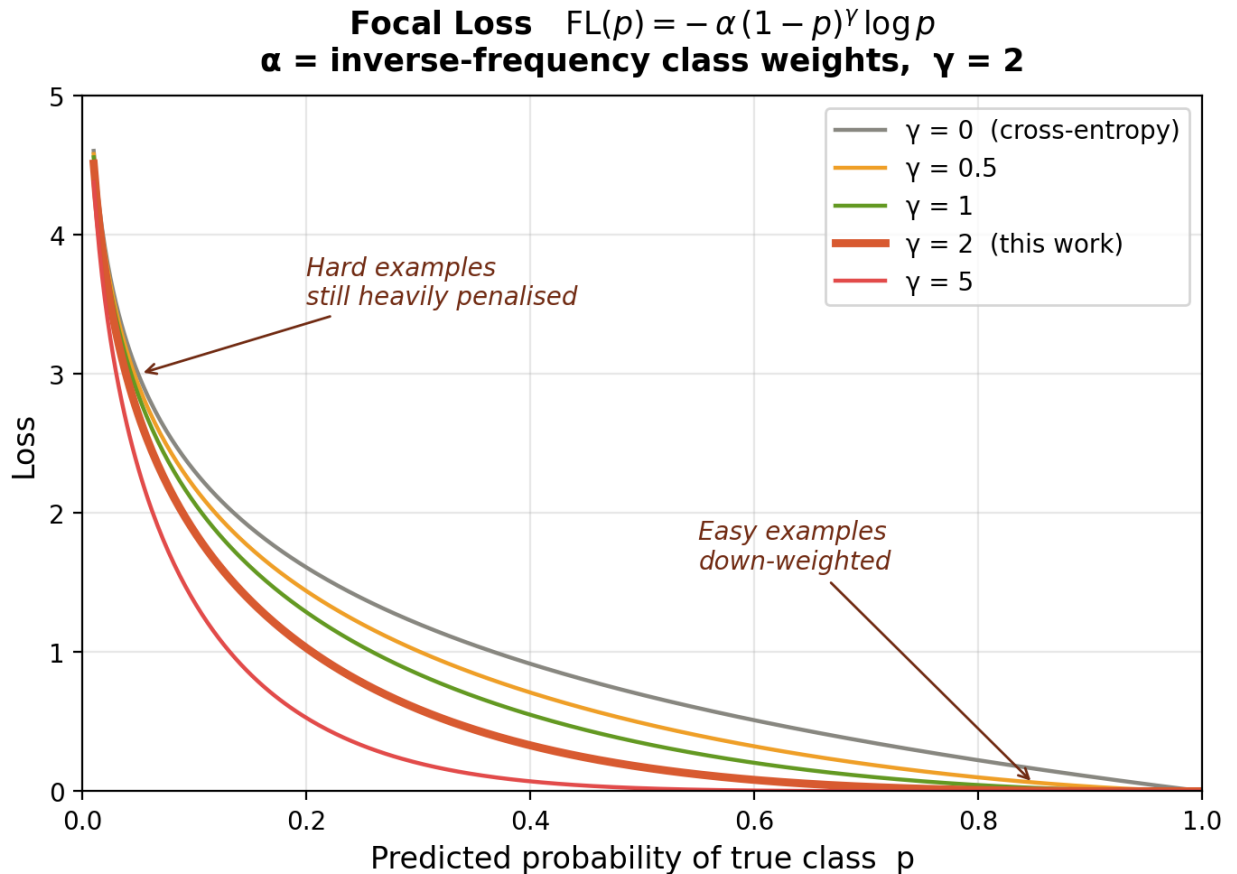


Figure 3.12 – focal-loss weighting schematic

This loss-side mechanism and the augmentation-side mechanism of §3.1.3 are the two levers by which the framework addresses imbalance (SC-1.4): the loss reshapes the objective surface, while augmentation reshapes the input distribution. Neither is asserted to resolve the imbalance, and the choice is made against a literature in which cost-sensitive learning, although an active direction, remains thinly validated — a systematic review of the field found that only two of one hundred seventy-three surveyed papers were validation studies [74], a survey-level observation that frames the design choice without supplying DR-specific evidence for it (SIR-3). The candidate's prior work on imbalanced DR classification (reported in the EfficientNetB0 transfer-learning thread, cited as prior own work and as one thread with its companion publication, SIR-4/SIR-5) similarly motivates rather than validates the present choice, and its EfficientNetB0 findings are not generalized to the present backbones (SIR-7).

The loss is coordinated with the evaluation framework rather than standing alone. Because the DR grades are ordinal, a misclassification by two grades is clinically worse than a misclassification by one, and this ordinal cost is captured at evaluation time by the quadratic-weighted Cohen's κ that the evidence hierarchy places among the primary metrics (EH-1). The weighted loss thus addresses the imbalance during training while the ordinal-sensitive metric penalizes ordinal error during evaluation; both are specified in §3.4.1. The loss is defined per its canonical primary source [75], now present in the assembled literature corpus (cited at first mention in §2.2.2), and per the dissertation's governance (RESEARCH_ARCHITECTURE §4.0); since Lin et al. introduced focal loss on the COCO object-detection benchmark rather than on retinal imagery, it is adopted here for its established loss-design rationale rather than as a DR-specific result (SIR-2). With the adaptation, initialization, fine-tuning schedule, and loss now all specified, §3.4 turns to the framework by which the trained models are evaluated.

§3.4.1 Multi-Metric Assessment Framework

The training procedure specified in §3.3 produces a trained model; this section specifies how that model is assessed. The governing principle is that diagnostic effectiveness is read from a hierarchy of metrics rather than from any single number, for a reason that follows directly from the data: the EyePACS distribution is severely imbalanced toward the no-DR class (SB-2.1), and under such imbalance overall accuracy is inflated by majority-class performance and is uninformative about the rare severe grades that matter most clinically — a distortion documented systematically for convolutional networks, in which class imbalance degrades performance and accuracy misleads [81]. The framework therefore ranks its metrics by evidentiary weight and requires the strongest evidence to rest on the metrics most robust to imbalance.

Primary metrics. The evidence hierarchy (EH-1) orders the four primary metrics in descending evidentiary weight: the weighted F1-score first, as it accounts for class imbalance and is directly interpretable under a skewed distribution; ROC-AUC second, as a threshold-independent measure of separability; the quadratic-

weighted Cohen's κ third, as it penalizes clinically significant ordinal misclassification — a two-grade error costs more than a one-grade error, the property that pairs this metric with the ordinal-aware focal loss of §3.3.4; and accuracy fourth, reported but never sufficient alone because of its inflation under imbalance. A configuration is deemed diagnostically effective only when it crosses the joint thresholds of the operational definition OD-5, summarized with the primary metrics in Table 3.2. These thresholds are not arbitrary: they were derived from the candidate's previously published empirical results (Sapakova et al.; weighted F1 = 0.91 and ROC-AUC = 0.9638 in one study, weighted average = 0.81 and accuracy = 0.80 in another), cited as prior own work and used here as reference anchors for an effectiveness floor, not as performance values transferable to the present experiments (SIR-4); the markedly higher figures reported elsewhere on small datasets are explicitly not transferable (SB-1.2). A qualification applies to every metric below: each is computed against the reference grades supplied with the dataset, and the reliability of those grades is itself an evaluation variable — grader variability can materially shift apparent performance, and adjudicated reference standards reduce but do not eliminate it [98]; the dissertation inherits the public datasets' provided labels and their reference-standard limitations (SIR-2).

TAB-3.2. Multi-metric evaluation framework and diagnostic-effectiveness thresholds (EH-1, EH-2, OD-5).

Tier	Metric	Evidentiary role	Effectiveness threshold (OD-5)
Primary (EH-1)	Weighted F1-score	Highest weight; accounts for class imbalance	≥ 0.80
Primary (EH-1)	ROC-AUC	Threshold-independent separability	≥ 0.90
Primary (EH-1)	Cohen's κ (quadratic weights)	Penalizes ordinal misclassification	≥ 0.70
Primary (EH-1)	Accuracy	Reported; inflated under imbalance, not sufficient alone	≥ 0.80
Secondary (EH-2)	Per-class precision / recall	Informative; unstable for minority classes	—
Secondary (EH-2)	Macro-average P/R/F1	Reported alongside weighted average	—
Secondary (EH-2)	Training-set metrics	Overfitting diagnosis only (>15 pp train–test gap, OD-4)	—
Clinical screening	Sensitivity / Specificity / PPV /	Reported for Exp 1, 3, 5	—

	NPV (referable DR, grade ≥ 2)		
Calibration	Expected Calibration Error (ECE), Brier score	Empirical reliability of probabilities, not clinical reliability (SB-1.10)	—
Explainability	ALO (primary), IoU (secondary)	Attention–lesion overlap; not clinical localization (NC-14)	—
Generalization	G = F1_external / F1_EyePACS (OD-4)	Cross-dataset transfer	(H-4: G ≥ 0.85)

Secondary metrics and overfitting. The secondary metrics (EH-2) are reported for completeness but cannot independently establish or refute a hypothesis. Per-class precision and recall are informative but unstable for the minority severe grades; macro averages are reported alongside the weighted averages to expose any divergence between them; and training-set metrics are used solely to diagnose overfitting, operationalized as a train-to-test gap exceeding fifteen percentage points on any primary metric (OD-4). Reporting these alongside the primary metrics guards against a configuration appearing effective on the weighted aggregate while failing on a minority grade.

Clinical-screening and calibration metrics. For the screening-oriented analyses (Experiments 1, 3, and 5), sensitivity, specificity, positive predictive value, and negative predictive value are reported for the referable-DR threshold (grade ≥ 2). Calibration is assessed by the Expected Calibration Error and the Brier score [97], which quantify whether the model’s predicted probabilities match observed frequencies. Calibration is reported as an empirical diagnostic property only; it does not establish the clinical reliability of the predicted probabilities, and no such claim is made (SB-1.10).

Image-quality metrics. To quantify the effect of preprocessing independently of downstream classification — closing the first link of the causal chain, preprocessing $\rightarrow$ improved feature visibility — four image-quality metrics are measured at pipeline stages, summarized in Table 3.3. These metrics evaluate the preprocessing stage on its own terms rather than only through the classifier, and one of them, the structural similarity index [96], serves a specific guard role: a high SSIM between the processed and original image confirms that the enhancement stages did not introduce destructive structural artifacts, which is the measurable form of the no-destructive-artifact design objective stated for the pipeline in §3.1.1.

TAB-3.3. Image-quality metrics for preprocessing evaluation (RESEARCH_ARCHITECTURE §3.3).

Metric	Measures	Expected effect of
--------	----------	--------------------

		preprocessing
Contrast-to-Noise Ratio (CNR)	Signal quality of vessel structures vs. background	Higher after flat-field correction and CLAHE
Vessel Visibility Index (VVI)	Detectability of retinal vasculature	Improved after flat-field correction and CLAHE
Image Entropy	Information content of the image	Increased after contrast enhancement
Structural Similarity (SSIM)	Preservation of structure relative to the original	High SSIM confirms no destructive artifacts introduced

The directional expectations in Table 3.3 are stated as the design intent of the corresponding stages, not as demonstrated outcomes; the measured values are reported for Experiment 2 (§4.3.3), and the contingency of contrast operations established in §3.1.2 applies to them as well.

Explainability metrics. Where preprocessing's effect on the model's attention is assessed (Experiment 4), two overlap metrics relate Grad-CAM activation regions to the pixel-level lesion masks of IDRiD. The primary metric is the Attention–Lesion Overlap, the fraction of lesion area covered by the attention region,

$$\text{ALO} = \frac{\text{area}(\text{GradCAM} \cap \text{lesion_mask})}{\text{area}(\text{lesion_mask})},$$

and the secondary metric is the Intersection-over-Union,

$$\text{IoU} = \frac{\text{area}(\text{GradCAM} \cap \text{lesion_mask})}{\text{area}(\text{GradCAM} \cup \text{lesion_mask})},$$

the symmetric overlap criterion borrowed from object-detection evaluation [94], used here as an overlap measure rather than as a detection benchmark. Both quantify the spatial agreement between model attention and annotated lesions; neither constitutes clinical localization of pathology, and Grad-CAM activation is an interpretability signal, not a diagnostic delineation of lesion boundaries (NC-14).

Generalization and the aggregation criteria. Cross-dataset transfer is measured by the generalization ratio $G = F1_{\text{external}} / F1_{\text{EyePACS}} \text{ (OD-4)}$, the ratio of external to primary-dataset weighted F1 under the same trained model without retraining; it is the quantity against which the pre-registered H-4 criterion ($G \geq 0.85$) is later tested. Finally, these metrics do not act in isolation at the level of the dissertation's central claim: the empirical-dominance criterion (EH-3) requires a simultaneous improvement of at least five percentage points in weighted F1, at least 0.02 in ROC-AUC, and no degradation in Cohen's κ , and sufficient validation (EH-4) further requires this to replicate across an external dataset and both architectures. These compound criteria are named here as the standard the metrics feed; their application to results belongs to §4.2. With the metrics defined, §3.4.2 specifies the cross-validation and statistical-reliability protocols by which they are computed with quantified uncertainty.

§3.4.2 Cross-Validation and Statistical Reliability Protocols

The metrics specified in §3.4.1 are estimates, and an estimate without a quantified uncertainty cannot support the compound dominance and validation criteria the dissertation relies on. This section specifies how the metrics are computed across data partitions, how their uncertainty is quantified, how configurations are compared statistically, and under what reproducibility conditions — the protocols that allow the EH-3 and EH-4 verdicts to rest on quantified evidence rather than on single point estimates.

Cross-validation and leakage control. All experiments use five-fold cross-validation with a patient-level stratified split: the data are partitioned into five folds, each fold serving once as the test partition while the remaining four train the model, and the process is repeated across all five folds (RESEARCH_ARCHITECTURE §2.2, §5.0). The split is shown in Figure 3.13. The defining property is that the partition is made at the level of the patient, not the image: no patient's images may appear in both the training and the test partition of any fold. This is not a cosmetic choice. Fundus datasets routinely contain multiple images of the same eye and both eyes of the same patient, which are strongly correlated; splitting at the image level would place correlated images of one patient on both sides of the partition and inflate the apparent test performance by leakage. Patient-level grouping removes that leakage path, and it composes with the train-only status of Stage-6 augmentation (§3.1.3) — no augmented image enters any test partition — so that the test estimate reflects generalization to unseen patients rather than memorization of seen ones. Every primary metric is reported as the mean $\pm$ standard deviation across the five folds, so that fold-to-fold variability is visible alongside the central estimate.

5-Fold Patient-Level Stratified Cross-Validation
(no patient appears in train and val of the same fold)

Legend: ■ Train ($\approx 80\%$) ■ Validation ($\approx 20\%$)

	Patient group 1	Patient group 2	Patient group 3	Patient group 4	Patient group 5
Fold 1	Val	Train	Train	Train	Train
Fold 2	Train	Val	Train	Train	Train
Fold 3	Train	Train	Val	Train	Train
Fold 4	Train	Train	Train	Val	Train
Fold 5	Train	Train	Train	Train	Val

Figure 3.13 – 5-fold patient-level cross-validation split

Statistical-test suite. Differences between configurations are assessed by a suite of tests, each matched to the comparison it serves (RESEARCH_ARCHITECTURE §6.8). The McNemar test is used for paired classification comparison in Experiment 1, where the same test cases are classified by competing configurations. The DeLong test is used to compare ROC-AUC values, in

Experiments 1 and 3. Ninety-five-percent confidence intervals are computed by bootstrap resampling with at least one thousand resamples for all experiments, which is the mechanism that attaches an uncertainty interval to every reported metric — and the basis on which a measured difference can be judged to exceed the EH-3 margins rather than to fall within noise. For Experiment 1 a mixed-effects model is fitted across folds with the fold treated as a random effect, separating the configuration effect of interest from fold-level variance. Because Experiments 1 and 2 each make many simultaneous comparisons across configurations and ablation levels, a Bonferroni/Holm multiple-comparison correction is applied to control the family-wise error rate, so that the number of comparisons does not manufacture spurious significance.

Relation to the dominance and validation criteria. These tests are the machinery behind the compound criteria restated in §3.4.1. The empirical-dominance criterion (EH-3) requires a simultaneous weighted-F1 improvement of at least five percentage points, a ROC-AUC improvement of at least 0.02, and no degradation in Cohen's κ ; the bootstrap intervals and paired tests are what determine whether an observed difference clears those margins reliably. The sufficient-validation criterion (EH-4) further requires the same direction of effect on at least one external dataset and replication across both architectures; the per-fold, per-architecture, per-dataset estimates produced by this protocol are precisely what that criterion is evaluated against. Both criteria are restated here to fix the role of the statistics; their application to results is reserved for Chapter 4.

Reproducibility configuration. All experiments are run under a single standardized training configuration, summarized in Table 3.1, with a fixed random seed and deterministic execution, fixed augmentation parameters, and a fixed learning-rate schedule (RESEARCH_ARCHITECTURE §4.0, §9.3). The optimizer is Adam [99], held constant with the other hyperparameters of Table 3.1 so that comparisons isolate the configuration effect rather than tuning differences. Fixing these removes run-to-run nondeterminism as a confound, so that a measured difference between configurations is attributable to the configuration rather than to seed variation.

TAB-3.1. Standardized training configuration (RESEARCH_ARCHITECTURE §4.0).

Parameter	Value
Optimizer	Adam (lr = 1×10^{-4} , weight_decay = 1×10^{-4})
Batch size	16
Maximum epochs	20 (early stopping, patience = 5)
Loss function	Focal loss ($\gamma = 2$, α = inverse-frequency class weights)
Input resolution	512×512
Input channels	3 (baseline) or 4 (full with FOV mask)
Mixed precision (fp16)	Enabled for ResNet-50; disabled for EfficientNet (fp16 overflow)

Cross-validation	5-fold, patient-level stratified split
Seed	42, deterministic = true

The reproducibility this configuration affords is bounded to the documented hardware: claims about computational efficiency or training behaviour are specific to the hardware on which the experiments are run and do not transfer to substantially different compute contexts without re-evaluation (DGL-2), and results dependent on the private clinical data carry the further reproducibility limit noted in §3.1.4 (SB-2.2). With the cross-validation, statistical, and reproducibility protocols specified, the methodological body of Chapter 3 is complete; §3.C consolidates the conditions under which the framework is reproducible.

§3.C Conclusions to Chapter 3

This chapter has specified the integrated preprocessing–CNN model and the framework by which it is evaluated, in sufficient detail for both to be reproduced and to serve as precisely defined objects in the experiments that follow. The specification was organized around the central commitment of this work — that the diagnostic model is a two-stage system, `model = preprocessing + CNN`, in which preprocessing is an integral model component rather than ancillary data preparation — and each of the chapter's four parts is the engineering realization of one face of that commitment.

The four parts compose a complete pipeline from raw image to evaluated prediction. Section 3.1 specified the preprocessing component: the eight ordered, deterministic stages of the pipeline — canonical flip, optic-disc–fovea rotation normalization, field-of-view crop and isotropic resize, field-of-view mask generation, adaptive flat-field correction, dual-constraint CLAHE, train-time augmentation, and dataset-specific normalization — together with the external-image ingestion protocol that defines the model's input contract for clinical data. Section 3.2 specified the network component: two established backbones from distinct architecture families, ResNet-50 and EfficientNet-B3, with the historical complexity references that bound the architectural-complexity definition. Section 3.3 specified how the network is prepared and trained: a common five-class classification head, an arm-specific initialization (ImageNet for the baseline arm, ophthalmology-specific self-supervised pretraining for the integrated arm), a two-stage fine-tuning schedule, and an inverse-frequency-weighted focal loss for the imbalanced ordinal task. Section 3.4 specified the measurement: a multi-metric assessment framework ordered by evidentiary weight, and a validation protocol built on patient-level five-fold cross-validation with a matched statistical-test suite and multiple-comparison correction.

Taken together, these specifications fix the conditions under which the framework is reproducible. The training configuration is standardized and the execution is deterministic under a fixed seed; the data partition is made at the patient level to preclude leakage; the preprocessing parameters are determined by validation within the dissertation's own framework rather than imported from external studies; and every reported metric is to carry a quantified uncertainty. These conditions are

what allow a measured difference between configurations to be attributed to the configuration rather than to nuisance variation.

The chapter has also been explicit about what its methodology deliberately bounds, and these commitments constrain how the results of Chapter 4 may be read. The independent variable of the primary experiment is the composite factor of preprocessing and pretraining source, so any observed effect is attributable only to the integrated configuration and never to preprocessing alone (CFC-2.8). The self-supervised initialization of the integrated arm is specified here but has not yet been trained; it is presented as a methodological design and a candidate contribution. The in-domain self-supervised methods it draws on are now represented in the corpus (acquired in the v6.1.0 expansion, and cited in §1.3.2 and §3.3.2), but the specific CNN-native integrated-arm configuration has not been trained and its efficacy remains to be established empirically rather than inherited from those sources. The reliability of the classical optic-disc-fovea detector was measured and reported honestly, including the finding that fovea localization is unreliable, and the pipeline's conservative fallbacks were motivated by that finding rather than concealing it. The CLAHE parameters are not assumed portable across datasets, transfer from any pretraining source is treated as an empirical question rather than a guarantee, no stage configuration is claimed to be universally optimal, and the dependence of some validation on private clinical data limits external reproducibility. These are not deficiencies disguised as caveats but the epistemic discipline that will make the eventual claims defensible. With the model and its evaluation framework so specified and so bounded, the dissertation proceeds to Chapter 4, which establishes the datasets and experimental configuration and then evaluates the hypotheses H-1 through H-7 against the dominance and sufficient-validation criteria defined here.

4 EXPERIMENTAL RESEARCH — PREPROCESSING IMPACT ON CNN DIAGNOSTIC PERFORMANCE

§4.1.1 Dataset Architecture

Chapter 3 specified the diagnostic model and its evaluation framework as an object of study — the eight-stage preprocessing pipeline with its input contract, two CNN backbones, the arm-specific initialization and fine-tuning protocol, and the multi-metric, patient-level cross-validated measurement system. That specification fixes *what* is to be evaluated and *how* the evaluation is to be read; it does not yet fix the data on which the evaluation is performed. The present chapter supplies that data and configuration before any hypothesis is tested. This first section specifies the dataset architecture, §4.1.2 the class distribution and partitioning of the training corpus together with the label harmonization the architecture requires, and §4.1.3 the hardware and reproducibility protocol. None of these three sections reports an experimental outcome; they establish the experimental substrate, and the empirical evaluation of H-1 through H-7 is reserved for §4.2 onward.

The substrate is deliberately not a single corpus. It is a *functionally tiered architecture* of eight datasets, in which membership of a tier is determined by the role a dataset plays in the argument rather than by its size. Four functional tiers are distinguished. The **TRAINING** tier holds a single dataset, EyePACS, which supplies all model fitting and the within-dataset evaluation of Experiments 1 and 2. The **EXTERNAL** tier holds the datasets used to probe behaviour on data the model was never trained on: APTOS 2019, for zero-shot cross-dataset transfer (Experiment 3), and Messidor-2, for the clinical-degradation comparison (Experiment 5). The **CLINICAL** tier holds the two datasets that carry clinically meaningful annotations beyond an image-level grade: IDRiD, whose pixel-level lesion masks make the quantitative explainability analysis of Experiment 4 possible, and a Kazakh medical-centre Clinical set used for qualitative explainability and as the held-out test corpus of Experiment 7. The **DEVICE** tier holds the three datasets — RFMiD, DDR, and ODIR-5K — across which the device domain-shift evaluation of Experiment 6 is performed; its purpose is to expose the model to camera hardware unrepresented in the training set, and the cross-device results it yields are empirical observations of performance variability, not a claim of device certification or regulatory compliance (NC-16). The complete architecture is set out in Table 4.1.

TAB-4.1. Tiered dataset architecture (RESEARCH_ARCHITECTURE §2.1). Sizes and camera models are dataset attributes as reported by the dataset descriptors; they are not experimental results.

Dataset	Tier	Primary role (experiments)	Approx. size	DR taxonomy	Camera model(s)	Public
EyePACS	TRAINING	Primary training + within-dataset evaluation (Exp 1, 2; source for 3–6)	~35,126 labeled images	5-class ICDR (0–4)	Canon CR-1	Yes (Kaggle)
APTOS 2019	EXTERNAL	Zero-shot cross-dataset transfer (Exp 3)	~3,662 images	5-class ICDR (0–4)	Mixed	Yes (Kaggle)
IDRiD	CLINICAL	Quantitative explainability + clinical degradation + small-data training	516 images (81 pixel-annotated)	5-class ICDR + pixel lesion masks	Kowa VX-10α	Yes

		(Exp 4, 5, 7)				
Messidor-2	EXTERNAL	Clinical degradation resistance (Exp 5)	~1,748 images	Referable/non-referable + grade	Topcon TRC-NW6	Yes (on registration)
RFMiD	DEVICE	Device domain shift (Exp 6)	~3,200 images	Multi-disease + DR subset (mapping documented)	Topcon, Kowa	Yes
DDR	DEVICE	Device domain shift (Exp 6)	~13,673 images	5-class DR	Canon, Topcon	Yes
ODIR-5K	DEVICE	Device domain shift (Exp 6)	~5,000 patients (bilateral)	Multi-disease + DR subset (mapping documented)	Canon, Zeiss	Yes
Clinical (Kazakhstan)	CLINICAL	Qualitative explainability + held-out test (Exp 4, 5, 7)	60 images (30 patients × 2 eyes), balanced	5-class ICDR (0–4)	Institutional	No (institutional agreement)

The EyePACS training corpus is documented by the telemedicine-system descriptor of Cuadros & Bresnick [100] [100], which establishes the ETDRS/ICDR-based five-class grading and the Canon nonmydriatic acquisition (CR-DGi / CR-1); the approximately 35,126 labeled fundus images relied upon here are an attribute of the Kaggle Diabetic Retinopathy Detection partition rather than a figure reported by Cuadros & Bresnick themselves, and are cited as such, not amplified. The Messidor-2 descriptor is likewise now carded [101], which distinguishes the original Messidor (1,200 images) from Messidor-2 (~1,748 images) used here. The remaining device-tier descriptors — RFMiD (#49), DDR (#50), and ODIR-5K (#51), the last recorded in the index as not yet authoritatively identified — are still indexed without dedicated literature cards; for each, only the index-level attributes reproduced in Table 4.1 are relied upon, and no further descriptor detail is asserted. The independent scale of the EyePACS partition is corroborated by Sharma et al. [32], who report a comparably sized EyePACS subset of 30,262 images; that source is cited here for dataset scale only, as its internally inconsistent class-level performance figures are not admitted as

evidence (source #41; LITERATURE_INDEX note 21). The IDRiD descriptor, by contrast, is fully carded: Porwal et al. [6] document 516 images, of which 81 carry pixel-level lesion masks for four lesion types, graded on the ICDR scale and acquired at a single Indian eye clinic with a single Kowa VX-10 α camera [6]. That single-centre, single-camera provenance, and the absence of any reported class-distribution statistic, are limitations inherited by every IDRiD-derived result and are carried forward to the explainability and degradation analyses that rely on this set (SIR-2).

The architecture is constructed so that its tiers span, by design, a heterogeneity that a single training set could not. That heterogeneity has two axes. The first is taxonomic: EyePACS, APTOS 2019, IDRiD, DDR, and the Clinical set use the five-class ICDR grading directly, Messidor-2 is distributed with a referable/non-referable distinction alongside a severity grade, and RFMiD and ODIR-5K are multi-disease datasets in which the diabetic-retinopathy labels form a subset whose mapping onto the five-class scale must be documented before use. The second is device-related: the corpus collectively spans Canon, Topcon, Kowa, and Zeiss fundus cameras, which is precisely the property that makes the DEVICE tier informative about acquisition-hardware shift. This twofold heterogeneity is a deliberate and bounded feature of the design rather than an uncontrolled confound, but it imposes a discipline that the dissertation observes throughout: any comparison drawn across datasets must carry with it the differences in imaging equipment, patient population, grading protocol, and disease taxonomy that distinguish them (SIR-3; SB-2.3). Where DeepDRiD is cited as a contemporary benchmark adopting the same ICDR taxonomy, the same discipline applies — that benchmark is single-centre and reports no confidence intervals, and those properties bound what it can be taken to establish [15] (SIR-2). The per-grade visual character of the datasets is shown in Figure 4.2, and the cross-dataset and cross-camera composition is summarized in Figure 4.3.

Representative fundus images per DR grade across the dataset architecture

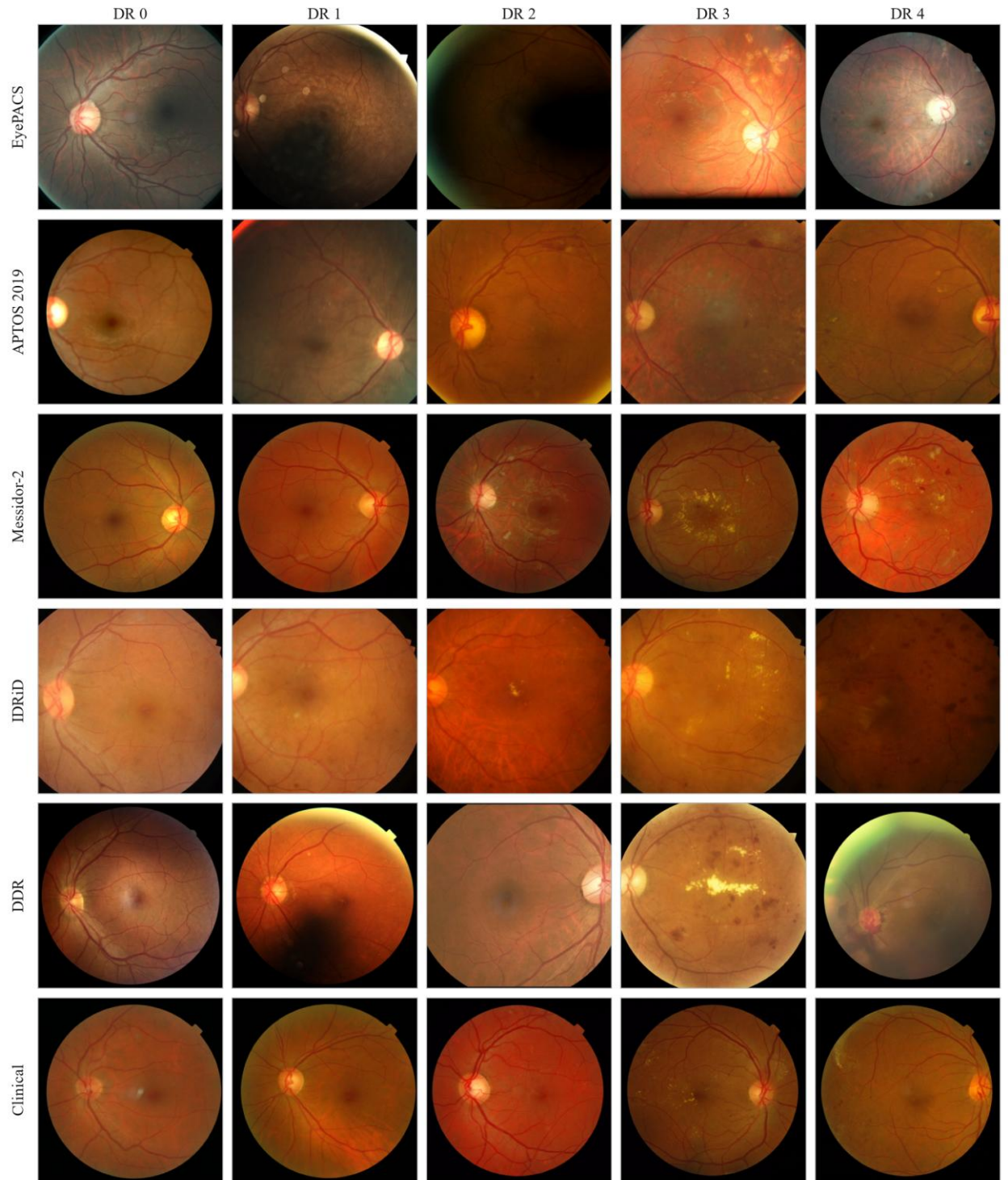


Figure 4.2 – representative fundus images per DR grade across the dataset architecture

Эксперименттер бойынша деректер жиындары · Datasets per experiment

	EyePACS APTOS 2019 n = ~35 126	IDRiD n = 3 662	Messidor-2 n = 516	DDR n = 1 748	ODIR-5K n = 13 673	ODIR-5K n = ~5 000	RFMiD n = ~3 200	Clinical (KZ) n = 60	
									Train (source) Eval (target / zero-shot)
Exp 1 H-1	Train								Преоброцессинг доминанттылығы
Exp 2 H-2	Train								Компоненттер абляциясы
Exp 3 H-4	Train	Eval							Кросс-дәтасет жалпылау
Exp 4 H-5	Train		Eval						Түсіндірмелілік · ALO
Exp 5 H-7	Train		Eval	Eval					Деградацияға төзімділік
Exp 6 H-6	Train				Eval	Eval	Eval		Камералық домен ауысуы
Exp 7 small-data			Train					Eval	Кіші деректермен оқыту

Figure 4.3 – cross-dataset and camera-alignment matrix

This tiered construction is what gives operational form to the dissertation's bounded-generalization commitment. Every performance claim the later experiments advance is bound to the dataset architecture defined here: to EyePACS as the training and primary-evaluation source, to APTOS 2019, IDRiD, and Messidor-2 as the external and clinical evaluation sources, and to RFMiD, DDR, and ODIR-5K as the device-shift sources. Extension to fundus datasets outside this architecture, to imaging devices unrepresented in these corpora, or to clinical populations not sampled by them is not claimed and would require independent experimental validation not undertaken here (DGL-1). Two boundary conditions on the substrate itself are registered at the outset. The EyePACS grade distribution is severely imbalanced, a property that conditions both the choice of primary metrics and the partitioning strategy and is treated in detail in §4.1.2 (SB-2.1). The Clinical set is held under an institutional data-use agreement and is not publicly available, so results dependent on it carry a structural reproducibility limit (SB-2.2).

The use of these data sources also stands in a definite relation to the candidate's prior work. In previously published studies, the candidate and co-authors evaluated transfer-learning behaviour on APTOS 2019 together with a private clinical set, using a single EfficientNetB0 backbone (a single experimental thread reported across `yesmukhamedov-conf.md` and `yesmukhamedov-kbtu.md`, and analysed further in `yesmukhamedov-scopus-q3.md`; SIR-4, SIR-5). Those studies acknowledged that dataset-specific training limited the generalization of their models and that the private data underpinning them was not publicly available (`yesmukhamedov-scopus-q3.md` §II.6; SIR-2). The dataset architecture specified in this section extends that prior work along exactly those two axes: it re-bases training on the large, public EyePACS corpus and embeds the external, clinical, and device-shift evaluation that the earlier single-dataset design could not support. With the substrate fixed and bounded, §4.1.2 turns to the class distribution of the training tier and the patient-level partitioning and label-harmonization protocol the architecture requires.

§4.1.2 Class Distribution and Data Partitioning

The dataset architecture fixed in §4.1.1 carried forward one boundary condition that bears directly on every design choice in the evaluation: the EyePACS grade distribution is severely imbalanced (SB-2.1). This section makes that imbalance precise enough to act on, traces the three downstream choices it determines — the primary-metric hierarchy, the loss function, and the partitioning strategy — and then specifies the label harmonization that the architecture's taxonomic heterogeneity requires before any dataset can be used for fitting or evaluation. The treatment remains at the level of experimental setup; no classification outcome is reported here.

The imbalance in EyePACS is structural rather than incidental. The five-class International Clinical Diabetic Retinopathy (ICDR) grading produces a distribution that is heavily concentrated in the no-apparent-retinopathy grade and falls away steeply across the more advanced grades, so that the severe and proliferative classes are represented by comparatively small numbers of images. The grade distribution is shown in Figure 4.1; the present text characterizes its shape rather than restating per-grade counts, which the figure reports. This shape is the typical profile of a population-screening corpus, in which most imaged eyes are unaffected or early-stage, and it is the property that the governance documents flag as requiring that all performance claims be interpreted in the context of distributional asymmetry (SB-2.1).

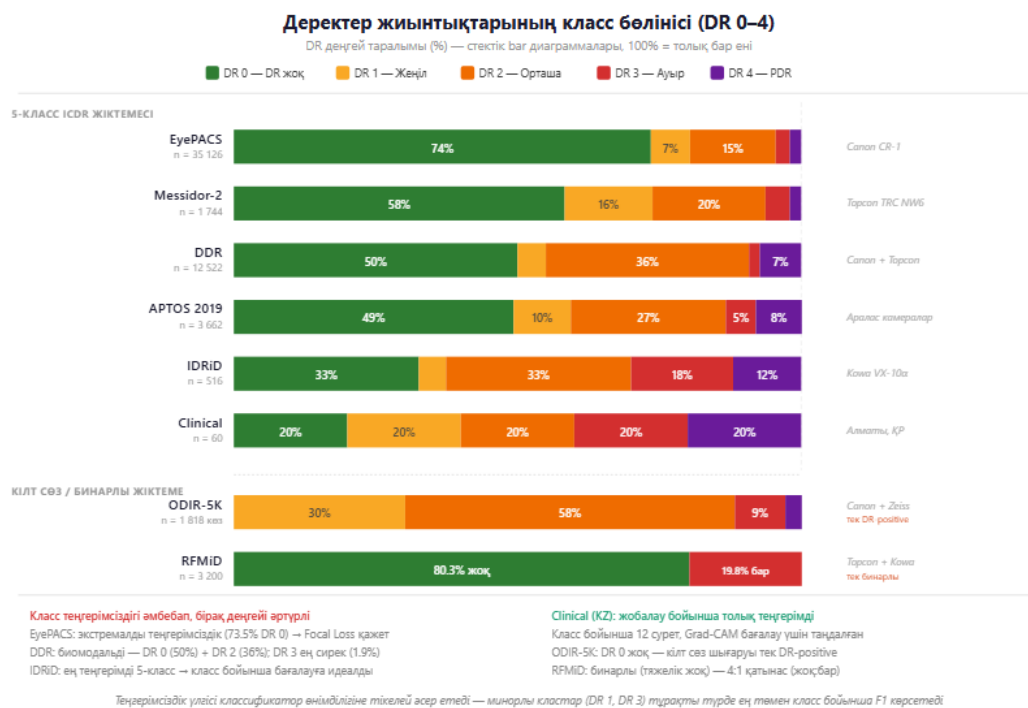


Figure 4.1 – EyePACS class-distribution across DR grades 0–4

That asymmetry is the principal lever behind the measurement and training choices already specified in Chapter 3, and naming the lever explicitly is the analytical purpose of this section (SC-1.4). It is why the evaluation does not lead with accuracy: on a distribution this skewed, a classifier can attain a high accuracy simply by favouring the majority grade, so accuracy is demoted to a supporting role and the primary-metric hierarchy is led by weighted F1-score and Cohen's κ with quadratic weights, with ROC-AUC as the threshold-independent measure (§3.4.1, EH-1; TAB-3.2). Weighted F1 is sensitive to per-class performance under skew, and quadratic-weighted κ penalizes the clinically consequential errors — those that misgrade across several severity steps — more heavily than adjacent-grade confusions. The same imbalance is why the loss is not unweighted cross-entropy but the inverse-frequency-weighted focal loss specified in §3.3.4, which raises the contribution of the rare severe and proliferative grades to the gradient, with the train-only augmentation of Stage 6 (§3.1.3) acting as a secondary lever that increases the effective variety of minority-class examples without altering the test distribution. These responses are restated here only to fix their common origin in the class distribution; their formal definitions are not reproduced.

The architecture's taxonomic heterogeneity, established in §4.1.1, imposes a second setup obligation: the grading schemes of the constituent datasets must be harmonized to a common target before they can be compared. The common target is the five-class ICDR scale on which EyePACS, APTOS 2019, IDRiD, DDR, and the Clinical set are already graded. Messidor-2, distributed with a referable/non-referable distinction alongside a severity grade, and the multi-disease datasets RFMiD and ODIR-5K, in which the diabetic-retinopathy labels form a subset of a broader ocular-disease taxonomy, require an explicit mapping onto that scale. Two conventions govern the mapping, and both are documented rather than assumed (SB-2.3; SIR-3). First, the multi-disease device-tier datasets are reduced to their diabetic-retinopathy content for Experiment 6, which uses DR labels only: non-DR disease labels are excluded or mapped to a non-DR category, consistent with the hypothesis under test (H-6). Second, where a referable/non-referable axis is used, the correspondence to the ICDR grades is stated so that any cross-dataset comparison carries its taxonomic context with it rather than presupposing label equivalence. This harmonization is the price of spanning the heterogeneity that makes the external and device tiers informative, and it is paid transparently.

Partitioning is governed by a single protocol applied uniformly across the architecture, specified in §3.4.2 and reused here without re-derivation. All experiments use five-fold cross-validation with a patient-level stratified split: the partition is made at the level of the patient, not the image, so that no patient's images appear in both the training and the test partition of any fold, which removes the leakage path that correlated multi-image and bilateral acquisitions would otherwise open (§3.4.2; (Figure 3.13)). Stratification by grade preserves the class proportions across folds, so that the rare severe grades are represented in every test partition rather than concentrated in one, and the protocol composes with the train-only status of Stage-6 augmentation, so that no augmented image enters any test partition. The

result is that every fold's test estimate reflects generalization to unseen patients under a distribution matched to the corpus as a whole.

One inherited limitation constrains the partitioning of the clinical tier. The IDRiD descriptor does not report a class-distribution statistic, and its release is a single-centre, single-camera acquisition [6] (SIR-2); its native 413/103 training–test division is the descriptor's own, whereas Experiment 7 re-partitions the 516 IDRiD images under the common patient-level five-fold protocol with the Clinical set held out as the test corpus. Because the IDRiD grade distribution is not published, stratification on that set proceeds from the labels as supplied without an external distributional reference, a constraint that is noted where IDRiD-derived results are later reported. With the training tier characterized, the metric and loss choices anchored to its distribution, the cross-dataset labels harmonized, and one leakage-controlled partitioning protocol fixed for all datasets, §4.1.3 specifies the hardware and the reproducibility protocol under which the experiments are run.

§4.1.3 Hardware and Reproducibility Protocol

With the datasets fixed (§4.1.1) and the class distribution, label harmonization, and partitioning protocol specified (§4.1.2), one element of the experimental setup remains before the experiments themselves: the execution environment under which models are fitted and evaluated, and the conditions that make the results reproducible. This section specifies the documented hardware, renders the standardized training configuration that governs every experiment, and states the reproducibility controls together with the bound — fixed by the hardware — within which the resulting computational statements hold.

The experiments are run on a single consumer-grade graphics processor, an NVIDIA RTX 3060 with 12 GB of video memory, within a WSL2 Ubuntu environment on Windows and a fixed conda environment that pins the software dependencies. This configuration is stated as the documented experimental setup rather than offered as a generalizable specification: per the hardware-specific reproducibility constraint, any statement about computational efficiency, training time, or inference throughput is bounded to this configuration and does not transfer to substantially different compute contexts — mobile inference on ARM processors at one extreme, or server-class GPU clusters at the other — without re-evaluation (DGL-2). The constraint is not merely a caveat appended after the fact; it is a design parameter that shapes the configuration. The training batch size of sixteen at a 512×512 input resolution is a direct consequence of the 12 GB memory bound, and the decision to enable mixed-precision (fp16) computation for ResNet-50 while disabling it for EfficientNet — where half-precision induced numerical overflow — is a hardware-driven engineering choice rather than a free hyperparameter. The single-device, consumer-GPU setting is also the operational instantiation of the resource-limited framing under which this work is positioned (OD-6); it does not, however, demonstrate real-time clinical inference, and no such capability is claimed here.

Every experiment is run under one standardized training configuration, reproduced in Table 4.2 (identical to the configuration introduced as Table 3.1 in §3.4.2). Fixing a single configuration across all experiments is what allows a measured difference between two configurations to be read as an effect of the factor under test rather than of an incidental change in optimizer, schedule, or batch size.

TAB-3.1 (rendered here as Table 4.2). Standardized training configuration (RESEARCH_ARCHITECTURE §4.0).

Parameter	Value
Optimizer	Adam ($\text{lr} = 1 \times 10^{-4}$, $\text{weight_decay} = 1 \times 10^{-4}$)
Batch size	16
Maximum epochs	20 (early stopping, $\text{patience} = 5$)
Loss function	Focal loss ($\gamma = 2$, $\alpha = \text{inverse-frequency class weights}$)
Input resolution	512×512
Input channels	3 (baseline) or 4 (full with FOV mask)
Mixed precision (fp16)	Enabled for ResNet-50; disabled for EfficientNet (fp16 overflow)
Cross-validation	5-fold, patient-level stratified split
Seed	42, deterministic = true

Reproducibility is treated as a property to be engineered rather than assumed, and each control is included for the specific source of variation it removes. Execution is deterministic under a single fixed random seed (42), which removes run-to-run nondeterminism — random initialization, data-ordering, and stochastic-operation variation — as a confound, so that a difference between two runs is attributable to their differing configuration rather than to seed variation. The augmentation parameters and the learning-rate schedule are likewise fixed across runs, removing two further degrees of freedom that could otherwise mask or manufacture an apparent effect (RESEARCH_ARCHITECTURE §9.3). The software stack is pinned — Python 3.11 with PyTorch, Torchvision, OpenCV, NumPy, and Scikit-learn (RESEARCH_ARCHITECTURE §8.5) — and the preprocessing pipeline and training code are versioned and reproduced in Appendix A, so that the transformation applied to every image and the procedure applied to every model are recoverable rather than described only in prose. This disciplined fixing of seed, schedule, augmentation, and code is the same reproducible-configuration practice established in the candidate's prior published work, which the present protocol formalizes and extends to the full multi-dataset architecture (`yesmukhamedov-kbtu.md`, ♦ prior own work; SIR-4).

Two limits bound the reproducibility this protocol affords, and both are stated rather than deferred. The first is the hardware bound already named: the configuration is reproducible on equivalent hardware, but the efficiency characteristics it exhibits are specific to the documented setup (DGL-2). The second concerns the data. Results that depend on the private Kazakh Clinical set carry a structural reproducibility limit,

because that set is held under an institutional data-use agreement and is not publicly available; an external party can reproduce the public-dataset experiments in full but cannot independently re-run the Clinical-dependent evaluations without the data (SB-2.2). With the hardware documented, the training configuration fixed, and the reproducibility conditions and their limits stated, the experimental setup of §4.1 is complete. The datasets, their distribution and partitioning, and the conditions of execution are now defined precisely enough to support the evaluation of H-1 through H-7, which the subsequent sections of this chapter undertake.

6 ARCHITECTURE OF AN AUTOMATED DR SCREENING SYSTEM FOR RESOURCE-LIMITED ENVIRONMENTS

§6.1.1 Functional and Non-Functional Requirements Specification

The preceding chapters specified the integrated preprocessing-CNN model and the framework against which its diagnostic effectiveness is to be measured: the eight-stage preprocessing pipeline and the two CNN backbones were fixed as a defined object of study, and the metric hierarchy, cross-validation scheme, and statistical tests were fixed as the apparatus that measures it. This chapter turns from the model as an object of measurement to the model as a component to be deployed. It asks what an automated diabetic-retinopathy screening system built around that model must *do*, and under what conditions it must do it. Before any requirement is stated, the chapter's epistemic status must be fixed and held throughout: the architecture presented in Chapter 6 is a **design specification only**. No prototype has been implemented, no field test has been conducted, and no projected deployment outcome is stated as a finding of this research (SB-4.1; DGL-4). Every requirement, component, and protocol that follows is a specification to be satisfied, not a property demonstrated to hold.

The requirements have two determinants. The first is the model itself, already specified. The second is the environment into which it must be placed, which fixes the conditions any realization must respect. The architecture is not introduced here *de novo*. In prior work by the candidate, Yesmukhamedov et al. (2025) proposed a modular, AI-driven information-system architecture for ophthalmological screening, decomposing it into components for image capture, image processing, recognition, diagnosis, reporting, user interaction, data storage, and error handling, and formalizing it through UML and entity-relationship modelling (Fig. 1, p. 83; Tables 2–3, pp. 82–83). That publication is prior own work and is cited as such; it presented the architecture as a design and reported neither a prototype implementation nor any quantitative validation of it (its acknowledged limitation, §II.6). The present section extends that prior design in two specific ways: it binds the requirements to the integrated model fixed in Chapter 3, and it anchors the non-functional envelope on the operational definition of a resource-limited environment that governs this dissertation (OD-6). The requirements set out below is therefore a formalization and tightening of a previously published architecture, not a new system claim.

Functionally, the system must realize the screening pipeline end to end, from a captured fundus image to a graded, physician-reviewable result. The functional requirements (FR), derived from the component decomposition of Yesmukhamedov et al. (2025) and mapped onto the Chapter-3 model, are summarized in Table 6.1.

Table 6.1 — Functional requirements (design specification; FR traced to originating model stage / #22 component).

ID	Functional requirement	Origin
FR-1	Ingest fundus images from heterogeneous capture sources, including portable and smartphone-based cameras	#22 Image Capture component (p. 83)
FR-2	Apply the eight-stage preprocessing pipeline as a configurable processing module prior to inference	Chapter 3 (OD-3); #22 Image Processing component
FR-3	Produce a five-class DR grade (DR 0–4) from a trained CNN backbone	Chapter 3 model; #22 Recognition/Diagnosis components
FR-4	Emit an interpretable decision-support output (grade plus a post-hoc Grad-CAM overlay) for clinician review	#22 Reporting component; NC-14 framing
FR-5	Present results to a clinician who interprets, audits, and may override them (physician-in-the-loop)	#22 Doctor-AI feedback loop (pp. 79, 85)
FR-6	Persist images, grades, and metadata and link them to patient records via PACS/EHR interfaces	#22 Data Storage; DMP/interoperability (Table 4, p. 86)
FR-7	Detect and handle malformed, low-quality, or out-of-contract inputs without silent failure	#22 Error Handling component

The functional set fixes *what* the system does; it does not by itself fix whether a given realization is admissible in the target setting. One requirement is worth isolating: FR-3 carries an acceptance target rather than a demonstrated property. The diagnostic-effectiveness profile defined operationally for this dissertation (OD-5) is the criterion the deployed classifier is *required* to meet; it is stated here as a design

target, and whether any trained model attains it is an empirical question reserved for the experimental chapters, not a result asserted in this design.

The non-functional requirements (NFR) are governed by the deployment context. A resource-limited environment is defined for this dissertation as a setting characterized by at least two of the following: absence of GPU acceleration for inference; available memory below 16 GB; throughput constraints requiring inference within a near-real-time clinical workflow; and network connectivity insufficient for continuous cloud reliance (OD-6). This definition, together with the modularity, interoperability, and security attributes of the prior architecture, yields the envelope in Table 6.2.

Table 6.2 — Non-functional requirements (design specification; NFR traced to governing constraint).

ID	Non-functional requirement	Governing constraint
NFR-1	Inference feasible without dedicated GPU acceleration	OD-6 (a)
NFR-2	Operation within a constrained memory budget (sub-16 GB host)	OD-6 (b)
NFR-3	Near-real-time response within clinical screening throughput	OD-6 (c)
NFR-4	Tolerance of intermittent connectivity, including store-and-forward operation	OD-6 (d)
NFR-5	Modular, independently replaceable components for adaptation across clinical contexts	#22 modular architecture (pp. 74–75, 90)
NFR-6	Interoperability with hospital systems via standardized exchange (FHIR/HL7)	#22 DMP/interoperability (Table 4, p. 86)
NFR-7	Data security and patient-data protection by design (encryption, authentication; GDPR/HIPAA-aligned)	#22 security (Table 3, p. 83); SB-4.2
NFR-8	Reliability and graceful degradation under faults	#22 Error Handling (p. 83)

The analytical weight of this specification rests not on the functional list, which a screening system of any kind would share, but on the non-functional envelope, which is what distinguishes a design for a resource-limited setting from a design for a well-provisioned clinic. The two requirement sets are not independent: the non-functional constraints prune the admissible realizations of the functional requirements. A cloud-only realization of inference (FR-3) is inadmissible under NFR-4, because intermittent connectivity cannot guarantee a round trip; a realization of FR-3 that presupposes server-class acceleration is inadmissible under NFR-1; and FR-2's preprocessing module, being computationally non-trivial, must be realizable within NFR-2 and NFR-3 rather than offloaded. The binding design problem of this chapter is therefore not the enumeration of functions but their realization inside the OD-6 envelope, and it is the envelope that the modular decomposition of the following section must be shown to respect.

Two boundaries close the specification. First, every entry in Tables 6.1 and 6.2 is a design requirement, not an attained property: the diagnostic target (OD-5) is a target, the security requirement (NFR-7) is a GDPR/HIPAA-*aligned* design specification and not a certified or tested compliance status (SB-4.2), and the prior publication's projected benefits for a Kazakhstan deployment — access for rural residents, reductions in late-stage complications and cost (Yesmukhamedov et al., 2025, p. 88) — are third-party projections cited for context only and are not attributed to this system as demonstrated outcomes (SIR-8). Second, the realization and validation of this requirements set are reserved for future work; the contribution of this chapter is the specification itself (CFC-1.5). With the functional and non-functional requirements fixed, the specification that the architecture must satisfy is now in place. The next section decomposes the system into modules and shows, at the level of a conceptual UML architecture, how those modules are arranged to satisfy this envelope, beginning with integration into PACS and EHR infrastructure.

§6.1.2 Modular Architecture with PACS and EHR Integration

The preceding section fixed the specification that the architecture must satisfy: a functional set (Table 6.1, FR-1...FR-7) describing what the screening system does, and a non-functional set (Table 6.2, NFR-1...NFR-8) describing the resource-limited envelope under which it must do it, with the analytical conclusion that the non-functional envelope — not the functional list, which any screening system shares — is the binding design driver because it prunes the admissible realizations of each function. This section discharges the obligation that conclusion creates: it decomposes the system into modules and shows that the chosen arrangement, including its integration with hospital Picture Archiving and Communication Systems (PACS) and Electronic Health Records (EHR), is an admissible realization of that envelope. The design-only status established for the chapter is carried unchanged: the decomposition below is a **design specification**, not an implemented or field-tested system — no module has been built and integrated, and no integration behaviour is reported as a finding (SB-4.1; DGL-4).

The module set is derived from the component decomposition of the candidate's prior published architecture. In prior work by the candidate, Yesmukhamedov et al. (2025) specified an AI-driven healthcare information system as a set of components — Image Capture, Image Processing, Recognition Model, Diagnosis, Reporting, User Interface, Data Storage, and Error Handling — connected through a UML component diagram and formalized through entity-relationship and data-management models (Fig. 1, p. 83; Tables 2–4, pp. 82–86). That publication is prior own work; it presented the architecture as a design and reported neither a prototype implementation nor any quantitative validation of it (its acknowledged limitation, §II.6). The present section does not merely restate that component list. Its extension is analytical: it binds the components to the integrated preprocessing-CNN model fixed in Chapter 3, and — the move absent from the prior publication — it constrains their *arrangement* by the OD-6 envelope, so that the topology, not only the membership, of the module set is determined by the deployment context. Table 6.3 records the resulting modules, the functional requirement each realizes, the non-functional constraints that govern its arrangement, and the section in which its internal realization is specified.

Table 6.3 — Module decomposition (design specification; each module traced to its originating FR and governing NFRs, with realization deferred as noted).

Module	Realizes (FR)	Governing NFR(s)	Internal realization
Ingestion	FR-1, FR-7	NFR-4, NFR-8	this section (boundary); §6.3.1 (capture/telemedicine)
Preprocessing Engine	FR-2	NFR-2, NFR-3, NFR-5	§6.2.1
Inference	FR-3	NFR-1, NFR-2, NFR-3	§6.2.2 (design-only)
Decision-Support / Reporting	FR-4	NFR-3, NFR-5	§6.2.2, §6.3.2
Clinician Interface	FR-5	NFR-5	§6.3.2
Data-Management / PACS-EHR Integration	FR-6	NFR-4, NFR-6, NFR-7	this section; §6.4.1 (security/compliance)
Orchestration / Error-Handling	FR-7	NFR-4, NFR-8	this section

The first design property the decomposition must secure is modularity itself (NFR-5). The modules are specified as loosely coupled: each communicates with its neighbours through a defined interface rather than through shared internal state, so that any single module can be replaced without re-architecting the others. This property is what licenses the system's intended adaptation across clinical contexts — a site with a different capture device replaces only the Ingestion module, a site with

stricter local data-residency rules replaces only the Data-Management module — and it is therefore the structural counterpart of the resource-limited-deployment goal rather than a generic software-engineering preference. Loose coupling is also the precondition for the deferrals recorded in Table 6.3: because the Inference module is bounded by its interface, the model-selection logic internal to it can be specified later (§6.2.2) without disturbing the surrounding topology, and the Preprocessing Engine's eight-stage internals can be specified in §6.2.1 as the realization of FR-2.

The analytically load-bearing module is the Data-Management / PACS-EHR integration layer, because it is where the non-functional envelope most sharply prunes the design. Functionally, this layer must persist images, grades, and metadata and link them to patient records through PACS and EHR interfaces (FR-6), and it must interoperate with hospital systems through standardized exchange (NFR-6). The prior architecture specifies the mechanism for the latter: a Data Management Platform (DMP) mediating storage, integration, and interoperability, with interoperability defined as data exchange through the FHIR and HL7 standards, and an entity-relationship model fixing the persisted entities — Patient, Medical Record, Doctor, Appointment, Telemedicine Session, AI Model, Diagnostic Result, and Device (Yesmukhamedov et al., 2025, Table 4, p. 86; Fig. 5, p. 88). The constraint the envelope imposes is on *how* this exchange is scheduled. Under NFR-4, connectivity is intermittent and store-and-forward operation is required; under NFR-1 and NFR-2 the inference path must run within a local, non-GPU, sub-16 GB host. Together these make any topology in which a grade can only be produced after a synchronous round trip to a hospital PACS or EHR inadmissible: such a topology would couple the inference critical path to a network resource the envelope does not guarantee. The admissible realization is therefore an *asynchronous* integration layer — inference and preprocessing complete locally, results and images are queued, and the FHIR/HL7 exchange with PACS/EHR is reconciled opportunistically when connectivity permits. This pruning of the topology is dissertation-original analysis; it is the specific sense in which the non-functional envelope, and not the component inventory, shapes the architecture. That AI-assisted HL7/FHIR interoperability is an actively studied and prototyped integration approach is corroborated externally by Nandal [102], who implemented an AI-enhanced HL7/FHIR exchange and evaluated it on the MIMIC-III database and synthetic HL7 messages in a sandbox setting; that study reports internal evaluation only, without external or clinical validation, so it is cited here as evidence that the integration approach is feasible and current, not as a measure of this system's behaviour.

The same integration boundary is where data security is concentrated (NFR-7). The persisted entities of the data-management layer are precisely the patient-identifying records, so the security mechanisms the prior architecture specifies — a security gateway for inter-component transfer, encryption, and user authentication (Yesmukhamedov et al., 2025, Table 3, p. 83; p. 90) — are positioned at this boundary by design. Consistent with the chapter's epistemic status, this is a security *requirement* expressed in the architecture, not an attained or certified property: the GDPR/HIPAA framing is a design specification and not a tested or certified

compliance status (SB-4.2; NC-9), and its detailed protocol-level treatment is deferred to §6.4.1.

The component-level view of this module set is given by DIA-6.1, the modular system-architecture diagram showing the capture → preprocessing → inference → reporting flow with parallel data-storage and error-handling components [DIA-6.1: Modular system architecture (capture→preprocess→inference→report) — defense/presentation/assets/architecture/06_system/02_system_architecture.png]. The formal UML treatment of the decomposition — component, sequence, class, activity, and ER diagrams elaborating the interfaces named above — is a deferred conceptual asset [DIA-6.3: System architecture UML — ASSET TO BE CREATED]; its absence does not affect the prose, which carries no data that the diagram would supply.

The decomposition is thus fixed as a design specification: a set of loosely-coupled, independently-replaceable modules whose arrangement — and in particular whose asynchronous PACS/EHR integration layer — is an admissible realization of the OD-6 envelope rather than an arbitrary partition of functions. As with the requirements of §6.1.1, the realization and validation of this design are reserved for future work; the contribution here is the requirement-driven decomposition itself (CFC-1.5). The following sections specify the internals the table defers: §6.2.1 details the Preprocessing Engine as the configurable realization of FR-2, §6.2.2 the Inference module and its model-selection logic, §6.3 the clinician interface and the telemedicine deployment of the Ingestion module, and §6.4.1 the security and data-protection protocols of the integration layer.

§6.2.1 Preprocessing Engine with Configurable Pipeline

The module decomposition of §6.1.2 fixed seven modules and deferred their internals; Table 6.3 assigned the realization of the preprocessing function (FR-2) to the Preprocessing Engine and remitted its specification to this section. That specification is given here as a design specification only: the engine described below is an architectural design, not an implemented or benchmarked component, and its diagnostic contribution is a target to be met rather than a property demonstrated to hold (SB-4.1; DGL-4).

The analytical core of the module is the eight-stage preprocessing pipeline specified in Chapter 3 (§3.1) — canonical flip, optic-disc/fovea rotation normalization, field-of-view crop and isotropic resize, FOV-mask generation, adaptive flat-field correction, dual-constraint CLAHE, augmentation, and dataset-specific normalization. That pipeline is incorporated here by reference and is not re-derived. A precise qualification governs how it is used in this chapter: the pipeline is the module core *as specified*, not as validated. Its diagnostic effectiveness is the acceptance target defined operationally for this dissertation (OD-5), and whether any configuration of the engine attains that target is an empirical question reserved for the experimental chapters; it is not asserted here as an established result. The Preprocessing Engine is therefore the architectural housing of a specified pipeline,

and the design contribution of this section is the housing — the way the pipeline is exposed as a module — rather than any claim about how well the pipeline performs.

This framing is the architectural expression of the central thesis that the model is the composition of preprocessing and a CNN backbone. In the candidate's prior published architecture, image processing was already identified as a distinct first-class component of the system rather than an implicit step folded into data handling (Yesmukhamedov et al., 2025, Image Processing component, Fig. 1 and Table 3, p. 83); that publication presented the component as a design and reported no prototype or quantitative validation of it (its acknowledged limitation, §II.6). The present section sharpens that component into a *configurable* engine and makes the configurability the load-bearing property, so that preprocessing enters the architecture as a parameterized module on the inference path, not as ancillary preparation upstream of it.

Configurability is specified along three axes. First, stage selection: each stage is an independently switchable element, so the engine can be operated at the full eight-stage configuration or at a reduced configuration in which a stage is bypassed. Second, parameterization: the stages that carry tunable parameters — the flat-field correction scale and the CLAHE clip-limit among them — expose those parameters at the module boundary rather than fixing them internally, with their specified values and formulations inherited from §3.1. Third, mode: the engine distinguishes train-time from inference-time behaviour, because two stages are mode-dependent by the Chapter-3 specification — the augmentation stage is applied only during training, and the CLAHE stage is applied stochastically during training but deterministically at inference. At inference, then, the engine resolves to a deterministic, augmentation-free transform from a captured fundus image to the four-channel tensor (RGB plus FOV mask) consumed by the inference module.

The distinction between train-time and inference-time behaviour also fixes the engine's interface contract. At inference the transform is deterministic and augmentation-free, so the same captured image maps to the same four-channel tensor on every invocation; this reproducibility is the precondition for the asynchronous, store-and-forward topology fixed in §6.1.2, in which the engine and the inference module complete locally and a result is queued for later reconciliation with hospital systems. The contract also assigns the engine a defined response to the malformed or out-of-contract inputs that the ingestion boundary admits (FR-7): an image that cannot be carried through the always-on stages — for instance one in which the field of view cannot be resolved — is rejected at the engine rather than silently producing a degraded tensor, keeping the failure explicit rather than propagating it to inference.

This configurability is what makes FR-2 admissible inside the OD-6 envelope, and it is the reason the engine is specified as configurable rather than fixed. The preprocessing computation is non-trivial, and the envelope fixed in §6.1.1 requires it to run within a constrained memory budget (NFR-2) and within a near-real-time clinical workflow (NFR-3) on a host that may lack GPU acceleration (NFR-1). Exposing stage selection and parameters at the module boundary lets a deployment budget the engine to its host — selecting the stage configuration and parameter settings that fit the available memory and latency envelope — rather than committing

every host to a single fixed cost. The same module boundary realizes NFR-5: because the engine communicates through a defined image-in, tensor-out interface, it is independently replaceable, so a site with different capture characteristics can re-tune or substitute the engine without disturbing the inference module downstream. It must be stressed that this is a statement about admissibility and adaptability, not optimality: the engine does not embed a preprocessing configuration claimed to be optimal, and the question of which configuration best serves a given setting is an empirical one bounded to the tested architectures and datasets (NC-17), not resolved by this design.

The component-level view of the engine — the staged image-in, tensor-out flow — is the pipeline-overview diagram reused from Chapter 3 [DIA-6.2: Preprocessing engine — configurable eight-stage pipeline — `defense/presentation/assets/preprocessing/10_input/04_preprocessing_pipeline_vertical.png`]. With the preprocessing function specified as a configurable module that produces the four-channel tensor under the OD-6 envelope, the next section specifies the module that consumes that tensor: the inference module and its model-selection logic.

§6.2.2 Inference Module with Model Selection Logic

The Preprocessing Engine of §6.2.1 resolves a captured fundus image into a deterministic four-channel tensor; this section specifies the module that consumes it. As with the rest of the chapter, the inference module is presented as a design specification: no trained model is deployed and no inference behaviour is reported as a finding here (SB-4.1). The module's effectiveness — whether a model it hosts attains the diagnostic-effectiveness target OD-5 — is an empirical question reserved for the experimental chapters and is not asserted in this design.

The module has two outputs, corresponding to two functional requirements. From the four-channel tensor it produces a five-class diabetic-retinopathy grade (DR 0–4) using a trained CNN backbone (FR-3), and alongside the grade it produces a post-hoc Grad-CAM overlay (FR-4) intended to make the basis of the grade legible to the reviewing clinician. The status of this overlay must be fixed precisely: it is an interpretability artifact computed after the grade, indicating the image regions to which the network's prediction was most sensitive, and it is not a clinical localization of pathology — it does not detect, delineate, or certify lesions (NC-14). For this reason the module's outputs are specified as inputs to a clinical decision, not as a decision: the system is a decision-support component operating with a physician in the loop, not a standalone diagnostic instrument (SB-1.3). The grade and overlay are the payload that the physician-in-the-loop interface of §6.3.2 presents for interpretation, audit, and possible override.

The distinctive element of the module is its model-selection logic, and the design intent of that logic must not be mistaken for a performance ranking. The dissertation fixes two CNN backbones in Chapter 3 — ResNet-50 and EfficientNet-B3 — which differ in their computational profile: the residual network is the higher-capacity option, while the EfficientNet backbone is the more parameter-efficient. The

selection logic chooses between them on the basis of the deployment host's resource profile — its available memory, its compute, and the latency its clinical workflow permits — so that the inference function fits the non-functional envelope of §6.1.1 (NFR-1's possible absence of GPU acceleration, NFR-2's sub-16 GB budget, NFR-3's near-real-time requirement). The criterion is therefore resource fit, conditioned on the specific hardware of the deployment (DGL-2); it is explicitly not a claim that one backbone is more accurate than the other. No performance ordering of the two backbones is asserted in this design, and none could be, because the comparison is the object of the experimental chapters and no result is imported here.

Two boundaries keep this logic within its design-only register. First, selecting a backbone by its resource profile is not selecting a *validated* or *optimal* model: the dissertation makes no claim that either backbone is the optimal architecture for the task (NC-6), and results obtained for one architecture are not generalized to the broader class of efficient CNNs (SIR-7). Second, the two backbones are design options whose diagnostic adequacy remains the OD-5 acceptance target; the selection logic decides which option a given host can run, not which option performs better. The architecture is, in this sense, deliberately agnostic to the eventual experimental verdict — it is designed to accommodate whichever backbone the resource envelope admits, and it would extend to a further backbone on the same resource axis without alteration of the selection principle.

So specified, the inference module is the resource-side counterpart of the configurable Preprocessing Engine: where the engine budgets the preprocessing cost to the host, the selection logic budgets the model cost to the host, and together they keep the entire grade-producing path local — the precondition for the asynchronous, store-and-forward operation fixed in §6.1.2, under which a grade and its overlay are produced on-site and queued for later reconciliation with hospital systems. The candidate's prior published architecture already placed a Recognition/Diagnosis component and a clinical-decision-support function at this point in the system (Yesmukhamedov et al., 2025, Fig. 1 and Table 3, p. 83), but presented them as a design without prototype or quantitative validation (its acknowledged limitation, §II.6); the present section's contribution is the resource-conditioned selection logic that binds that component to the OD-6 envelope. The deployed interface through which the grade and overlay reach the clinician is illustrated by the system's dashboard (Figure 6.1), shown here as an illustration of the designed interaction, not as evidence of measured performance. With the inference function fixed as a design specification, §6.3 turns to the deployment of this local inference path in telemedicine and portable settings, and to the physician-in-the-loop interface that consumes its outputs.

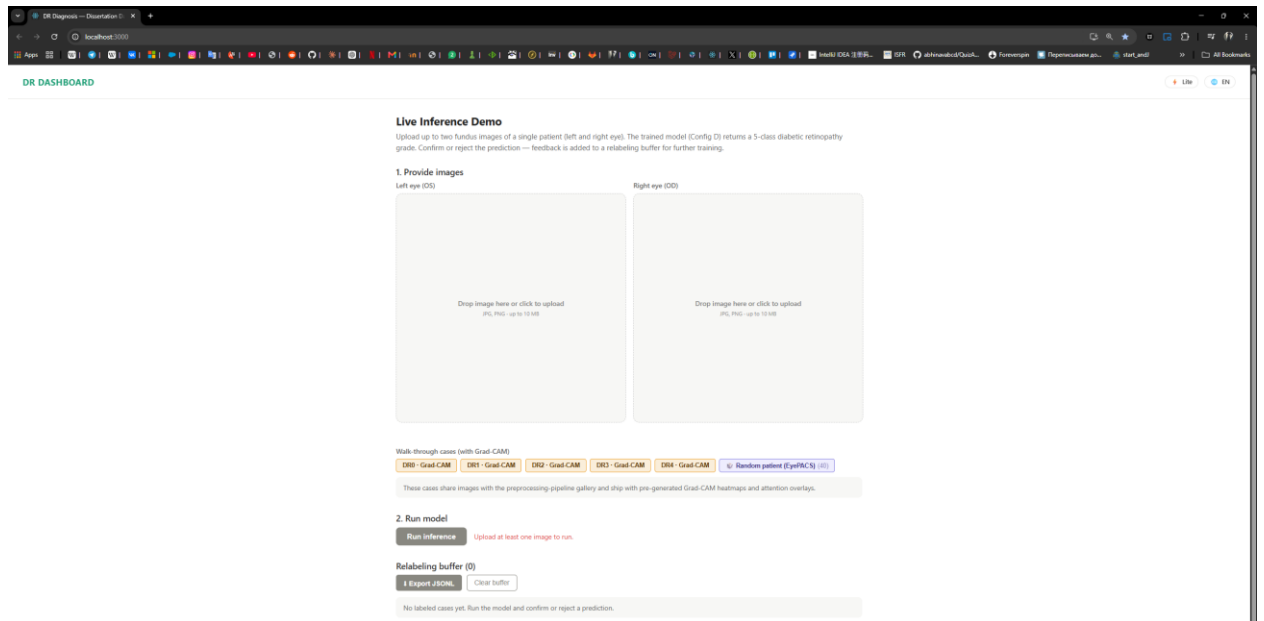


Figure 6.1 – Deployed web-application / decision-support dashboard

§6.3.1 Telemedicine and Portable Device Support for Rural Deployment

The two preceding sections fixed a local grade-producing path: a configurable Preprocessing Engine (§6.2.1) and a resource-budgeted Inference Module (§6.2.2) that together produce a DR grade and a decision-support overlay on the deployment host. This section specifies how that path is deployed in the setting the whole architecture targets — distributed, connectivity-limited, and rural. The deployment is a design specification: no system has been installed, no screening campaign has been run, and the applicability of the design to any particular healthcare context, including Kazakhstan's, is bounded by the absence of field testing (SB-4.1; SB-4.3). External screening programmes are cited in what follows as evidence that the *deployment modality* is established in practice, never as measurements of this system.

6.3.1.1 Deployment in distributed telemedicine systems

A telemedicine deployment distributes image capture to peripheral sites while concentrating interpretation and oversight elsewhere, and the architecture supports this distribution without modification because the design choices of §6.1.2 were made for it. The module set admits two natural placements: the Ingestion, Preprocessing Engine, and Inference modules sit at the peripheral capture site, while the physician-in-the-loop review surface (§6.3.2) sits at a reading centre that may serve many sites. The Ingestion module (FR-1) admits images from heterogeneous and portable capture sources at the peripheral site; the local inference path produces the grade and overlay on-site; and the asynchronous, store-and-forward integration layer queues the result for reconciliation with central systems when connectivity permits, rather than requiring a live link during acquisition. This is the operational meaning of the topology pruned in §6.1.2: distributed telemedicine over intermittent links is admissible precisely because no step on the critical path depends on synchronous connectivity. The error-handling responsibility of the Ingestion module (FR-7) is

load-bearing in this placement: an input that cannot be carried through the always-on preprocessing stages is flagged at the peripheral site, so that an unscreenable acquisition is surfaced for re-capture locally rather than discovered only after transmission to the reading centre. That deep-learning DR screening can function within such telemedicine and primary-care workflows is supported by external evidence. A meta-analysis of deep-learning screening in primary care reported pooled sensitivity of 0.87 and specificity of 0.90 across ten studies [62]; the same analysis, however, found extreme between-study heterogeneity ($I^2 > 99\%$) and a positive predictive value of only 49% at the 10% disease prevalence typical of a primary-care population, leading its authors to conclude that the evidence does not yet allow a general recommendation for routine clinical use. Prospectively, a deep-learning system has been integrated into a national screening programme across nine primary-care sites in a middle-income country, with real-time grading at the point of care and task-shifting to non-physician personnel [59]. A deep-learning DR system has likewise been deployed for screening within a sub-Saharan African programme [57], and a human-centered field evaluation across eleven clinics in Thailand documented that socio-environmental factors and image-quality gating materially shaped real-world deployment [14] — the latter a reminder that the modality is feasible but its field performance is contingent on conditions a design specification cannot guarantee. These studies, together with technical-and-clinical reviews of deep learning in ophthalmology [64], establish that the telemedicine screening modality is feasible and current; they are cited for that precedent only, with their own datasets and validation designs, and not as performance attributable to the system designed here.

6.3.1.2 Integration with national eHealth platforms

Where a national or regional eHealth platform exists, the system integrates with it through the data-management layer specified in §6.1.2 rather than through a bespoke interface: patient linkage, image archiving, and result return are carried over the standardized FHIR/HL7 exchange (NFR-6) against the platform's PACS and EHR services. The candidate's prior published architecture situated exactly this integration within the Kazakhstan context, describing the PACS, cloud, and electronic-record components a national deployment would interoperate with (Yesmukhamedov et al., 2025, pp. 86–87); that work was a design and reported no implementation or validation of the integration, and explicitly noted that successful adoption would require investment in diagnostic equipment and infrastructure (its acknowledged limitation, §II.6). The dissertation inherits that prerequisite rather than dissolving it: integration with a national platform is a design specification conditioned on the platform and connectivity being present (DGL-4), and the store-and-forward design is the architecture's accommodation of the case where they are intermittently absent. The unit of exchange across this boundary is the telemedicine encounter, and the entity-relationship model carried from §6.1.2 already fixes its persisted form — Patient, Telemedicine Session, Device, Diagnostic Result, and AI Model are the entities the platform integration reads and writes. Because these are patient-identifying records traversing a distributed link, the integration boundary is also where data-protection obligations concentrate; their protocol-level treatment, and the

qualification that the GDPR/HIPAA framing is a design requirement and not a certified status, are specified in §6.4.1. Any projected national-scale benefit associated with such integration in the prior publication — figures for rural population reached or cost reduction — is a third-party projection cited for context and is not a finding of this work (SIR-8).

6.3.1.3 Real-time remote DR screening in low-resource regions

The low-resource regime is not a peripheral case for this architecture but its defining target: the OD-6 envelope — possible absence of GPU acceleration, a sub-16 GB memory budget, near-real-time throughput, and intermittent connectivity — is the operational definition of exactly the regions this sub-section concerns. Because the inference path is budgeted to run locally on such a host (§6.2) and the integration is asynchronous (§6.1.2), producing a grade in or near real time at a remote, low-resource site is a feasibility the design is constructed to admit. It is, however, a design feasibility and not a demonstrated capability: no real-time remote screening has been performed or measured in this research, and the claim made here is one of architectural admissibility under the OD-6 envelope, not of attained throughput or accuracy in the field (SB-4.1; SB-4.3). The motivation for the design is the documented screening gap — only an estimated 35–55% of diabetic patients comply with annual ocular screening [2] — which remote, portable-capture screening is intended to narrow. Portability enters through the Ingestion module: the module is specified to accept input from portable and smartphone-based fundus cameras, the class of device the prior publication identified as the affordable capture front-end for rural deployment (Yesmukhamedov et al., 2025, p. 87, where a representative portable camera cost is cited). Portable capture is precisely the case in which the Preprocessing Engine's role is most pronounced: the field-of-view normalization, masking, and flat-field correction stages specified in Chapter 3 are designed to reduce the acquisition variability that inexpensive, non-mydratic, and operator-dependent devices introduce, so that the inference module receives a comparably conditioned tensor regardless of the capture front-end. Device-independent analysis is itself an established direction — De Fauw et al. [65] decoupled a retinal referral system from specific scanner hardware through an intermediate representation, albeit for OCT rather than fundus photography — but whether the conditioning is sufficient to preserve grading performance across fundus-camera classes is an empirical question reserved for the experimental chapters and is not asserted here. The architecture therefore pairs an inexpensive, mobile capture front-end with a host-local inference path and an asynchronous return channel — a configuration designed for the low-resource setting, with its real-world performance reserved for future field evaluation.

The deployment design across these three settings is unified by a single property carried from §6.1.2 and §6.2: the grade-producing path is local and the hospital-system exchange is asynchronous, which is what makes distributed, infrastructure-light, intermittently-connected operation admissible by design. The interaction through which a clinician receives the grade and overlay at the reviewing end of this deployment is illustrated by the system's dashboard (Figure 6.1), shown as an illustration of the designed interface rather than as evidence of field performance.

The next section specifies that interface as a physician-in-the-loop decision-support surface.

§6.3.2 Physician-in-the-Loop Decision Support Interface

The telemedicine deployment of §6.3.1 placed a review surface at the reading centre, where a clinician receives the output of the local inference path. This section specifies that surface. As with the rest of the chapter it is a design specification: the interface is described as an interaction design, and no usability study, clinician-acceptance evaluation, or decision-quality measurement is reported here (SB-4.1).

The interface presents two artifacts for each screened eye: the five-class diabetic-retinopathy grade produced by the inference module (FR-3), and the post-hoc Grad-CAM overlay that accompanies it (FR-4). The overlay is presented to the clinician as a model-sensitivity map — an indication of the image regions to which the prediction was most responsive — and its interpretive status is fixed accordingly: it is read as support for understanding why the model produced a given grade, not as a clinical localization of pathology, and it does not detect, delineate, or certify lesions (NC-14). The interface thus surfaces both a decision proposal and a legibility aid for that proposal, but it surfaces them as material for a clinician to interpret rather than as a conclusion.

This is the architectural expression of the binding scope of the whole system: it is a decision-support component operating with a physician in the loop, not a standalone diagnostic instrument (SB-1.3). The interface is designed so that the clinician is the decision-maker — the grade and overlay are inputs the clinician interprets, audits against the image, and may override — and the system issues no autonomous diagnosis. This arrangement realizes the role the candidate's prior published architecture assigned to the physician: not a passive consumer of model output but the system's "tuner" and auditor, who interprets results and can adjust or correct the model's behaviour, mediated by a Doctor-AI feedback loop through which the clinician requests clarification or correction after receiving an analysis (Yesmukhamedov et al., 2025, pp. 78–79, 85). That publication specified this interaction as a design and reported no prototype or evaluation of it (its acknowledged limitation, §II.6); the present section carries the same interaction forward as the interface contract of the screening system, again as a specification rather than a tested artifact.

The override-and-audit channel is not an optional convenience but the architectural counterpart of the decision-support scope boundary. By requiring that every grade pass through a clinician who can accept, amend, or reject it, the interface keeps clinical accountability with the physician rather than transferring it to the model, which is the design condition under which an automated screening aid can be introduced into a workflow without claiming autonomous diagnostic authority. The feedback channel additionally lets the clinician register disagreement or request clarification, so that the interface is bidirectional — carrying the model's proposal to the clinician and the clinician's judgement back — rather than a one-way display. This bidirectionality has a persistence consequence the data layer must honour: the

Diagnostic Result entity carried from the entity-relationship model of §6.1.2 records not only the model's grade but the clinician's disposition of it — acceptance, amendment, or override — so that the authoritative screening outcome stored against the patient record is the reviewed result rather than the raw model output. The interface is therefore the point at which the model's proposal becomes a clinician-owned decision, and the design fixes that ownership in what is persisted, not merely in what is displayed. In a distributed deployment this surface is also where the screening gap the system addresses is ultimately closed or not: the design positions the clinician's review, not the model's grade, as the act that yields a screening decision, consistent with the system's role as an aid to clinicians managing a screening burden that ordinarily exceeds specialist capacity [2].

So specified, the interface fixes the system as a physician-in-the-loop decision-support surface whose every output is subject to clinician interpretation and override, with usability and clinical-acceptance evidence reserved for future evaluation. The records that this interface reads and writes — the grade, the overlay, the patient linkage — are patient data traversing the system's storage and exchange layer; the next section specifies the data-management and protection protocols that govern them.

§6.4.1 GDPR/HIPAA-Aligned Data Management Protocols

The interface of §6.3.2 and the integration layer of §6.1.2 read and write patient-identifying records — fundus images, grades, Grad-CAM overlays, patient linkages, and the Diagnostic Result that now carries the clinician's disposition — across a distributed, intermittently-connected deployment. This section specifies the data-management and protection protocols that govern those records, discharging the security treatment deferred from §6.1.2 and §6.3.1.2. One boundary must be fixed before any protocol is described and held throughout: the protocols below are a design specification framed to *align with* the GDPR and HIPAA data-protection regimes; this alignment is a requirement the architecture is designed to satisfy, not a certified, audited, or tested compliance status (SB-4.2; SB-4.1).

The protection mechanisms are concentrated at the data-management layer, consistent with the security architecture of the candidate's prior published design. That work specified a security gateway mediating inter-component transfer, together with encryption, user authentication, and — within its Data Management Platform — data anonymization and protected exchange (Yesmukhamedov et al., 2025, Table 3, p. 83; Table 4, p. 86); it presented these as a design and reported no implementation or security testing of them (its acknowledged limitation, §II.6). The present specification organizes these mechanisms into the protocols the screening system requires: encryption of records both at rest and in transit; authentication of every actor and service that touches patient data; role-based access control restricting records to the clinician and roles entitled to them; de-identification of data used outside the immediate care context; and audit logging of access and of the clinician disposition recorded against each Diagnostic Result.

Mapped onto the principles the two regimes share, these protocols address confidentiality and integrity through encryption and authentication, access limitation and purpose limitation through role-based access control and de-identification, and accountability through audit logging — the design-level realization of the security and patient-data-protection requirement fixed as NFR-7 in §6.1.1. Two further principles are specified as design requirements at the same level. Data minimization and retention bound the records the system holds to those needed for the screening purpose and to a defined retention period, rather than accumulating images indefinitely; and the design reserves a channel for data-subject requests — access to, and where applicable erasure of, a patient's records — to be serviced through the same authenticated access layer. The mapping is deliberately stated at the level of principle rather than of specific statutory provision: which articles of the GDPR or controls of the HIPAA Security Rule a given deployment must satisfy depends on the jurisdiction and the processing context, and naming a particular provision as satisfied would exceed what this design establishes.

The store-and-forward topology fixed in §6.1.2 sharpens these requirements rather than relaxing them. Because results and images are held locally on the deployment host while connectivity is absent and reconciled with hospital systems opportunistically, the protection design cannot assume a trusted, continuous channel: data queued at rest on a peripheral host must be encrypted in place, and the asynchronous exchange with PACS and EHR services must be mutually authenticated and encrypted in transit when it occurs. The intermittently-connected, distributed character that makes the system deployable in resource-limited settings is therefore also what makes its data-protection design non-trivial, and the protocols are specified to hold under that character rather than in spite of it.

Finally, the epistemic boundary must be made explicit, because the prior publication did not. That work concluded that its proposed system "is GDPR and HIPAA-compliant" (Yesmukhamedov et al., 2025, p. 90); the present dissertation does not carry that conclusion forward as stated, and reframes it as a design requirement. No compliance audit, formal certification, data-protection impact assessment, or penetration test has been conducted in this research, and none is reported (SB-4.2; NC-9). The protocols specified here are the design the system would have to implement and then have independently assessed to claim compliance; the claim of this section is the specification, not the compliance. Jurisdiction-specific conformance — including the national data-protection regime applicable to a Kazakhstan deployment — is a design target whose applicability is taken up, within the same design-only bound, in §6.4.2.

§6.4.2 Applicability to Kazakhstan Healthcare Infrastructure

The preceding section reserved jurisdiction-specific conformance — including the data-protection regime of a particular deployment country — as a design target. This section takes up the corresponding applicability question for the dissertation's named context, Kazakhstan, and answers it as a design argument under an explicit bound: the architecture is argued to be applicable to Kazakhstan's healthcare

infrastructure by design, but no deployment, pilot, or field evaluation has been conducted in any Kazakhstan clinical setting, and the applicability claimed here is therefore one of design fit, not of demonstrated effect (SB-4.1; SB-4.3).

The context against which applicability is judged is the one documented in the candidate's prior published work, which characterized Kazakhstan's ophthalmological-care setting as one of marked specialist scarcity and rural dispersion — on the order of 1,200 ophthalmologists for the national population, approximately 40% of which is rural, against a diabetes prevalence of roughly 8% and limited rural access to specialist eye care (Yesmukhamedov et al., 2025, pp. 74, 77, 86–87). That publication reported this context as a design motivation and did not validate any system against it, noting that adoption would require investment in equipment and infrastructure, adaptation to local data, and the development of national standards (its acknowledged limitation, §II.6). These figures are cited here as the documented context the design addresses, not as findings of this dissertation.

The applicability argument is that the constraints of this context are precisely the constraints the architecture was built to respect, so that its design choices map onto Kazakhstan's needs rather than merely being compatible with them. Specialist scarcity is the condition under which a physician-in-the-loop decision-support system — which extends the reach of a small number of specialists by allowing review of screening proposals rather than primary acquisition by each, and which is compatible with task-shifting to non-specialist personnel — is most useful (§6.3.2). Rural dispersion and intermittent connectivity are the conditions the store-and-forward topology and portable-capture ingestion were designed for (§6.3.1), and the cost barrier to rural deployment is what the inexpensive capture front-end and the non-GPU, sub-16 GB inference host address (§6.2). The OD-6 envelope, in other words, is not a generic resource-limited abstraction here: it was operationalized to match the kind of setting Kazakhstan's rural ophthalmological-care context exemplifies, and the applicability of the design follows from that envelope-fit.

This applicability is, however, a claim about fit and not about benefit. The prior publication projected substantial national-scale benefits from a deployment of this kind — access to retinal care for several million rural residents, a 20–30% reduction in late-stage diabetic-retinopathy complications, a 15–20% reduction in associated healthcare cost, and screening of around two million people annually (Yesmukhamedov et al., 2025, pp. 87–88). These are third-party projections, cited here for context to indicate the scale at which the problem is framed; they are not outcomes of this dissertation, are not attributed to the system designed here, and no such coverage, cost, or complication figure is advanced as a result of this work (SIR-8; NC-3). Whether a deployment of this architecture in Kazakhstan would approach any of these figures is exactly the empirical question that field testing — not undertaken in this research — would have to answer.

Regulatory applicability is treated at the same design-target level established in §6.4.1: a Kazakhstan deployment would have to satisfy the national data-protection regime in addition to the GDPR/HIPAA-aligned protocols specified there, and the architecture is designed to accommodate such conformance, but no specific statutory conformance is claimed as established. The applicability of the design to Kazakhstan

is thus a bounded design argument — a demonstrated fit between the system's envelope and a documented national context, with realization, benefit, and certification all reserved for work beyond this dissertation. With this argument made, the chapter's design specification is complete; the concluding section consolidates it and reaffirms the design-only status that has governed throughout.

§6.C Conclusions to Chapter 6

This chapter translated the integrated preprocessing-CNN model of the preceding chapters from an object of measurement into a component for deployment, and specified the architecture of an automated diabetic-retinopathy screening system built around it for resource-limited environments. The specification proceeded in a single direction. It began by fixing the requirements the system must satisfy — a functional set describing what the system does and a non-functional set describing the resource-limited envelope under which it must do it (§6.1.1) — and then decomposed the system into a set of loosely-coupled modules whose topology, including its integration with PACS and EHR infrastructure, was shown to be an admissible realization of that envelope (§6.1.2). It specified the internals of the two computational modules — the configurable Preprocessing Engine (§6.2.1) and the resource-budgeted Inference Module (§6.2.2) — and then the deployment of the resulting local screening path in distributed telemedicine and portable settings (§6.3.1) and through a physician-in-the-loop decision-support interface (§6.3.2). It closed by specifying the data-management and protection protocols governing the system's patient records (§6.4.1) and arguing the design's applicability to Kazakhstan's healthcare infrastructure (§6.4.2).

A single analytical thread runs through this progression. The non-functional envelope operationally defined for this dissertation — the resource-limited deployment context (OD-6) — was the binding design driver at every stage, not merely a constraint registered at the outset. It is what pruned the admissible topology to an asynchronous, store-and-forward arrangement rather than a cloud-dependent one; what motivated the Preprocessing Engine's configurability and the Inference Module's resource-conditioned model-selection logic; and what shaped the distributed, intermittently-connected deployment and the data-protection design that must hold under it. The chapter's contribution is therefore not the inventory of components, which a screening system of any kind would share, but the requirement-driven design that fits those components to the resource-limited envelope — a design that sharpens and binds the candidate's previously published architecture to the model and the deployment envelope of this dissertation.

That contribution is, and remains throughout, a design specification. No prototype has been implemented, no field test has been conducted, and no projected deployment outcome has been stated as a finding of this research (SB-4.1). The properties the chapter specifies are accordingly held as requirements and targets rather than attained results: the diagnostic effectiveness the inference path is required to meet (OD-5) is an acceptance target whose attainment is reserved for the experimental chapters; the GDPR/HIPAA framing of the data-management protocols

is an alignment by design and not a certified or tested compliance status (SB-4.2); the applicability to Kazakhstan is a demonstrated fit between the design and a documented context, not a demonstrated benefit, and the national-scale projections cited along the way remain third-party projections rather than outcomes of this work (SB-4.3; SIR-8). The formal UML elaboration of the architecture likewise remains a deferred asset. The realization, validation, certification, and evaluation of this specification are the work it makes possible, not the work it reports; the contribution of the chapter is the specification itself (CFC-1.5).

APPENDICES

Appendix A — Source Code of the Preprocessing Pipeline

Section 4.1.3 committed the preprocessing pipeline and training code to version control and promised their reproduction here, so that the transformation applied to every image is recoverable as source rather than described only in prose. This appendix discharges that promise. Its inclusion is not incidental documentation: under the central thesis of this work — that the model is the composition of preprocessing and the convolutional network, with the eight-stage pipeline an integral model component rather than ancillary data preparation — the preprocessing source is part of the model specification. Reproducing it is what makes that thesis auditable rather than merely asserted.

The code is organized as a Python package under `experiments/src/`. The preprocessing stack lives in the canonical package `experiments/src/preprocessing/`, whose public interface is exported from `experiments/src/preprocessing/__init__.py` and whose orchestrator, `PreprocessingPipeline` in `experiments/src/preprocessing/pipeline.py`, chains the eight stages in the fixed order specified in Chapter 3. The remaining apparatus required to reproduce the experiments is distributed across sibling subpackages of `experiments/src/`: `data/` (dataset loaders, the patient-level stratified k-fold splitter, unified augmentation, and label harmonization), `models/` (the ResNet-50 and EfficientNet factories and the two-stage fine-tuning utility), `training/` (the training loop, focal loss with inverse-frequency weights, and checkpoint management), `evaluation/` (metrics, calibration, and statistical tests), `explainability/` (Grad-CAM, IoU/ALO, and overlay rendering), `experiments/` (the seven experiment drivers), and `utils/` (YAML configuration handling, the reproducibility seed utility, and image-quality metrics). The catalogue below is confined to the preprocessing package, which implements the model component that distinguishes this work; the wider tree is named here only so that the reproduction is locatable in full.

Table A.1 maps each pipeline stage of Chapter 3 to the module that implements it. Every path is given relative to the repository root and corresponds to a file present in the versioned source tree.

Table A.1. Pipeline stage → implementing module (experiments/src/preprocessing/).

Stage	Description (Ch 3)	Implementing module
0	Canonical flip (left→right eye orientation)	canonical_flip.py, canonical_orientation.py
1	OD-fovea rotation normalization	od_fovea_detect.py, canonical_orientation.py
2	FOV crop + isotropic resize to 512×512 (centered zero-padding)	crop_resize.py
3	FOV mask generation (binary → 4th channel)	crop_resize.py (mask returned with the resized image)
4	Flat-field correction (adaptive $\sigma = 0.07 \cdot D$, inside mask)	flat_field.py
5	Dual-constraint CLAHE (LAB L-channel; stochastic at train)	upgraded_clahe.py, polar_clahe.py, clahe.py
6	Augmentation (unified affine + brightness/contrast + PCA colour; train only)	experiments/src/data/augmentation_unified.py
7	Dataset-specific normalize → tensor (always last)	imagenet_normalize.py
—	Configuration surface (baseline vs full-pipeline presets)	config.py (PreprocessingConfig, PIPELINE_PRESETS)
—	Stage orchestration	pipeline.py (PreprocessingPipeline)

	(fixed execution order)	
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Stage 5 is implemented by more than one module because the pipeline ships with a polar variant of the dual-constraint CLAHE as its current default (`polar_clahe.py`), alongside the rectilinear tile-based formulation (`upgraded_clahe.py`); both are present in the versioned source and are reproduced as catalogued. Stages 0–5 and 7 are applied identically at training and inference, except that the Stage 5 contrast enhancement is stochastic at training time; Stage 6 augmentation is applied only during training and is inserted before the Stage 7 normalization so that it operates on `uint8` images. The baseline configuration (Configs A and C) bypasses Stages 0–6 in favour of a simple stretch-resize to 512×512 followed by ImageNet normalization on three channels, selected through the configuration preset rather than by editing the pipeline; the full pipeline (Configs B and D) runs all eight stages and emits four channels (RGB plus the FOV mask).

To establish that this catalogue reproduces real source rather than a paraphrase of it, the Stage 4 flat-field module is reproduced verbatim from `experiments/src/preprocessing/flat_field.py`:

"""

Stage 4: Flat-Field Correction.

Reduces uneven illumination by subtracting a heavily blurred version of the image and re-centering at 128:

```
corrected = image - GaussianBlur(image,  $\sigma$ ) + 128
```

A large σ captures only the low-frequency illumination gradient, so the subtraction removes broad brightness variation while preserving local vessel and lesion detail.

σ is computed adaptively as $\sigma = 0.07 \times \text{FOV_diameter}$. Correction is applied only inside the FOV mask (padding pixels are left at zero).

Input/output images are RGB `uint8` NumPy arrays.

"""

```
from __future__ import annotations
```

```
import cv2
```

```
import numpy as np
```

```
def apply_flat_field(
    image: np.ndarray,
    sigma: float = 45.0,
```

```

    mask: np.ndarray | None = None,
) -> np.ndarray:
    """
    Apply flat-field correction to reduce uneven illumination.

    Algorithm::

        blur      = GaussianBlur(image,  $\sigma$ )
        corrected = image - blur + 128

    When *mask* is provided, correction is applied only inside the mask
    (``mask > 0``). Padding areas (``mask == 0``) are left at zero.

    Kernel size is derived automatically from *sigma* (passed as ``(0,
0)``
to :func:`cv2.GaussianBlur`).

    Args:
        image: RGB uint8 NumPy array of shape ``(H, W, 3)``.
        sigma: Gaussian blur  $\sigma$  controlling the spatial scale of the
            illumination estimate.
        mask: Optional binary mask of shape ``(H, W)`` (float32 or
uint8).
            When provided, only pixels where ``mask > 0`` are corrected;
padding regions remain zero.

    Returns:
        Corrected RGB uint8 NumPy array of shape ``(H, W, 3)``.
    """
    blur = cv2.GaussianBlur(image, (0, 0), sigma)
    corrected = image.astype(np.float32) - blur.astype(np.float32) +
128.0
    corrected = np.clip(corrected, 0, 255).astype(np.uint8)
    if mask is not None:
        mask_3ch = np.expand_dims(mask > 0, axis=-1).astype(np.uint8)
        corrected = corrected * mask_3ch # zero out padding
    return corrected

```

The remaining modules follow the same conventions established for the codebase — type-hinted signatures, Args/Returns docstrings, paths resolved from configuration rather than hardcoded, and `pathlib.Path` throughout — and are reproduced in full in the assembled document from the same package. The pipeline lineage descends from the candidate's prior published work on upgraded CLAHE and preprocessing-classification integration (yesmukhamedov-scopus-q2.md/yesmukhamedov-scopus-q3.md, yesmukhamedov-kbtu.md, and the conference paper, ♦ prior own work; SIR-4); the source reproduced here formalizes and consolidates that line into the single versioned eight-stage system specified in Chapter 3. Consistent with the hardware-specific reproducibility bound stated in §4.1.3, the source is reproducible on equivalent hardware, but the computational-

efficiency characteristics it exhibits remain specific to the documented setup (DGL-2); no claim of performance, accuracy, or deployment readiness is made by reproducing it. With the source catalogued and a representative module shown to be the real, on-disk implementation, the reproducibility loop opened in §4.1.3 is closed: the fixed configuration (Table 4.2), the documented hardware, and this versioned code together render the experimental pipeline recoverable.

Appendix D — Certificates and Publication Confirmations

This appendix records the prior peer-reviewed publications through which the components of this dissertation were progressively disseminated, together with the indexing and publication-confirmation assets held for them. It provides the documentary basis for the approbation and publications statements of the front matter (§0.11, §0.12). Every publication catalogued here is co-authored by the candidate and is treated throughout the dissertation as prior own work: where the main text draws on any of these outputs, it does so under explicit self-citation (SIR-4), so that the relationship between the published record and the present contribution is transparent. The appendix records the *existence*, *venue*, and *provenance* of each publication; it does not restate or endorse the performance figures reported inside those publications, which are re-examined — and, where necessary, bounded — under this dissertation's own governance rather than imported as findings.

Table D.1 lists the five distinct peer-reviewed outputs. They span the four strands the dissertation integrates: preprocessing-and-classification of fundus images, the two-stage transfer-learning adaptation, the thermal-optical laser model, and the AI-diagnostic system architecture. The first row corresponds to a single article in the *Eastern-European Journal of Enterprise Technologies*; two literature cards (yesmukhamedov-scopus-q2.md and yesmukhamedov-scopus-q3.md, indexed #23 and #24) were prepared from that one article from different analytical perspectives, and it is therefore counted once here (LITERATURE_INDEX note 5; SIR-5).

Table D.1. Candidate's prior peer-reviewed publications (all ♦ prior own work; SIR-4).

#	Authors (year)	Title	Venue	Indexing	DOI	Card ID
1	Sapakova, Yesmukhamedov & Sapakov (2025)	Development of an image quality enhancement approach for diabetic retinopath	<i>Eastern-European Journal of Enterprise Technologies</i> , 4(9(136)), 79–88	Scopus (journal)	10.15587/1729-4061.2025.335570	#23 / #24

		y diagnosis				
2	Sapakova, Yesmukhamedov, Sapakov, Yemberdiyeva & Kozhamkulova (2025)	Methods for pre-processing and analysis of fundus images for detection of diabetic retinopathy	<i>Procedia Computer Science</i> , 272, 496–501 (DS 2025, Istanbul)	Scopus (conference)	10.1016/j.procs.2025.10.237	#19
3	Yesmukhamedov, Sapakova, Kozhamkulova, Daniyarova & Armankyzy (2025)	Methods for Preprocessing and Analysis of Fundus Images for Diabetic Retinopathy Detection	<i>Herald of the Kazakh-British Technical University</i> , 4(75), 119–130	KBTU Herald	10.55452/1998-6688-2025-22-4-119-130	#21
4	Sapakova, Daniyarova, Yesmukhamedov, Armankyzy, Emberdieva & Kaldybaeva (2024)	Mathematical Modeling of Laser Exposure on Fundus Tissues in the Treatment of Diabetic Retinopathy	<i>Vestnik KazUTB</i> , 2(27-740)	KazUTB Vestnik	10.58805/kazutb.v.2.27-740	#20
5	Yesmukhamedov, Sapakova, Al-Haddad & Daniyarova (2025)	Development of an information system architecture for healthcare institution	<i>News of the National Academy of Sciences of the Republic</i>	NAS RK	10.32014/2025.2518-1726.345	#22

		s using artificial intelligence	of Kazakhstan, Physico- Mathematical Series, 2(354), 74–91			
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Two documentary notes attach to the catalogue and are recorded rather than suppressed. First, the two cards for the EEJET article (entry 1) are retained because the dissertation cites that article from more than one section; the duplication is bibliographic bookkeeping, not two publications (SIR-5). Second, the card yesmukhamedov-scopus-q2.md carries an internal unique-ID field (LC-Altimemy-2021) that does not match its own recorded citation; the anomaly is flagged here for corpus hygiene and has no bearing on the bibliographic entry, which keys on the recorded citation and DOI above. Consistent with the no-amplification rule, none of the performance figures stated inside these publications — including the high-accuracy result reported in the EEJET article's own abstract — is repeated or endorsed in this appendix or treated as evidence for the dissertation's hypotheses (SIR-1).

The publication and indexing confirmations held for these outputs are stored as image assets and are listed below by their real repository path; each is inserted as an image in the assembled document during final assembly (§11.2). The assets map to the catalogue as follows:

- [APP-D: Scopus journal indexing confirmation – defense/presentation/assets/publications/SCOPUS.png] → entry 1 (EEJET).
- [APP-D: Scopus conference indexing confirmation – defense/presentation/assets/publications/SCOPUS_CONF.png] → entry 2 (Procedia / DS 2025).
- [APP-D: KBTU Herald publication confirmation – defense/presentation/assets/publications/KBTU.png] → entry 3 (KBTU Herald).
- [APP-D: KazUTB Vestnik publication confirmation – defense/presentation/assets/publications/KAZTBU.png] → entry 4 (KazUTB Vestnik).
- [APP-D: NAS RK publication confirmation – defense/presentation/assets/publications/AKADEMY.png] → entry 5 (NAS RK News).
- [APP-D: Aggregate publications record – defense/presentation/assets/publications/PUBLICATIONS.png] → consolidated confirmation across the publication record.

Together, Table D.1 and these six confirmation assets constitute the approbation evidence underpinning §0.11 and §0.12: a peer-reviewed publication trail, co-authored by the candidate and identified as prior own work throughout, recorded here with its provenance intact and without importing any of its internal performance claims as findings of this dissertation.

LIST OF REFERENCES USED

> In order of first appearance (GOST 7.32-2001 §6.11). Entries derived from the literature cards; GOST 7.1-2003 punctuation refinement is the final typesetting step.

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> **Note — pending entries.** Self-citations of the candidate (Yesmukhamedov; Sapakova, Yesmukhamedov & Sapakov, 2024–2025) and the Appendix-D publication records are intentionally left in author-year form: the same surface form maps to different self-papers by section, so their [N] numbers require manual per-section disambiguation before the final bound submission. Their SIR-4 'prior own work' prose framing is preserved in the running text.